

Mathematical Glossary for Analysis/FabiusFunction

Complete updated edition for the current formal and documentation corpus

Terminology synthesized from Vladimir Reshetnikov’s ProveIt repository

Repository state inspected: 3 September 2026

Formal census checkpoint: 1 September 2026

Abstract

This edition is a ground-up corpus-level update of the mathematical glossary for Analysis/FabiusFunction. It contains **976 alphabetized headwords**, replacing the prior 437-headword edition. The audit covers the bounded and global Fabius functions, Rvachev’s up-function, inverse and computability theory, exact dyadic arithmetic, Thue–Morse analysis, probability and operator models, q -Pochhammer and Gaussian-binomial calculus, basic hypergeometric and theta functions, Fourier and spectral theory, splines and sampling, orthogonal polynomials, combinatorial coefficient calculus, Lambert- W and Mellin endpoint analysis, polynomial–logarithmic transseries, and the active frontier programs. Definitions give a characteristic formula, normalization or branch choice when material, and the role of the notion in the project. The typography and running-head design preserve the style of the repository’s principal Fabius–Rvachev documents.

Repository snapshot and coverage. The live formal census used by this edition records 675 source modules and 8,909 public declarations at its 1 September 2026 checkpoint. The active documentation/notation inventory inspected on 3 September 2026 records 172 active `.tex` sources including the notation catalogue, of which 73 are standalone documents; it also records 146 shared semantic notation commands and 30 validated document-local non-Fourier hat names. Historical sources below `docs/archive/` were used for provenance and alias control, not counted as current theorem-bearing duplicates.

Editorial unit of completeness. Completeness is concept-based rather than token-based. The glossary includes every mathematically substantive concept, named object, normalization, proof-status distinction, transform convention, asymptotic coordinate, and recurring construction found in the active corpus. It consolidates transparent spelling and inflection variants; it does not create separate headwords for elementary words such as “number” or “function,” local dummy variables, one-use theorem labels, or mechanically predictable specializations. A local symbol not promoted to the shared notation API has no project-wide meaning unless its defining document supplies one.

How to read the status badges

LEAN-PROVED

The named object or core theorem has an active Lean source and belongs to the formal facade. The badge does not claim that every expository sentence in the entry is itself a theorem statement.

CLASSICAL

Standard background mathematics used by the project.

PROJECT / MIXED	A classical notion with repository-specific normalization, a bridge between formal and analytic presentations, or a topic whose relevant subresults have mixed proof status.
FRONTIER	Active research material, conjectural structure, or a semi-formalized program. No kernel-checked status is implied.
CONVENTION	A notation, normalization, branch, equality mode, or editorial governance rule.
HISTORICAL	A source-faithful or retired usage retained solely for interpretation and provenance.

Contents

How to read the status badges	1
Headword index	4
1 A	11
2 B	18
3 C	26
4 D	39
5 E	47
6 F	54
7 G	64
8 H	72
9 I	77
10 J	84
11 K	86
12 L	87
13 M	97
14 N	104
15 O	109
16 P	112
17 Q	124
18 R	131

19 S	141
20 T	155
21 U	162
22 V	165
23 W	167
24 Z	169
A Normalization and notation guardrails	170
B Representative Lean crosswalk	171
C Coverage map and edition delta	172
C.1 Status distribution	173
D Audit method and maintenance contract	173
E Repository sources inspected	173
F Selected external reference works	174

Headword index

This index is hyperlinked to the alphabetized definitions. Status badges appear at each entry, not in the index.

Abel equation	Absolute continuity	Absolute convergence
Absolute Legendre summability	Activation deficit	Activation profile
Active documentation corpus	Adjacent Gaussian-binomial ratio	Admissible class
Affine covariance of fractional integrals	Affine mixed difference	Affine transformation
Algebraic function branch	Aliasing	All-orders asymptotic expansion
Almost everywhere	Almost-sure convergence	Amplitude
Analysis operator	Analytic continuation	Analytic germ
Analytic locus	Analytic moment	Analytic zero multiplicity
Analyticity at a point	Angular-frequency Fourier transform	Antidiagonal polynomial
Appell deconvolution	Appell sequence	Approximate identity
Approximation theory	Arbitrary-phase synthesis	Arbitrary-phase up synthesis
Arithmetic ray	Associahedron	Asymptotic coefficient jet
Asymptotic equivalence	Asymptotic expansion	Asymptotic scale
Atom of a measure	Atomic function	Atomless law
Automatic sequence	Averaging operator	B-spline
Bailey lemma	Bailey pair	Balanced basic hypergeometric series
Banach fixed-point theorem	Barnes G -function	Base- q digit
Base- q digit sum	Basic hypergeometric series	Basic hypergeometric series ${}_r\phi_s$
Bell–Hessenberg determinant	Bell number	Bell polynomial
Bell-polynomial asymptotic coefficient	Bernoulli number	Bernoulli recurrence
Berry–Esseen bound	Bessel inequality	Big-O and little-o notation
Bijection	Bilateral basic hypergeometric series	Bilateral Laplace transform
Binary block factorization	Binary digit sum	Binary expansion
Binary floor-digit formula	Binary reduction series	Binary reflection recursion
Binomial coefficient	Binomial polynomial basis	Binomial transform
Biorthogonal system	Bit extraction	Boolean-cube mixed difference
Boolean cumulant	Borel probability measure	Borel summability
Borel transform	Bose–Einstein kernel	Bose finite-part integral
Boundary value problem	Bounded Fabius candidate	Bounded Fabius function
Bounded variation	Box spline	Bromwich inversion integral
Bromwich saddle method	Bump function	C-infinity smoothness
Canonical clamped inverse	Canonical correlation	Canonical Fabius fixed point
Canonical Fabius specification	Canonical inverse Fabius function	Canonical solution
Caputo fractional derivative	Cardinal interpolation	Carleman condition
Carleman linearization	Carleman matrix	Carry count
Catalan number	Cauchy formula for repeated integration	Cauchy integral formula
Cauchy product	Cauchy theorem	Cauchy transform of a measure
CDF layer-cake formula	Centered Appell sequence	Centered finite spline
Centered geometric uniform law	Centered periodic correction	Centered sinc partial product
Centered Thue–Morse shell	Central binomial valuation	Central Gaussian binomial coefficient
Central limit theorem	Chaos multi-index	Characteristic function
Christoffel–Darboux kernel	Christoffel function	Closed form
Coefficient extraction	Coercion and scalar cast	Compact support
Companion Pascal row	Compiled tree	Complementary Bell number
Complete binary block	Complete exponential Bell polynomial	Complete homogeneous symmetric polynomial
Complex exponential generating function	Complex moment-generating function	Complex shift

Composite-mesh sharpness	Computable inverse Fabius function	Computable real
Computable real function	Computable sequence	Conditional asymptotic theorem
Conditional law	Continuity of Gaussian coefficients in the base	Continuous Gaussian coefficient
Continuous-linear naturality	Contour deformation	Contour integral
Contraction constant	Contraction mapping	Convexity and concavity
Convolution	Convolution of measures	Convolution power
Cornish–Fisher expansion	Correction term	Cosine series
Coupling	Covariance kernel	Critical value
Cumulant	Cumulant-generating function	Cumulant scaling law
Cumulative distribution function	Cyclotomic inverse branch	Cyclotomic polynomial
Cyclotomic valuation	D -finite function	Decay estimate
Declaration arity	Declaration inventory	Delay differential equation
Delta series	Denominator bound	Denominator clearing
Dense analytic locus	Derivative scaling	Derivative-value distribution
Determinant–Bell identity	Determinate moment problem	Differential entropy
Differential equation	Differentially algebraic function	Differentially transcendental function
Differentiation under the integral sign	Digit character	Digit multiplicity profile
Digit power sum	Digitwise factorization	Dirac comb
Dirac measure	Dirichlet eta function	Discrete antiderivative
Discrete B-spline	Discrete convolution	Discrete convolution-power kernel
Discrete Fourier transform	Discrete limit	Distribution law
Division-free identity	Division-free recurrence	Dominated convergence theorem
Double factorial	Dyadic argument	Dyadic common denominator
Dyadic derivative cell	Dyadic derivative sign law	Dyadic direction profile
Dyadic factorized-product extrapolant	Dyadic grid	Dyadic harmonic sum
Dyadic name	Dyadic rational	Dyadic refinement invariance
Dyadic spine	Dyadic Taylor evaluator	Dyadic valuation
Dyadic-value denominator LCM	Dyadic zero classification	Edgeworth expansion
Effective flatness	Effective modulus	Effective uniform continuity
Elementary-function class	Elementary logarithmic asymptotic	Elementary symmetric polynomial
Empty sum, empty product, and zero-power conventions	Endpoint asymptotic	Endpoint convention
Endpoint derivative jet	Endpoint flatness	Entire function
Entire function of exponential type	Entire-product zero ledger	Entropy power
Equality in distribution	Equality of measures	Equicontinuity
Euler–Maclaurin formula	Euler–Mascheroni constant	Euler pentagonal theorem
Euler’s first q -exponential	Euler’s second q -exponential	Eulerian number
Eulerian polynomial	Exact arithmetic	Exact dyadic derivative
Exact rational moment table	Exact topological support	Expansive matrix
Expectation	Explicit non-claim	Explicit uniform error
Exponential formula	Exponential generating function	Exponential Riordan array
Exponentially tilted measure	Extended nonnegative reals	Extension by zero
Exterior constant tails	Faa di Bruno formula	Faa di Bruno partition propagation
Fabius CDF–density bridge	Fabius distribution	Fabius fractional shift
Fabius function	Fabius recurrence sequence	Fabius–Rvachev fold map
Fabius self-map dynamics	Factorial normalization	Falling factorial
Fast signed-dyadic name	Filter convergence	Finite difference
Finite-difference table	Finite field	Finite-part integral
Finite q -binomial theorem	Finite q -Pochhammer product	Finite shifted-up synthesis
Finite sinc product	Finite spline approximation	First-defect monomial
First nonzero shell	First saddle correction	Fisher information
Fixed point	Fixed-scale up synthesis	Flat function
Flat jet	Focused import	Formal asymptotic series

Formal germ	Formal logarithm	Formal normal form
Formal power series	Formal versus analytic identity	Formal versus realized transseries
Formalization crosswalk	Fourier coefficient	Fourier energy
Fourier inversion	Fourier-Legendre coefficient	Fourier-Legendre series
Fourier mode	Fourier series	Fourier shell
Fourier transform	Fourier-transform convention	Fourier transform of a distribution
Fractional CDF layer-cake formula	Fractional integration semigroup law	Fractional iterate
Fractional order lowering	Fractional order raising	Fractional part
Frame bound	Free cumulant	Free infinite divisibility
Fubini theorem	Full logarithmic expansion	Full moment sequence c_n
Functional-differential equation	Functional equation	Fundamental theorem of calculus
Gamma-derivative tower	Gamma function	Gamma ratio
Gamma-tower ratio	Gamma-zeta coefficient	Gärtner–Ellis theorem
Gauss–Lobatto quadrature	Gauss–Radau quadrature	Gaussian-binomial bounds
Gaussian binomial coefficient	Gaussian binomial palindromy	Gaussian binomial polynomial
Gaussian coefficients at $q = -1$	Gaussian contraction	Gaussian finite difference
Gaussian integral	Gaussian Pascal recurrence	Gaussian polynomial
Gaussian quadrature	Gaussian rational	Generalized quantile function
Generating function	Genocchi number	Genocchi polynomial
Geometric Bernoulli polynomial	Geometric interpolation nodes	Geometric Lagrange interpolation
Geometric Newton interpolation	Geometric uniform CDF	Geometric uniform density
Geometric uniform law	Geometric uniform random series	Germ
Global binary series	Global differential equation	Global Fabius object
Global first-jet reflection law	Global signed extension	Global uniform spline approximation
Gram determinant	Gram matrix	Gregory coefficient
Grid-based transseries	Grid histogram	Grzegorzczak computability
Hadamard finite part	Hahn series	Half moment
Half-moment numerator	Half-moment numerator G_n	Half-moment sequence d_n
Half-weighted interval indicator	Hamming weight	Hankel moment matrix
Harmonic number	HasSum and tsum	Hausdorff moment sequence
Head–tail decomposition	Heat trace	Heine transformation
Herglotz function	Highest nonzero dyadic derivative	Highest-set-bit decomposition
Highest-set-bit recursion	Hilbert-space projection	Hilbert transform
Historical notation scope	Hockey-stick identity	Holomorphic function
Holonomic function	Homothety	Horner scheme
Hypergeometric term	Hypothesis ledger	Identity theorem
Immutable recovery point	Improper integral	Inclusion–exclusion principle
Independent-copy equation	Independent random variables	Indicator function
Infinite convolution	Infinite product	Infinite q -binomial theorem
Infinite q -Pochhammer product	Infinite random series	Infinite weighted sum
Inflection point	Information density	Injective, surjective, and bijective
Integer composition	Integer numerator normalization	Integer partition
Integer-valued polynomial	Integrability	Integral transform
Integrality	Interior of the support	Interpolating polynomial
Interval partition	Invariant measure	Inverse Barnes- G asymptotic
Inverse-branch closure	Inverse-dyadic value	Inverse-Fabius self-map dynamics
Inverse function	Inverse-function condition number	Inverse-gamma asymptotic
Inverse-Lambert expansion	Inverse-logarithmic error	Inverse-Mellin residue expansion
Inverse modulus from a gap bound	Inverse-power table	Inversion formula
IsFabius characterization	Iterated derivative	Iterated prefix kernel
Iterated prefix sum	Jackson fundamental theorem of calculus	Jackson integration by parts
Jackson product rule	Jackson q -derivative	Jackson q -integral
Jacobi continued fraction	Jacobi cubic identity	Jacobi matrix
Jacobi theta function	Jacobi triple product	Jet
K-fold convolution	K-fold Thue-Morse sum	K-function (generalized hyperfactorial)

Kernel	Knot multiplicity	Kolmogorov distance
Koopman eigenfunction	Koopman operator	Kummer's theorem
L2 norm	Lagrange–Good multivariate inversion	Lagrange interpolation
Lagrange inversion formula	Lagrange up-atom	Lah number
Lambert polynomial	Lambert W branch point	Lambert W function
Lambert W Puiseux coordinate	Laplace transform	Large-deviation principle
Laurent expansion	Laurent–logarithmic series	Law as a pushforward measure
Layer-cake representation	Leading-coefficient functional	Least common multiple
Least Lipschitz constant	Least-squares approximation	Least-squares prediction error
Lebesgue integral	Legendre block	Legendre coefficient
Legendre coefficient parity completion	Legendre derivative identity	Legendre energy tail
Legendre–Fenchel transform	Legendre Gaunt coefficient	Legendre linearization coefficient
Legendre normalization	Legendre parity	Legendre partial sum
Legendre polynomial	Legendre's factorial-valuation formula	Legendre superconvergence
Leibniz rule	Levy continuity theorem	Lipschitz function
Local phase	Locally finite sum	Locally uniform convergence
Log-gamma function	Log-periodic asymptotic	Log-periodic function
Log-squared asymptotic	Logarithmic delay equation	Logarithmic height
Logarithmic scale	Logarithmic Stieltjes fixed point	Lower Lambert branch
Lower Lambert coordinate	Lower-Lambert phase	Lower-order term
Lucas's theorem	Maclaurin series	Mahler equation
Mahler function	Major arc	Mass expansion
Mathematical interface theorem	Matrix dilation	Mean value theorem
Measure with density	Measure-zero set	Mellin finite part
Mellin inversion	Mellin-pole asymptotics	Mellin strip
Mellin transform	Meromorphic continuation	Mersenne factor
Minimum mean-square error	Minkowski sum of supports	Minor arc
Minor-arc bound	Möbius function of a subspace lattice	Möbius function of the partition lattice
Mod-Gaussian convergence	Moderate-deviation regime	Module docstring
Modulus of continuity	Moment	Moment–cumulant transform
Moment functional	Moment-generating function	Moment Laurent expansion
Moment numerator	Moment numerator F_n	Moment recurrence
Moment sequence	Monic orthogonal polynomial	Monodromy
Monotone convergence theorem	Monotone function	Monotone order isomorphism
Multiple q -Pochhammer notation	Multiplicative periodic fluctuation	Multiresolution analysis
Multiresolution detail operator	Mutual information	Namespace
Negative-index Bell value	Negative-Laplace decomposition	Negative-Laplace product
Newton identities	Newton iteration	Newton series
Noise floor	Nome	Non-elementarity of the Fabius function
Non-elementarity of the inverse Fabius function	Nonconstant periodic correction	Noncrossing partition
Nondegenerate saddle	Normal convergence	Normal convergence of an infinite product
Normal convergence of q -products	Normalized reversion coefficient	Normalized reversion input
Normalized saddle mass	Normalized sinc functions	Normalized Volterra operator
Normative notation contract	Notation collision	Nowhere analytic
Nowhere-analytic Fabius iterate	Nowhere-analytic inverse-Fabius iterate	Null set
Numerator sequence	Numerical residual	Odd cosine series
Odd discrete Fourier transform	Odd part of an integer	One-periodic function
Operation count	Optimal truncation	Order of vanishing
Ordered composition	Ordered-composition coefficient	Ordinary Bell polynomial

Ordinary generating function	Original-bounded Fabius bridge	Original compact-support characteri- zation
Original Fabius predicate IsOriginalFabius	Orthogonal polynomial	Orthogonal projection
Orthogonality	Orthonormal polynomial	Oscillatory correction
P -recursive sequence	Padé approximant	Paley–Wiener theorem
Parity	Parity-power series	Parseval identity
Partial exponential Bell polynomial	Partition defect	Partition lattice
Partition of unity	Pascal row modulo a prime power	Peano kernel
Periodic correction	Periodic-correction Fourier coeffi- cient	Periodic correction Psi
Periodic mean	Periodicity	Periodization
Perron–Frobenius transfer operator	Perturbation expansion	Phase function
Physicists’ Hermite polynomial	Piecewise polynomial	Plancherel theorem
Pochhammer symbol	Poincaré asymptotic expansion	Pointwise convergence
Pointwise equality	Poisson summation	Pole
Polygamma function	Polynomial approximation	Polynomial cell
Polynomial commutator with inte- gration	Polynomial deconvolution by the up-law	Polynomial diagonal stabilization
Polynomial identity over a ring	Polynomial-logarithmic transseries	Polynomial reproduction
Polynomial reproduction order	Polynomial sequence of binomial type	Positive compositional iterate
Positive-definite kernel	Positive-definite moment functional	Positive-half-line Fourier energy
Positive-part kernel	Positive rational gap lower-bound sequence	Post-render source union
Power series	Power-sum symmetric polynomial	Powerset expansion
Prefix polynomial	Prefix-sum boundary convention	Primitive-recursive fold
Primitive-recursive function	Primitive-root block	Principal branch
Principal-Lambert phase	Probabilistic representation	Probabilists’ Hermite polynomial
Probability density	Probability measure	Product measure
Profile generating function	Proof-backed exposition	Proof-status boundary
Prouhet cancellation	Prouhet partition	Prouhet–Tarry–Escott problem
Provenance ledger	Public declaration	Public facade
Puiseux-logarithmic series	Puiseux series	Pushforward measure
Pythagorean identity	q -Bernstein polynomial	q -beta function
q -beta integral	q -binomial coefficient	q -binomial scalar extension
q -binomial theorem	q -calculus	q -Catalan number
q -digamma function	q -discrete limit	q -factorial
q -finite difference	q -gamma function	q -Gauss summation
q -integer	q -kernel of a sequence	q -Lucas theorem
q -multinomial coefficient	q -Pascal summation	q -Pfaff–Saalschütz summation
q -Pochhammer dissection	q -Pochhammer logarithmic deriva- tive	q -Pochhammer order derivative
q -Pochhammer reversal	q -Pochhammer shift law	q -Pochhammer symbol
q -product	q -Richardson extrapolation	q -series
q -shifted factorial	q -Taylor identity	$q \rightarrow 1$ limit
q -Vandermonde convolution	Quadratic logarithmic law	Quadrature weight
Quantile coupling	Quantile transport	Quantum binomial theorem
Quantum multinomial theorem	Quantum plane	Quotient-of-exponentials approxima- tion
r -Stirling number	Rademacher sign	Rademacher sine-product identity
Radius asymptotic	Radon–Nikodym derivative	Radon transform
Ramification index	Random series	Rapid decay
Rarefaction of a sequence	Rate-distortion function	Rate of convergence
Ratio set	Rational Legendre coefficient	Rational recurrence
Rationality	Real-analytic function	Reciprocal-base q -duality
Reciprocal gamma function	Reciprocal-power table	Recurrence relation

Recursive fast name	Recursive modulus	Reduced dyadic level
Reference tail	Reference weight	Refinement equation
Refinement operator	Reflection symmetry	Regular sequence
Relative entropy	Relative error	Relative saddle mass
Relative tail logarithm	Remainder estimate	Removable singularity
Removable-value convention	Report finite Fabius approximant	Representation independence
Representation operator	Reproducing-kernel Hilbert space	Reshetnikov integers
Residue	Resolvent-order hierarchy	Retired alias
Riemann–Liouville fractional integral	Riemann zeta function	Riesz basis
Rising factorial	Rodrigues formula	Rogers–Szegő polynomial
Root-of-unity collision	Rounding to nearest dyadic	Rvachev Appell polynomial
Rvachev atom	Rvachev decoder	Rvachev differential–difference law
Rvachev polynomial deconvolution	Rvachev up-function	Saddle central arc
Saddle coefficient recurrence	Saddle equation	Saddle Hessian
Saddle jet	Saddle mass	Saddle point
Saddle-point method	Saddle radius	Saddle tail bound
Saddlepoint density approximation	Sampling comb	Sampling identity
Scalar extension	Scaling law	Schröder equation
Schur complement	Schwartz function	Second-order Eulerian number
Second saddle correction	Sectorial asymptotic expansion	Self-convolution
Self-sampling identity	Self-similarity	Semantic API
Sequential computability	Series-reversion coefficient	Sharp asymptotic
Sharp constant	Sharp derivative bound	Sharp transfer
Sheffer sequence	Shifted spline	Signed global series
Signed measure	Signed-natural numerator	Signed Stirling numbers of the first kind
Signed Thue–Morse prefix	Signed-weight random series	Sinc function
Sinc-lobe sign	Sinc product	Small-argument asymptotic
Smooth compact support	Source checkpoint	Source module
Source–PDF parity	Spectral measure	Spectral profile inversion
Spectral zeta function	Spline	Square-summable activation profile
Stable endpoint evaluation	Standardized cumulant	Standardized geometric uniform law
Stars-and-bars principle	Stationary point	Steepest descent
Steepest-descent contour	Stein equation	Stein operator
Stieltjes constant	Stieltjes inversion formula	Stieltjes moment sequence
Stieltjes–Padé approximation	Stieltjes–Perron inversion	Stieltjes transform
Stirling formula	Stirling numbers	Stirling numbers of the second kind
Stokes line	Stopped primitive	Strang–Fix condition
Strict interior positivity	Strict interior range	Strict unimodality
Strip of convergence	Sturm–Liouville equation	Subgraph Fubini identity
Subspace lattice	Summability	Superasymptotic approximation
Supergraph Fubini identity	Support	Support-as-range principle
Support of an indexed family	Supremum norm	Survival function
Symmetric mixed difference	Symmetry	Tail estimate
Tail flatness	Tanh divided slope	Taylor block
Taylor polynomial	Taylor series	Terminating basic hypergeometric series
Terminating Taylor series	Termination index	Termwise differentiation
Termwise integration	Test-function space	Theta quasi-periodicity
Three-term recurrence for orthogonal polynomials	Thue–Morse bit	Thue–Morse block product
Thue–Morse gamma product	Thue–Morse sequence	Thue–Morse sign
Tightness	Toeplitz lemma	Toeplitz row
Tonelli theorem	Topological support	Total variation
Total-variation distance	Totalized definition	Touchard polynomial
Transfer theorem	Translate block	Translate sum

Translated Legendre series	Transmonomial	Transseries
Transseries composition	Transseries reversion	Triangle admissibility
Triangular cancellation	Triangular recurrence	Triangular reduction
Truncated-power basis	Truncation error	2-adic unit
2-kernel of a sequence	2π -frequency Fourier transform	Type-B Eulerian number
Typed map	Unconditional convergence	Uniform approximation
Uniform asymptotic expansion	Uniform convergence	Uniform distribution
Uniform Legendre convergence	Uniform norm	Uniform random variable
Uniform spline	Unimodality	Unique midpoint inflection
Uniqueness of the periodic correction	Unit-interval Fabius-up translate	Unsigned Stirling numbers of the first kind
Up-synthesis amplitude	Valuation	Valuation histogram
Vandermonde determinant	Vanishing moment	Varentropy
Variable substitution	Variance	Vertical Bromwich line
Volterra integral operator	Volterra resolvent	Walsh–Hadamard transform
Wasserstein distance	Weak composition	Weak convergence
Weak-star convergence	Weierstrass M-test	Weierstrass product
Weighted path expansion	Weighted random sum	Weighted-uniform law
Well-based support	Well-founded recursion	Well-poised basic hypergeometric series
Wiener or orthogonal chaos	Wigner $3j$ -symbol	Wright omega function
Zero-bias transform	Zero of an entire function	Zero run
Zeta regularization	Zeta-regularized determinant	Zonoid
Zonotope		

1 A

Abel equation.

FRONTIER

Abel's equation

$$\psi(f(x)) = \psi(x) + 1$$

conjugates iteration to translation. It is suited to parabolic dynamics and continuous iteration. Solutions are not unique without normalization and domain conditions; endpoint singularities of the Fabius inverse make a global Abel coordinate a frontier problem.

Absolute continuity.

CLASSICAL

A function $f : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous if, for every $\varepsilon > 0$, some $\delta > 0$ makes $\sum_j |f(b_j) - f(a_j)| < \varepsilon$ whenever the intervals (a_j, b_j) are pairwise disjoint and have total length below δ . Equivalently, f has an integrable derivative almost everywhere and $f(x) = f(a) + \int_a^x f'(t) dt$. Smooth functions are absolutely continuous on compact intervals. The concept underlies the passage between the Fabius cumulative distribution function and its density and justifies integral forms of the differential identities.

Absolute convergence.

CLASSICAL

A series $\sum_{n \geq 0} a_n$ is absolutely convergent when $\sum_{n \geq 0} |a_n| < \infty$. Absolute convergence permits rearrangement and regrouping and is the usual route to interchanging sums with bounded linear operations. For series of functions one distinguishes pointwise absolute convergence, uniform absolute convergence, and normal convergence on compact sets. In the current repository this language is used chiefly for transform products, q -series, translate blocks, and analytic parameter families. A locally finite translate sum, such as the signed global Fabius construction at a fixed real argument, needs no convergence theorem and should not be misdescribed as an infinite absolutely convergent series.

Absolute Legendre summability.

LEAN-PROVED

A Legendre expansion is absolutely summable when $\sum_n |c_n| < \infty$. Since $|P_n(x)| \leq 1$ on $[-1, 1]$, this implies uniform convergence by the Weierstrass test. Repeated integration by parts and endpoint flatness give stronger-than-polynomial coefficient decay for the up-function, hence absolute summability.

Activation deficit.

LEAN-PROVED

For a symmetric activation F , an activation deficit measures the distance from a plateau, such as $F(x)$ near zero or $1 - F(x)$ near one. Reflection identifies the two deficits, and the differential scaling law relates a deficit at one scale to values at the next. Deficit estimates feed flatness bounds, tail inequalities, and comparison with standard activations.

Activation profile.

PROJECT / MIXED

An activation profile is a monotone sigmoidal map used as a normalized nonlinear transition between two plateaus. The bounded Fabius function is a C^∞ compact-transition activation with exactly constant exterior tails and a symmetric derivative density. Unlike logistic or error-function activations, it is nowhere analytic on its transition interval and has exact dyadic arithmetic.

Active documentation corpus.

CONVENTION

The active documentation corpus is the set of current project-authored Markdown and LaTeX sources outside the archival provenance tree. It includes the proof-backed exposition, the Lean walkthrough, the notation catalogue, audits, and theorem-bearing frontier manuscripts. Inclusion in the active corpus means that shared notation and status conventions apply; it does not mean that every mathematical assertion has already been formalized.

Adjacent Gaussian-binomial ratio.

LEAN-PROVED

Where the denominator is nonzero,

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{1 - q^{n-k}}{1 - q^{k+1}}.$$

The cross-multiplied polynomial identity remains meaningful at roots of unity where the quotient itself may be undefined. Ratio formulas are therefore stored together with denominator hypotheses and separate cancellation lemmas.

Admissible class.

CLASSICAL

An admissible class is the collection of objects on which a defining operator is intended to act and within which existence or uniqueness is asserted. Conditions may include boundedness, continuity, prescribed endpoint values, symmetry, monotonicity, support restrictions, or differentiability. For the bounded Fabius function, a convenient admissible class consists of continuous maps with constant tails and the correct reflection law; the refinement or integration operator preserves this class and is contractive in the supremum norm. Stating the class explicitly prevents a local functional equation from being mistaken for a global characterization.

Affine covariance of fractional integrals.

PROJECT / MIXED

An affine change of variables transports a fractional integral with a power of the scale factor: $I^\alpha(f \circ A)$ is related to $(I^\alpha f) \circ A$ by $|A'|^{-\alpha}$ in the orientation-preserving case. The terminal and side of integration must transform as well. This covariance connects Fabius and up normalizations at noninteger orders.

Affine mixed difference.

LEAN-PROVED

An affine mixed difference evaluates a function at subset sums built from affinely transformed increments. Translation and scaling factors can be pulled out according to degree, so the operator is suited to dyadic grids and geometric interpolation. The current finite-difference modules use explicit powerset expansions to avoid ambiguous recursive conventions.

Affine transformation.

CLASSICAL

An affine transformation of one real variable has the form $x \mapsto ax + b$, with $a \neq 0$; it combines scaling, reflection when $a < 0$, and translation. Affine changes of variable transport support intervals, densities, derivatives, and Fourier transforms in controlled ways. The relations $\text{up}(x) = F(1 - |x|)$, $F(x) = \text{up}(x - 1)$ on $[0, 1]$, and the derivative identity $F'(x) = 2 \text{up}(2x - 1)$ are built from such transformations. Dyadic self-similarity uses the special dilation $x \mapsto 2x$.

Algebraic function branch.

LEAN-PROVED

An algebraic branch is a single-valued function $y = f(x)$ satisfying a nonzero polynomial relation $P(x, y) = 0$ on a domain where a branch has been chosen. Away from discriminant points, such a branch is analytic. The project formalizes this local analyticity principle and excludes the canonical Fabius function from algebraic-branch representations on the open unit interval.

Aliasing.

CLASSICAL

Aliasing is the identification of distinct frequencies after sampling on a lattice. Frequencies differing by a dual-lattice multiple have the same samples. Exact polynomial reproduction exploits transform zeros to suppress unwanted aliases; numerical Fourier reconstruction must instead bandlimit or bound the aliased tail.

All-orders asymptotic expansion.

CLASSICAL

An all-orders asymptotic expansion is a hierarchy of finite approximations valid to arbitrary prescribed order. Writing $f(x) \sim \sum_{j \geq 0} a_j \phi_j(x)$ in an asymptotic scale means that for every N , subtracting the first N terms leaves $o(\phi_{N-1}(x))$, or a stated sharper remainder. It does not generally assert convergence of the infinite formal series. For the Fabius endpoint, the repository proves expansions in inverse powers of the lower-Lambert coordinate $\lambda(x)$, with smooth one-periodic coefficient functions that encode the surviving dyadic oscillation.

Almost everywhere.

CLASSICAL

A statement holds almost everywhere, abbreviated a.e., when the exceptional set has measure zero. Equality almost everywhere is the natural equality in L^p spaces, but it is weaker than pointwise equality and does not preserve chosen endpoint values. The repository distinguishes a.e. identities arising from integration from the pointwise functional and differential laws satisfied by the canonical continuous Fabius and Rvachev representatives.

Almost-sure convergence.

CLASSICAL

Random variables X_n converge almost surely to X if $\mathbb{P}(\{\omega : X_n(\omega) \rightarrow X(\omega)\}) = 1$. This is a pathwise mode of convergence, stronger than convergence in probability but not identical to convergence in mean. For an infinite weighted random sum $\sum_j a_j U_j$, absolute summability of the deterministic weights together with boundedness of the independent uniforms gives almost-sure convergence directly. The resulting random variable furnishes the probabilistic representation of the Fabius distribution and its compactly supported density.

Amplitude.

CLASSICAL

In an integral treated by the saddle-point or steepest-descent method, the amplitude is the factor multiplying the dominant exponential phase, as in $I(\lambda) = \int A(z, \lambda) \exp(\lambda \Phi(z)) \, dz$. The phase determines the saddle and Gaussian localization, while derivatives of the amplitude contribute to correction coefficients. In the Fabius Bromwich analysis, factors originating in the Laplace product, the Jacobian of the local change of variables, and reciprocal powers of the saddle coordinate form the amplitude. Separating amplitude from phase organizes the all-orders coefficient algebra.

Analysis operator.

CLASSICAL

An analysis operator maps a function to coefficients, usually $Tf = (\langle f, \phi_k \rangle)_k$. For an orthonormal basis it is an isometry into ℓ^2 ; for a frame it is bounded with frame inequalities. Exact moment functionals and sample-value maps are analysis operators of different kinds and must not be conflated.

Analytic continuation.

CLASSICAL

Analytic continuation extends a holomorphic function from one domain to a larger connected domain by requiring agreement on their overlap. Uniqueness follows from the identity theorem: two holomorphic continuations that agree on a set with an accumulation point agree everywhere on the connected component. Mellin transforms and zeta expressions in the endpoint analysis are first defined in regions of absolute convergence and then continued meromorphically. The procedure must be distinguished from merely assigning a formal value to a divergent expression; poles and finite parts have to be tracked explicitly.

Analytic germ.

CLASSICAL

The analytic germ of a function at a point records its behavior on some unspecified neighborhood of that point, with two representatives identified when they agree on a smaller neighborhood. A smooth function has an analytic germ at x_0 precisely when it equals a convergent power series near x_0 . At dyadic interior points the Fabius formal Taylor series terminates, but the function does not agree locally with that polynomial; hence no analytic germ is present there. This distinction is central to proving nowhere analyticity despite exact finite Taylor data.

Analytic locus.

CLASSICAL

The analytic locus of a smooth real function is the set of points at which it is real analytic on some neighborhood. It is open by definition. A constant tail is analytic at every point strictly outside the transition interval, whereas flatness at a boundary point need not imply analyticity there. The repository identifies the exact analytic loci of the bounded Fabius function, the compactly supported up-function, and the signed global extension, carefully separating exterior constant regions, support endpoints, and the dense set of dyadic points where Taylor series terminate without representing the function.

Analytic moment.

CLASSICAL

An analytic moment is a moment regarded not only as an integral but as a Taylor coefficient of an analytic transform. If $M(z) = \int e^{zx} \mu(dx)$ is entire, then $m_n = M^{(n)}(0)$. This viewpoint gives Cauchy estimates, generating-function identities, and scalar extension to complex arguments. The

repository's analytic-moment modules connect the exact Fabius moment recurrences with the entire complex moment-generating function.

Analytic zero multiplicity.

CLASSICAL

A holomorphic function has a zero of multiplicity r at z_0 when $f(z) = (z - z_0)^r g(z)$ with $g(z_0) \neq 0$. Equivalently its first $r - 1$ derivatives vanish and the r -th does not. Product factorization makes multiplicities additive and supports exact jet-preserving Fourier extrapolants.

Analyticity at a point.

CLASSICAL

A real function is analytic at x_0 when it agrees with a convergent power series on some neighborhood of x_0 . Possessing derivatives of all orders at the point is insufficient. For Fabius functions, terminating or flat jets at dense dyadic configurations contradict neighborhood power-series representations, giving nowhere-analytic rather than merely endpoint nonanalytic behavior.

Angular-frequency Fourier transform.

CONVENTION

The angular convention is $\mathcal{F}_\omega f(\omega) = \int f(x) e^{-i\omega x} dx$. It is related to the 2π -frequency transform by $\omega = 2\pi\xi$. Product factors, inversion constants, derivative multipliers, and exponential type all change accordingly; formulas from the two conventions cannot be mixed termwise.

Antidiagonal polynomial.

FRONTIER

An antidiagonal polynomial records values of a two-parameter repeated-sum array along $k - n = r$ or $n + k = r$, depending on the indexing convention. After the vanishing region is removed, many such diagonals are polynomial in the remaining index. Explicit diagonal formulas enable constant-time evaluation far inside a repeated-summation table.

Appell deconvolution.

LEAN-PROVED

Given a random variable X with moment-generating function $M_X(z)$, the deconvolution Appell polynomials have generating function

$$\frac{e^{xz}}{M_X(z)}.$$

They satisfy $\mathbb{E}[A_n(x + X)] = x^n$. For the Rvachev law, reciprocal sinc or moment-series coefficients produce exact polynomial reconstruction formulas.

Appell sequence.

LEAN-PROVED

A polynomial sequence $A_n(x)$ is Appell when $A'_n(x) = nA_{n-1}(x)$, equivalently its exponential generating function is

$$A(z)e^{xz}$$

for a series A with nonzero constant term. Bernoulli and Euler polynomials are examples. Moment-based Appell sequences encode convolution and deconvolution with the Rvachev law.

Approximate identity.

CLASSICAL

An approximate identity is a family of kernels K_n whose mass is normalized, whose mass concentrates near the origin, and whose convolution with a suitable function tends to that function. In probability it is often realized by laws of random variables whose tails shrink to zero; in spline theory it appears through increasingly fine convolution kernels. The finite centered spline approximants to the Fabius density or distribution behave in this way: they are explicit piecewise-polynomial convolutions and converge uniformly after the support and centering are normalized correctly.

Approximation theory.

CLASSICAL

Approximation theory studies how functions are represented or approximated by simpler classes such as polynomials, splines, rational functions, or translates of a fixed kernel. It asks for convergence, rates, optimality, stability, and constructive algorithms. The up-function belongs to Rvachev's atomic-function approach, while the repository develops both centered spline approximations with explicit uniform error and Fourier-Legendre polynomial approximations with least-squares optimality.

Arbitrary-phase synthesis.

LEAN-PROVED

Arbitrary-phase synthesis reconstructs a polynomial for every real shift θ using samples at $(k + \theta)/M$ and correspondingly shifted up-atoms. Phase-uniform exactness is stronger than reconstruction on the unshifted lattice and is the property classified by the composite-mesh theorems.

Arbitrary-phase up synthesis.

LEAN-PROVED

Arbitrary-phase synthesis permits the centers of the wide up-atoms to be shifted by a common real phase while retaining exact polynomial reproduction on the target interval. The coefficients change through an explicitly invertible interpolation system. The phase is a geometric placement parameter, not the periodic endpoint phase from the Lambert asymptotic.

Arithmetic ray.

FRONTIER

An arithmetic ray is a sequence of frequencies or complex arguments selected by a fixed arithmetic pattern, such as odd multiples times powers of two. Evaluating a dyadic product along such rays isolates valuation classes, lobe signs, and extremal decay behavior. Ray asymptotics need not describe uniform behavior over the full real axis.

Associahedron.

CLASSICAL

The associahedron is a convex polytope whose face structure encodes parenthesizations or polygon dissections. Indexing conventions differ: K_n may denote dimension $n - 2$, $n - 1$, or triangulations of an n -gon. The notation catalogue replaces ambiguous historical K_n symbols by a dimension-explicit form before formulas are merged.

Asymptotic coefficient jet.

PROJECT / MIXED

An asymptotic coefficient jet is the finite list of phase and amplitude derivatives needed to compute a chosen number of saddle corrections. Bell polynomials and Wick/Gaussian moments turn the jet into explicit coefficients. The order of the required jet grows with the requested expansion depth.

Asymptotic equivalence.

CLASSICAL

For nonzero comparison functions, $f(x) \sim g(x)$ as $x \rightarrow a$ means $f(x)/g(x) \rightarrow 1$. This multiplicative statement is stronger than equality of leading logarithmic orders. If $\log f = \log g + o(1)$, then asymptotic equivalence follows after exponentiation; an $O(1)$ logarithmic discrepancy generally does not suffice. In the Fabius endpoint problem, the periodic correction must be included in the exponential main term: omitting a nonconstant bounded function changes the ratio by a nonconvergent oscillatory factor and destroys the claimed equivalence.

Asymptotic expansion.

CLASSICAL

An asymptotic expansion expresses a function by successively smaller comparison terms near a limiting regime. If $\phi_{j+1} = o(\phi_j)$, the notation $f \sim \sum_{j \geq 0} a_j \phi_j$ means that every finite truncation approximates f to the scale of its last retained term. Coefficients may be constants, polynomials, or periodic functions. The Fabius development uses logarithmic expansions, saddle expansions, and inverse-Lambert expansions; their remainders are stated quantitatively rather than inferred from a formal infinite series.

Asymptotic scale.

CLASSICAL

An asymptotic scale is an ordered family (ϕ_j) with $\phi_{j+1}(x) = o(\phi_j(x))$ in the limit under study. It supplies the notion of successive order needed for an asymptotic expansion. Powers of $1/\lambda$, where $\lambda \rightarrow \infty$, form the main scale in the corrected Fabius saddle analysis; powers of $1/|\log x|$ are a coarser equivalent scale after expanding the lower Lambert coordinate. Logarithmic and log-periodic factors can modify the scale and must not be silently absorbed into constants.

Atom of a measure.

CLASSICAL

An atom is a measurable set of positive measure containing no smaller measurable subset of intermediate positive measure; for a real law, a point mass at a is measured by $\mathbb{P}(X = a)$. A distribution with a continuous density is atomless. Infinite weighted sums can lose atoms even when finite approximants have piecewise-polynomial densities.

Atomic function.

CLASSICAL

In Rvachev's terminology, an atomic function is a compactly supported smooth solution of a functional-differential or refinement equation from which broader systems of localized basis functions can be generated. The up-function is the canonical example. Its support is finite, all derivatives vanish at the support boundary, and dilates and translates inherit exact algebraic relations. Atomic functions are used in approximation theory and numerical methods because they combine localization with very high regularity, although unlike polynomials or analytic kernels they are typically nowhere analytic in the interior.

Atomless law.

CLASSICAL

A probability law is atomless when every singleton has mass zero. Continuity of its CDF is equivalent to atomlessness for real distributions. The geometric uniform and Fabius laws are atomless and in fact have smooth densities, allowing ordinary inverse functions rather than quantile intervals on their strict support interiors.

Automatic sequence.

CLASSICAL

A sequence is k -automatic if a finite automaton reading the base- k digits of n outputs its n -th term. The Thue–Morse sequence is the fundamental 2-automatic example: the automaton tracks the parity of the number of one bits. Automatic structure yields digit recurrences, finite block factorization, and efficient generation. These properties drive the Fabius signed translate and dyadic cancellation formulas.

Averaging operator.

CLASSICAL

An averaging operator replaces a value by an integral or finite mean over translated or rescaled values. A typical probability operator is $(Tf)(x) = \int f(x - y) \mu(dy)$, namely convolution with a probability measure. Such operators preserve positivity and constants, contract the supremum norm, and smooth irregularities. The fixed-point construction of the Fabius law can be formulated with an operator induced by a scaled uniform random variable; iterating it corresponds to adding independent dyadic-scale summands and produces finite spline approximants.

2 B**B-spline.**

CLASSICAL

A B-spline is a compactly supported piecewise polynomial obtained from repeated convolution of interval indicators, with degree and smoothness increasing under convolution. Uniform B-splines use equally spaced knots; centered forms are symmetric about the origin. Finite convolutions of the dyadically scaled uniform densities in the probabilistic Fabius construction are nonuniformly scaled box splines, and after a common-grid reformulation they yield explicit centered finite splines. These approximants have rational dyadic data, controlled support, and a certified uniform error.

Bailey lemma.

FRONTIER

The Bailey lemma maps one Bailey pair to another by multiplying and summing with additional q -Pochhammer factors. Iteration produces a Bailey chain and many Rogers–Ramanujan type identities. In the Fabius project it is a frontier organizing principle for repeated triangular transforms, not a substitute for checking convergence and denominator hypotheses.

Bailey pair.

FRONTIER

A pair of sequences (α_n, β_n) is a Bailey pair relative to a when β_n is a specified triangular q -hypergeometric transform of $\alpha_0, \dots, \alpha_n$, commonly

$$\beta_n = \sum_{r=0}^n \frac{\alpha_r}{(q; q)_{n-r} (aq; q)_{n+r}}.$$

The normalization varies. Bailey pairs convert finite coefficient data into families of product and series identities.

Balanced basic hypergeometric series.

CLASSICAL

A terminating basic hypergeometric series is balanced when the product of its lower parameters is a prescribed power of q times the product of its upper parameters, with the exact exponent determined by the ${}_r\phi_s$ normalization. Balance is what makes the q -Pfaff–Saalschütz product evaluation possible. It is stronger than mere termination.

Banach fixed-point theorem.

CLASSICAL

The Banach fixed-point theorem states that a strict contraction T on a complete metric space has a unique fixed point. If $d(Tx, Ty) \leq qd(x, y)$ with $0 < q < 1$, then the iterates $T^n x_0$ converge geometrically to that fixed point and satisfy explicit a posteriori and a priori error bounds. In the Fabius construction, the ambient space is a closed class of bounded continuous functions with the supremum metric, and the defining averaging/refinement operator is contractive. The theorem simultaneously proves existence, uniqueness, and a certified approximation scheme.

Barnes G -function.

FRONTIER

Barnes' G -function satisfies $G(z+1) = \Gamma(z)G(z)$ and $G(1) = 1$. It interpolates products of factorials and has a large- z logarithmic expansion led by $\frac{1}{2}z^2 \log z$. Inverting G therefore requires a polynomial–logarithmic transseries on a square-root logarithmic scale.

Base- q digit.

PROJECT / MIXED

For an integer base $q \geq 2$, the j -th digit of n is $d_j(n) = \lfloor n/q^j \rfloor \bmod q$. Then $n = \sum_{j \geq 0} d_j(n)q^j$ with finite support. Base- q digit formulas generalize binary weight, support carry theorems, and enter q -Lucas and prime-power valuation arguments.

Base- q digit sum.

CLASSICAL

The base- q digit sum is $s_q(n) = \sum_j d_j(n)$. For prime p , Legendre's formula connects it to $v_p(n!)$, and Kummer's theorem connects digit carries to binomial valuations. At $q = 2$, its parity is the Thue–Morse bit and its value is the Hamming weight.

Basic hypergeometric series.

CLASSICAL

A basic hypergeometric series is a q -series whose consecutive coefficient ratio is rational in q^n . In standard notation, ${}_r\phi_s$ is built from products $(a_i; q)_n / (b_j; q)_n$, the factor $(q; q)_n$, and a convention-dependent power of $(-1)^n q^{\binom{n}{2}}$. Bilateral analogues sum over $n \in \mathbb{Z}$. The current repository fixes these normalizations explicitly and formalizes selected transformation and summation theorems, including Heine, q -Gauss, and terminating q -Pfaff–Saalschutz forms. Parameter denominators and convergence regions are part of every statement.

Basic hypergeometric series ${}_r\phi_s$.

LEAN-PROVED

With one standard normalization,

$${}_r\phi_s \left[\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right] = \sum_{n \geq 0} \frac{(a_1, \dots, a_r; q)_n}{(q, b_1, \dots, b_s; q)_n} \left((-1)^n q^{n(n-1)/2} \right)^{1+s-r} z^n.$$

The exponent convention is part of the notation. Denominator parameters, termination, and convergence are stated explicitly in every theorem.

Bell–Hessenberg determinant.

FRONTIER

A Bell–Hessenberg determinant is a lower-Hessenberg determinant whose entries are a moment, cumulant, or coefficient sequence and whose superdiagonal contains simple integer factors. Expanding by the last row reproduces a Bell recurrence. The exact signs and factorial normalization depend on whether ordinary or exponential series are used.

Bell number.

CLASSICAL

The Bell number B_n counts set partitions of an n -element set: $B_n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$. Its exponential generating function is $\exp(e^z - 1)$. Bell numbers organize moments, composition derivatives, determinants, and asymptotic reversion coefficients.

Bell polynomial.

CLASSICAL

Bell polynomials encode the combinatorics of differentiating a composite exponential or logarithm. The complete exponential Bell polynomial satisfies $\exp(\sum_{m \geq 1} x_m t^m / m!) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) t^n / n!$; partial Bell polynomials refine the number of factors. They provide a compact form of the Faà di Bruno formula. In saddle expansions, they convert phase and amplitude jets into coefficients of the exponential series, while logarithmic coefficients can also be expressed through ordered compositions. The repository often spells out these finite sums to facilitate formal verification.

Bell-polynomial asymptotic coefficient.

PROJECT / MIXED

When an exponent or logarithm has a formal expansion, complete Bell polynomials convert its coefficient sequence into the coefficients of the exponential. Partial Bell polynomials organize a fixed number of factors. This gives finite, explicit formulas for saddle corrections, gamma derivative jets, and moment–cumulant expansions.

Bernoulli number.

CLASSICAL

The Bernoulli numbers B_n are rational numbers defined by $t/(e^t - 1) = \sum_{n \geq 0} B_n t^n / n!$, with the convention $B_1 = -1/2$. They govern sums of powers, Euler–Maclaurin summation, and the Taylor expansions of trigonometric and Bose-type kernels. The Fabius arithmetic development derives Bernoulli recurrences related to moment sequences and reciprocal-sine products. Because two sign conventions for B_1 occur in the literature, a formal development must fix the generating function rather than rely on the name alone.

Bernoulli recurrence.

CLASSICAL

A Bernoulli recurrence is a triangular identity determining Bernoulli numbers from earlier values, classically $\sum_{k=0}^n \binom{n+1}{k} B_k = 0$ for $n \geq 1$. Equivalent recurrences arise by multiplying or differentiating the generating function $t/(e^t - 1)$. In the repository, Bernoulli-type relations connect coefficients in reciprocal trigonometric expansions with Fabius moment recurrences. Triangularity means the newest coefficient appears with a nonzero scalar and can therefore be solved exactly over the rationals.

Berry–Esseen bound.

FRONTIER

A Berry–Esseen bound quantifies the Kolmogorov distance to a Gaussian by a third absolute moment divided by a variance power, summed over independent terms. It gives an $O(\epsilon)$ -type rate under the appropriate triangular array normalization. Symmetry may improve formal cumulant order but does not automatically improve the classical absolute-moment bound.

Bessel inequality.

CLASSICAL

Bessel’s inequality states that for an orthonormal system (e_n) in a Hilbert space, $\sum |\langle f, e_n \rangle|^2 \leq \|f\|^2$. Equality holds for a complete basis and becomes Parseval’s identity. Applied to normalized Legendre polynomials, it bounds coefficient squares and the L^2 error of partial sums. Stronger coefficient estimates are needed for the repository’s absolute uniform convergence theorem.

Big-O and little-o notation.

CONVENTION

The relation $f(x) = \mathcal{O}(g(x))$ means that $|f(x)| \leq C|g(x)|$ near the specified limit for some constant C ; $f(x) = o(g(x))$ means $f(x)/g(x) \rightarrow 0$. Uniform qualifiers indicate that the constant or convergence is independent of auxiliary variables. These symbols describe error scales, not hidden equalities. In the corrected Fabius asymptotic, an $\mathcal{O}(1/\lambda(x))$ logarithmic remainder is meaningful only after the nonconstant periodic main term has been retained.

Bijection.

CLASSICAL

A bijection is a map that is both injective and surjective, so every target element has exactly one preimage. A continuous strictly monotone function from an interval onto an interval is a bijection and admits a continuous inverse. The bounded Fabius function is constant outside $[0, 1]$, but its restriction to $[0, 1]$ is a strictly increasing bijection onto $[0, 1]$. This fact licenses the inverse Fabius function and makes inverse-dyadic questions meaningful: for a dyadic ordinate one asks for the unique abscissa with that value.

Bilateral basic hypergeometric series.

PROJECT / MIXED

A bilateral series ${}_r\psi_s$ sums over all $n \in \mathbb{Z}$ rather than $n \geq 0$. Negative-index q -Pochhammer symbols are interpreted through reciprocal finite products. Convergence typically requires an annulus rather than a disk, and zeros of denominator factors must be excluded. Bilateral series appear in theta and dissection identities but are not interchangeable with unilateral ${}_r\phi_s$ series.

Bilateral Laplace transform.

CLASSICAL

The bilateral Laplace transform of a suitable function $f : \mathbb{R} \rightarrow \mathbb{C}$ is

$$\mathcal{L}_{\text{bil}} f(s) = \int_{-\infty}^{\infty} e^{-sx} f(x) \, dx.$$

It differs from the one-sided transform by integrating over the whole real line, and its convergence domain is generally a vertical strip. Compact support makes the transform entire. The distinction matters when passing between the compactly supported up-function, the bounded CDF, and transforms of global signed extensions.

Binary block factorization.

PROJECT / MIXED

When an index is split as $2^m a + r$ with $0 \leq r < 2^m$, nonoverlap of binary digits gives $\varepsilon_{2^m a + r} = \varepsilon_a \varepsilon_r$. This block factorization separates coarse and fine scales and is the key multiplicative identity behind complete-block cancellation and self-similar prefix arrays.

Binary digit sum.

CLASSICAL

The binary digit sum $\text{wt}_2(n)$, also called the Hamming weight or popcount, is the number of ones in the base-two expansion of a nonnegative integer n . Its parity generates the Thue–Morse sign $\varepsilon_n = (-1)^{\text{wt}_2(n)}$. It obeys $\text{wt}_2(2n) = \text{wt}_2(n)$ and $\text{wt}_2(2n + 1) = \text{wt}_2(n) + 1$, which yield the fundamental two-block recursions. The Fabius development uses both the exact integer $\text{wt}_2(n)$ and its parity in dyadic cancellation, signed translate sums, and logarithmic identities.

Binary expansion.

CLASSICAL

A binary expansion represents a nonnegative real number as $x = \sum_j b_j 2^{-j}$ or, on the unit interval, as $x = \sum_{j \geq 1} b_j 2^{-j}$ with digits $b_j \in \{0, 1\}$. Dyadic rationals have two infinite expansions, one terminating and one ending in infinitely many ones, so algorithms must impose a convention or prove representation independence. The global binary reduction in the repository peels off scale blocks according to the set bits and produces an absolutely convergent series whose value is independent of harmless encoding choices.

Binary floor-digit formula.

LEAN-PROVED

A binary floor-digit formula expresses a bit, digit sum, or Thue–Morse value entirely through floor functions. Such formulas extend naturally to a real variable between integer jumps and can be inserted into finite sums, integrals, and interpolation identities. Their discontinuity set and endpoint convention must be stated explicitly.

Binary reduction series.

PROJECT / MIXED

A binary reduction series is a scale-by-scale expansion obtained by iterating an identity that reduces the argument according to its binary digits. In the Fabius setting, a Taylor block is extracted at each active binary scale, and the residual argument is shifted and rescaled. The finite reductions telescope; passing to the limit yields an absolutely convergent representation of the signed global extension. Unlike a mere digit algorithm, the construction is an analytic identity and includes explicit coefficients, convergence estimates, and a proof that the series agrees with the bounded Fabius function on $[0, 1]$.

Binary reflection recursion.

LEAN-PROVED

A binary reflection recursion reduces an index in the upper half of a power-of-two block to its reflected lower-half index, multiplying by a Thue–Morse sign or adding a correction. It is the finite-sequence analogue of Fabius reflection and underlies fast prefix, dyadic-value, and automatic kernel algorithms.

Binomial coefficient.

CLASSICAL

The binomial coefficient $\binom{n}{k} = n!/(k!(n-k)!)$ counts k -element subsets of an n -element set and is the coefficient of x^k in $(1+x)^n$. Pascal's recurrence and the binomial theorem make it ubiquitous in derivative scaling, moment recurrences, Taylor expansions, and finite differences. The Fabius formulas combine ordinary binomial coefficients with Gaussian q -binomial coefficients; keeping the two notations distinct is essential because the latter depend on a base parameter and encode geometric rather than equally spaced nodes.

Binomial polynomial basis.

CLASSICAL

The polynomials $\binom{x}{k} = x^k/k!$ form a basis for polynomials in x . Forward difference acts simply: $\Delta \binom{x}{k} = \binom{x}{k-1}$. Integer-valued polynomials have integer coefficients in this basis, making it natural for repeated prefix sums and exact diagonal formulas.

Binomial transform.

CLASSICAL

The binomial transform sends a sequence (a_k) to $b_n = \sum_{k=0}^n \binom{n}{k} a_k$, with signed variants obtained by inserting $(-1)^k$. It is inverted by the corresponding signed transform. Such transforms arise when moments are shifted, when an exponential generating function is multiplied by e^z , or when a variable is replaced by $1-x$. In the Fabius arithmetic, the full moments and half moments are connected by finite even-index binomial sums produced by the symmetry and translation between the bounded and centered forms.

Biorthogonal system.

FRONTIER

Two families (ϕ_n) and (ψ_n) are biorthogonal when $\langle \phi_n, \psi_m \rangle = \delta_{nm}$. Such pairs support coefficient extraction without orthogonality of either family alone. Appell deconvolution and translated-kernel systems suggest natural biorthogonal questions in the Fabius/Rvachev setting.

Bit extraction.

LEAN-PROVED

For $n \in \mathbb{N}$, the j -th bit can be extracted by

$$b_j(n) = \left\lfloor \frac{n}{2^j} \right\rfloor - 2 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor.$$

This equals the remainder of $\lfloor n/2^j \rfloor$ modulo two. It links Lean's bit-testing operations to floor formulas used in analytic sums and makes binary digit identities available without choosing a string representation.

Boolean-cube mixed difference.

PROJECT / MIXED

For increments $h_1, \dots, h_m$, the mixed difference is an alternating sum over the Boolean cube,

$$\Delta_{h_1} \cdots \Delta_{h_m} f(x) = \sum_{S \subseteq [m]} (-1)^{m-|S|} f\left(x + \sum_{j \in S} h_j\right).$$

It annihilates polynomials of degree below m and is the continuous-node analogue of Prouhet cancellation.

Boolean cumulant.

FRONTIER

Boolean cumulants linearize Boolean convolution and are related to moments through interval partitions. Their generating functions satisfy a simple reciprocal relation. Free, Boolean, and classical cumulants share low-order patterns but have different partition lattices and independence notions.

Borel probability measure.

CLASSICAL

A Borel probability measure on a topological space assigns total mass one to the sigma-algebra generated by open sets. On $\mathbb{R}$, it is determined by its cumulative distribution function and, when absolutely continuous, by a density. The law of the infinite dyadically weighted sum of independent uniform variables is a compactly supported Borel probability measure. Its distribution function is the bounded Fabius function, while a centered affine image has density equal to the Rvachev up-function after the chosen normalization.

Borel summability.

FRONTIER

A formal series is Borel summable in a direction when its Borel transform has suitable analytic continuation and growth there and the directional Laplace integral exists. The resulting analytic function has the original series as an asymptotic expansion. Borel summability is stronger than formal manipulability and is not claimed for a repository transseries unless an analytic continuation theorem is supplied.

Borel transform.

FRONTIER

For a factorially divergent formal series $\sum_{n \geq 0} a_n x^{-n-1}$, its formal Borel transform is $\sum_{n \geq 0} a_n \xi^n / n!$. If this series can be analytically continued and Laplace transformed along a direction, the result is a Borel sum. Singularities of the Borel transform encode exponentially small sectors and Stokes jumps.

Bose-Einstein kernel.

CLASSICAL

A Bose-Einstein kernel is a function built from $(e^t - 1)^{-1}$, often $t/(e^t - 1)$ or $(e^t - 1)^{-1} - t^{-1}$, whose Mellin transform produces Gamma and zeta factors. Near zero it has a Bernoulli expansion and usually a pole that must be subtracted before integration. In the Fabius negative-Laplace analysis, a related dyadic harmonic sum is decomposed into polynomial-logarithmic terms plus a periodic fluctuation. The finite-part treatment of the kernel isolates the Mellin pole and makes the Gamma-zeta Fourier coefficients explicit.

Bose finite-part integral.

CLASSICAL

A Bose finite-part integral is a regularized integral involving a Bose-Einstein kernel whose endpoint singular terms have been explicitly removed. If an integrand has an expansion $a_{-m} t^{-m} + \dots + a_{-1} t^{-1} + O(1)$, one subtracts the singular part, integrates the remainder, and adds the canonically determined finite contributions. In the repository this construction supplies constants and periodic corrections in the Mellin analysis of the negative-Laplace product. It is a Hadamard finite part, not an ordinary improper integral when the unsubtracted integral diverges.

Boundary value problem.

CLASSICAL

A boundary value problem seeks a solution of a differential or functional-differential equation satisfying conditions at boundary points or on the boundary of a domain. Compact support can be viewed as an infinite collection of exterior boundary conditions, while endpoint flatness enforces smooth gluing to zero. Rvachev atomic functions were developed for nonclassical approximation methods in boundary value problems because their localized dilates and translates can encode domain geometry.

Bounded Fabius candidate.

LEAN-PROVED

A bounded Fabius candidate is a function $f : \mathbb{R} \rightarrow \mathbb{R}$ to which the predicate `IsFabius` may apply. It has constant exterior tails, reflection $f(x) + f(1 - x) = 1$, regularity, and the left-half differential-dilation law encoded by the predicate. The word “candidate” keeps the variable function distinct from the canonical fixed point F .

Bounded Fabius function.

PROJECT / MIXED

The bounded Fabius function is the normalization $F : \mathbb{R} \rightarrow [0, 1]$ that equals zero for $x \leq 0$, one for $x \geq 1$, satisfies $F(1 - x) = 1 - F(x)$, and obeys $F'(x) = 2F(2x)$ on $[0, 1/2]$. The adjective bounded distinguishes it from the signed global extension $\mathcal{F}$, which satisfies the pure dilation equation everywhere but is not a CDF. It is the normalization used for dyadic values, inverse values, and computability.

Bounded variation.

CLASSICAL

A real or complex function on an interval has bounded variation if the supremum of $\sum_j |f(x_j) - f(x_{j-1})|$ over all partitions is finite. For a finite row of weights, total variation means $\sum_j |\omega_{n,j}|$. Uniform bounded variation prevents cancellation from concealing unbounded coefficients and is the stability hypothesis in a Toeplitz convergence argument. The complex q -binomial discrete limit uses rows whose mass is one and whose total variation is uniformly bounded by an explicit finite product ratio.

Box spline.

FRONTIER

A box spline is the density obtained by pushing forward uniform measure on a cube by a linear map, or equivalently convolving uniform segment measures in specified directions. It is piecewise polynomial on a hyperplane arrangement and supported on the associated zonotope. The univariate finite sinc-prefix splines are rank-one instances.

Bromwich inversion integral.

CLASSICAL

The Bromwich inversion integral recovers a function from its Laplace transform along a vertical line in the complex plane. In a common form, $f(x) = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} e^{sx} \mathcal{L}f(s) \, ds$, where c lies to the right of all singularities. Analytic decay and contour justification are essential. For the Fabius endpoint, the contour is moved through a real negative saddle after rewriting the transform by an exact negative-Laplace product; the resulting integral supports quantitative saddle analysis.

Bromwich saddle method.

PROJECT / MIXED

The Bromwich saddle method applies steepest-descent ideas directly to a Laplace inversion contour. One first identifies a large real parameter and a saddle solving the derivative equation for the logarithmic phase, then deforms or localizes the vertical contour, expands the phase near the saddle, and bounds the tails and minor arcs. In the Fabius analysis the saddle is naturally expressed through the lower branch of Lambert W . The method yields the main exponential scale, the log-periodic correction, the Gaussian mass, and systematic inverse-saddle corrections.

Bump function.

CLASSICAL

A bump function is a C^∞ function with compact support. Nonzero bump functions cannot be real analytic on all of $\mathbb{R}$, because an analytic function vanishing on an open set vanishes identically. Rvachev's up-function is a distinguished bump: it is even, nonnegative, supported on $[-1, 1]$, strictly unimodal, infinitely flat at ± 1 , and satisfies an exact refinement-type differential equation. Its special algebraic structure makes it useful far beyond the generic existence of smooth cutoffs.

3 C**C-infinity smoothness.**

CLASSICAL

A function is C^∞ if derivatives of every finite order exist and are continuous. Smoothness is weaker than real analyticity: the Taylor series may fail to converge to the function, or may even be identically zero at a flat point. The Fabius and up-functions are canonical examples of smooth but nowhere-analytic behavior on their nonconstant regions. Their differential scaling law gives exact formulas and sharp supremum bounds for all derivatives, while compact support forces infinite-order vanishing at the boundary.

Canonical clamped inverse.

CONVENTION

A clamped inverse extends an inverse on its natural range by fixing the lower endpoint below the range and the upper endpoint above it. This gives a total monotone map on $\mathbb{R}$ while preserving the ordinary inverse on $[0, 1]$. Clamping is a convention, not analytic continuation; it creates constant tails and nondifferentiable transition behavior unrelated to the interior inverse.

Canonical correlation.

FRONTIER

Canonical correlation maximizes correlation between linear combinations of two families of square-integrable variables. It is determined by covariance operators after quotienting null directions. Pairwise correlation formulas do not automatically solve the multi-family canonical-correlation problem.

Canonical Fabius fixed point.

LEAN-PROVED

The canonical bounded Fabius function is obtained as the unique fixed point of a refinement or integration operator on an admissible complete metric space. Its specification theorem packages the defining Fabius predicate, while uniqueness lets later results quantify either over arbitrary witnesses

or specialize to the named function. The fixed-point construction supplies a mathematical object before exact arithmetic and transform representations are developed.

Canonical Fabius specification.

LEAN-PROVED

The canonical specification theorem states that the named function `fabius` satisfies the abstract predicate `IsFabius`. It is the principal interface between the existence construction and downstream theorems. Rather than unfolding the fixed-point definition, later proofs use the specification's tail, symmetry, derivative, and smoothness consequences.

Canonical inverse Fabius function.

LEAN-PROVED

The canonical inverse Fabius function is the inverse of $F : [0, 1] \rightarrow [0, 1]$, extended at endpoints according to the repository's clamped convention. On $(0, 1)$ it is strictly increasing and continuous, but endpoint flatness of F produces singular derivative behavior for the inverse. It is not the inverse of the signed global extension and should not be assigned branches outside the declared range without a separate construction.

Canonical solution.

CLASSICAL

A canonical solution is a solution selected uniquely by natural normalization and regularity conditions, rather than by an arbitrary parameter choice. A differential or functional equation alone may admit many local or distributional solutions; support, symmetry, endpoint values, positivity, and smoothness can isolate one. The bounded Fabius function is canonical as the unique smooth distribution function with the stated tails, symmetry, and dyadic differential equation. Rvachev's up-function is the corresponding canonical compactly supported even density.

Caputo fractional derivative.

CLASSICAL

For $m - 1 < \alpha \leq m$, the Caputo derivative is ${}^C D_{a+}^\alpha f = I_{a+}^{m-\alpha} f^{(m)}$. It differs from the Riemann–Liouville derivative by explicit initial-value terms and is often better adapted to classical boundary conditions. Neither operator may be substituted for the other in Fabius fractional extensions without checking endpoint jets.

Cardinal interpolation.

CLASSICAL

Cardinal interpolation uses a kernel L satisfying $L(k) = \delta_{0k}$ on an integer or scaled lattice, so that $f(x) \approx \sum_k f(kh)L(x/h - k)$. Exactness depends on the function class and kernel. Rvachev atoms yield finite cardinal systems after polynomial deconvolution and a sufficient dyadic mesh.

Carleman condition.

FRONTIER

For a Hamburger moment sequence, Carleman's condition

$$\sum_{n \geq 1} m_{2n}^{-1/(2n)} = \infty$$

is sufficient for determinacy. It is automatic for compactly supported probability measures because $m_{2n}^{1/(2n)}$ stays bounded. The condition also appears in ultradifferentiable and quasi-analytic discussions, where a different derivative-weight sequence must be distinguished from moments.

Carleman linearization.

FRONTIER

Carleman linearization embeds nonlinear iteration into the linear action of the Carleman matrix on monomials. It can encode integer iterates and suggest fractional iteration through matrix powers. Infinite-dimensional domains, spectral branches, and convergence of truncations are substantial additional requirements.

Carleman matrix.

FRONTIER

For an analytic map $f(z) = \sum_{n \geq 0} a_n z^n$, its Carleman matrix is

$$C(f)_{mn} = [z^n]f(z)^m, \quad m, n \geq 0.$$

Formally $C(f \circ g) = C(f)C(g)$ when the coefficient sums are finite or convergent. A finite principal truncation is an approximation to the infinite operator, not an exact linearization theorem.

Carry count.

CLASSICAL

A carry count records how many digit positions produce carries when adding integers in a fixed base. Kummer's theorem identifies the number of base- p carries in $k + (n - k)$ with $v_p\binom{n}{k}$. Carry statistics also control cyclotomic exponents in Gaussian coefficients and prime-power Pascal patterns.

Catalan number.

CLASSICAL

The Catalan number $C_n = \frac{1}{n+1}\binom{2n}{n}$ counts binary trees, polygon triangulations, Dyck paths, and many equivalent objects. Its generating function is the distinguished zero-constant solution of $C(z) = 1 + zC(z)^2$, or after a shift a quadratic inverse germ. The repository also studies q -Catalan polynomials and quarter-scaled Catalan germs.

Cauchy formula for repeated integration.

CLASSICAL

For an integer $n \geq 1$,

$$I^n f(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) \, dt.$$

The identity collapses an n -fold ordered integral to one integral over a simplex volume. It is the integer prototype of the Riemann–Liouville fractional integral and explains the polynomial kernels in iterated Fabius primitives.

Cauchy integral formula.

CLASSICAL

If f is holomorphic inside and on a positively oriented simple closed contour γ , then $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\gamma} f(z)/(z - z_0)^{n+1} \, dz$. It implies analyticity, derivative bounds, and coefficient extraction. Transform functions in the Fabius development are entire, so Cauchy estimates can control Taylor coefficients and justify analytic scalar manipulations.

Cauchy product.

CLASSICAL

The Cauchy product of $\sum a_n z^n$ and $\sum b_n z^n$ is $\sum_{n \geq 0} (\sum_{k=0}^n a_k b_{n-k}) z^n$. Under absolute convergence it represents the product of the sums; for formal power series it is the definition of multiplication without analytic convergence. Moment and Bernoulli recurrences arise by equating Cauchy-product coefficients in generating-function identities. In formalized proofs, finite coefficient sums are preferable to informal series multiplication because every index range and normalization is explicit.

Cauchy theorem.

CLASSICAL

Cauchy's theorem says that the integral of a holomorphic function around a closed contour is zero under standard domain hypotheses. Consequently, integrals along homotopic contours agree unless singularities are crossed. It underlies the deformation of Bromwich and Mellin contours used in the endpoint analysis. Quantitative tail bounds are still required because the contours are unbounded.

Cauchy transform of a measure.

LEAN-PROVED

For a finite measure μ ,

$$G_\mu(z) = \int_{\mathbb{R}} \frac{1}{z - x} d\mu(x), \quad z \notin \text{supp } \mu.$$

The sign convention $z - x$ is fixed. Expanding at infinity gives the moment series, while boundary imaginary parts recover the density under Stieltjes inversion.

CDF layer-cake formula.

LEAN-PROVED

A CDF layer-cake formula rewrites an integral involving $F(x)$ or $1 - F(x)$ as an expectation or a moment of the underlying random variable. For example, integrating a survival function gives a first moment. The formalized versions state support and integrability hypotheses so that boundary terms and totalized integrals are controlled.

Centered Appell sequence.

LEAN-PROVED

A centered Appell sequence is normalized relative to a centered probability law, usually so that applying its moment functional annihilates every positive degree: $\mathbb{E}[A_n(x + X)] = x^n$. Centering removes the first cumulant and imposes parity when the law is symmetric. The sequence acts as an exact polynomial deconvolution basis.

Centered finite spline.

PROJECT / MIXED

A centered finite spline in this development is the density or distribution obtained from a finite sum of centered, dyadically scaled uniform random variables, normalized so that its support is symmetric. It is a compactly supported piecewise polynomial computable from finitely many rational operations. As the number of summands grows, it converges uniformly to the Rvachev up-function or, after translation and integration, to the bounded Fabius function. The centering correction is essential: an uncentered approximation carries a systematic half-mesh displacement.

Centered geometric uniform law.

LEAN-PROVED

Centering subtracts the mean from a geometric uniform series, producing an even law supported on a symmetric interval. The characteristic function then loses its phase and becomes a product of sinc factors. Even cumulants remain, odd moments vanish, and the $q = 1/2$ density is the up-function under the project's normalization.

Centered periodic correction.

PROJECT / MIXED

A centered periodic correction is a periodic function from which its mean value has been subtracted. If $P(u + 1) = P(u)$, then $\Psi(u) = P(u) - \int_0^1 P(v) dv$ has period one and zero average. Centering moves the constant Fourier mode into the nonoscillatory main constant and leaves only nonzero modes. In the corrected Fabius endpoint asymptotic, $\Psi(\lambda(x))$ is a genuine nonconstant log-periodic term whose Fourier coefficients are explicit Gamma-zeta values.

Centered sinc partial product.

LEAN-PROVED

A centered partial product is the phase-free Fourier product obtained after centering each uniform summand. Removing the phases makes it real and even on the real axis and exposes sign changes only through sinc lobes. It is the transform counterpart of a centered finite spline approximant.

Centered Thue–Morse shell.

LEAN-PROVED

The centered Thue–Morse shell is a standalone finite sinc-product model in which dyadic lobe signs are encoded by centered binary data. Its formal theorems concern the product model itself; they do not automatically identify every shell formula with a Fourier transform of a separately constructed finite inverse object.

Central binomial valuation.

PROJECT / MIXED

The valuation of $\binom{2n}{n}$ is governed by carries in $n + n$. For $p = 2$, $v_2\left(\binom{2n}{n}\right) = s_2(n)$. This exact bridge between central binomial coefficients and binary weight appears in Legendre/Gaunt formulas, denominator estimates, and $q = -1$ specializations.

Central Gaussian binomial coefficient.

LEAN-PROVED

A central Gaussian coefficient is $\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q$, or more generally the coefficient with k nearest $n/2$. It has degree n^2 , is palindromic, and specializes at $q = 1$ to the central binomial coefficient. The current reduction theory factors central objects into cyclotomic and product pieces suited to divisibility, asymptotic, and Catalan analyses.

Central limit theorem.

PROJECT / MIXED

A central limit theorem states that a centered, variance-normalized sequence of random variables converges in distribution to a standard Gaussian. For geometric-uniform families approaching the dense-scale boundary, cumulants and characteristic functions yield an explicit CLT. It is a weak limit and does not alone imply density, entropy, or tail asymptotics.

Chaos multi-index.

FRONTIER

A chaos multi-index $\alpha = (\alpha_1, \alpha_2, \dots)$ has finite support and labels a tensor product of one-coordinate orthogonal polynomials. Its total degree is $|\alpha| = \sum_j \alpha_j$. The symbol and admissible zero components are declared so it cannot be confused with a valuation or asymptotic exponent.

Characteristic function.

CLASSICAL

The characteristic function of a real random variable X is $\varphi_X(t) = \mathbb{E}(e^{itX})$. It always exists, is positive definite and uniformly continuous, and determines the probability law uniquely. Independence turns sums into products: $\varphi_{X+Y} = \varphi_X \varphi_Y$. For the centered dyadically weighted uniform series underlying the up-function, each summand contributes a sinc factor, so the characteristic function becomes an infinite sinc product. Under the repository's Fourier convention this is precisely the Fourier transform of the density.

Christoffel–Darboux kernel.

PROJECT / MIXED

For orthonormal polynomials $p_0, \dots, p_n$,

$$K_n(x, y) = \sum_{k=0}^n p_k(x)p_k(y).$$

The Christoffel–Darboux identity rewrites this sum as a quotient involving p_n, p_{n+1} . On the diagonal $K_n(x, x) > 0$; its reciprocal is the Christoffel function. The kernel is the reproducing kernel for polynomials of degree at most n in the moment inner product.

Christoffel function.

PROJECT / MIXED

The Christoffel function is

$$\lambda_n(x) = K_n(x, x)^{-1} = \min \left\{ \int |p|^2 d\mu : \deg p \leq n, p(x) = 1 \right\}.$$

It measures the cost of point evaluation and controls quadrature weights, least-squares stability, and local polynomial concentration. Its normalization depends on whether the orthogonal basis is monic or orthonormal.

Closed form.

CLASSICAL

A closed form is a finite expression assembled from an agreed collection of standard operations or special functions. The term has no universally fixed boundary: a finite sum involving moments and Thue–Morse signs may be considered explicit even when no elementary simplification exists. In the Fabius development, closed forms include exact dyadic finite sums, explicit recurrences, Gaussian q -binomial formulas, and Lambert-based asymptotic coordinates. The phrase should not imply that evaluation is numerically stable or computationally optimal.

Coefficient extraction.

CLASSICAL

Coefficient extraction is denoted $[z^n]A(z)$ and returns the coefficient of z^n in a polynomial, formal power series, or convergent analytic series. It converts product identities into finite convolution sums and turns interpolation or generating-function statements into arithmetic formulas. In the q -binomial part of the Fabius theory, a weighted Thue–Morse block annihilates all lower-degree node polynomials, leaving a leading coefficient that equals the desired dyadic value after normalization.

Coercion and scalar cast.

CONVENTION

A coercion transports an element along a canonical embedding, such as $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$, or views a polynomial over a smaller coefficient ring in a larger one. A scalar-cast theorem states that an exact rational or integer identity survives this transport. The cast changes the ambient type, not the mathematical normalization; explicit bridge lemmas keep arithmetic proofs separate from analytic consequences.

Compact support.

CLASSICAL

The support of a function is the closure of the set where it is nonzero; compact support means this closed set is compact. On $\mathbb{R}$, such a function vanishes outside a bounded interval. The up-function has support exactly $[-1, 1]$. Compact support implies that its Fourier transform extends to an entire function of exponential type, while C^∞ smoothness and endpoint flatness give rapid decay along the real axis. Support is topological: endpoints belong to it even though the function value there is zero.

Companion Pascal row.

FRONTIER

A companion row is an auxiliary binomial or valuation row chosen so that a recurrence, symmetry, or convolution becomes closed. It may track neighboring indices, reciprocals after denominator clearing, or carry corrections. Because the term is report-local, its exact normalization is always printed with the defining formula.

Compiled tree.

CONVENTION

The compiled tree is the set of Lean sources that successfully build under the declared import graph and toolchain. A theorem present only in a draft file outside that graph is not part of the formal library. Corpus census, public-facade, and proof-backed-exposition claims refer to the compiled tree at a recorded checkpoint.

Complementary Bell number.

FRONTIER

A complementary Bell number is a signed or transformed Bell-family value defined in the relevant report, often by replacing $e^z - 1$ with $1 - e^z$ or by a binomial involution. Because several conventions exist, the defining generating function is authoritative. The family appears in derangement and inverse-coefficient asymptotics.

Complete binary block.

LEAN-PROVED

A complete binary block is an interval of 2^m consecutive integers aligned at a multiple of 2^m . Its lower m bits run through all binary words exactly once. Thue–Morse signs factor across aligned blocks, so Prouhet cancellation and prefix recursions can be transferred from the initial block to every aligned one.

Complete exponential Bell polynomial.

CLASSICAL

The complete exponential Bell polynomial is $B_n(x_1, \dots, x_n) = \sum_k B_{n,k}(x_1, \dots)$. It satisfies

$$\exp\left(\sum_{j \geq 1} x_j \frac{t^j}{j!}\right) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) \frac{t^n}{n!}.$$

It converts cumulants to moments and reciprocal moment data to Appell coefficients.

Complete homogeneous symmetric polynomial.

LEAN-PROVED

The complete homogeneous symmetric polynomial $h_m(x_1, \dots, x_r)$ is the sum of all monomials of total degree m . Its generating function is $\prod_i (1 - x_i t)^{-1}$. Evaluating it on a geometric progression produces Gaussian-binomial expressions used in geometric Lagrange moment identities.

Complex exponential generating function.

CLASSICAL

An exponential generating function has the form $A(z) = \sum_{n \geq 0} a_n z^n / n!$. When $z \in \mathbb{C}$ and the series converges, it is a complex analytic function; formally, it packages binomial convolutions especially efficiently. Moment generating functions are exponential generating functions of moments. For the Fabius and up distributions, complex EGFs connect the moment recurrences with products of elementary exponential factors and permit coefficient identities to be proved uniformly over $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$.

Complex moment-generating function.

CLASSICAL

For a bounded real random variable X , the complex moment-generating function is $M_X(z) = \mathbb{E}(e^{zX})$ for $z \in \mathbb{C}$. Boundedness makes it entire, and $M_X^{(n)}(0) = \mathbb{E}(X^n)$. Independence converts sums into products. Evaluating at imaginary arguments gives the characteristic function, while negative real arguments lead to Laplace asymptotics. The Fabius development uses this single entire object to relate moments, the sinc product, the negative-Laplace product, and Bromwich inversion.

Complex shift.

CLASSICAL

A complex shift replaces an argument or interpolation node by $z + Q$ with a fixed $Q \in \mathbb{C}$. Translation usually changes finite expressions, but a sufficiently high-order finite-difference functional annihilates all lower-degree contributions introduced by the shift. The repository proves that each normalized finite q -binomial Taylor block is independent of a common complex shift, even though individual summands depend on it. In the associated discrete-limit approximants, dependence may remain at finite level and disappears only in the limit.

Composite-mesh sharpness.

LEAN-PROVED

Composite-mesh sharpness classifies all nonzero integer meshes achieving a given universal reproduction degree. The formal result equates exactness through degree d with $2^d \mid M$, hence identifies 2^d as the least mesh. It is a whole polynomial-space theorem and does not assert minimality for one particular target polynomial.

Computable inverse Fabius function.

LEAN-PROVED

Computability of the inverse means that from a name for $y \in [0, 1]$ and a precision request one can produce a rational approximation to $F^{-1}(y)$ with certified error. Monotonicity, computability of F , and an effective separation or modulus condition provide the algorithm. This is a statement in computable analysis, not a practical complexity bound.

Computable real.

CLASSICAL

A computable real is a real number supplied by an algorithm that, given a precision request p , returns a rational or signed dyadic approximation within 2^{-p} . The algorithm must be uniform and terminating, and the error guarantee is part of the representation. A computable sequence requires one algorithm uniform in both the sequence index and the precision. The repository uses redundant pairs of natural-number numerators to keep the coding primitive recursive and to avoid relying on hidden integer arithmetic.

Computable real function.

CLASSICAL

In the Grzegorzczuk-style formulation used here, a real function is computable when it is sequentially computable and effectively uniformly continuous. Sequential computability means that it maps every computable real sequence to a computable sequence, uniformly in indices and requested precision. Effective uniform continuity supplies a recursive modulus controlling how accurate the input must be. The bounded Fabius function is proved computable by evaluating explicit finite splines and using a certified uniform approximation error together with the sharp global Lipschitz bound.

Computable sequence.

CLASSICAL

A sequence $(x_i)_{i \geq 0}$ of real or complex numbers is computable if a single algorithm, given i and a precision parameter p , produces a rational approximation to x_i with a certified error such as 2^{-p} . Uniformity in i distinguishes a computable sequence from a list of individually computable numbers with unrelated algorithms. Moment sequences, dyadic values, and coefficients of spline approximants in the repository are represented by exact recurrences, yielding stronger information than approximation alone.

Conditional asymptotic theorem.

FRONTIER

A conditional asymptotic theorem derives a limiting formula under an explicit hypothesis not yet proved in the surrounding development, such as a parity assertion, a tail estimate, or a nonvanishing condition. Its logical status is weaker than an unconditional theorem even when the derivation is correct. Several repository modules retain conditional intermediate results as reusable transfer lemmas, while later modules discharge the hypotheses. The glossary distinguishes this proof-theoretic adjective from conditional convergence of series, which is a different analytic notion.

Conditional law.

CLASSICAL

A conditional law describes the distribution of a random variable after conditioning on a sub- σ -algebra or event. In geometric random series, conditioning on the first uniform summand leaves an independent scaled tail, which yields integral equations for the CDF and density. Formal statements avoid pointwise conditional densities when a regular conditional probability has not been constructed.

Continuity of Gaussian coefficients in the base.

LEAN-PROVED

Because $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is a polynomial, it defines a continuous and entire function of q , including $q = 1$, $q = 0$, and roots of unity. At $q = 1$ it equals $\binom{n}{k}$. This polynomial continuation is the canonical meaning at singular-looking bases; a quotient of q -factorials is only a derived formula on its nonvanishing domain.

Continuous Gaussian coefficient.

FRONTIER

For $0 < q < 1$ and suitable complex or positive real x, y , a continuous extension is

$$\begin{bmatrix} x \\ y \end{bmatrix}_q = \frac{\Gamma_q(x+1)}{\Gamma_q(y+1)\Gamma_q(x-y+1)}.$$

It agrees with the Gaussian polynomial at integral $x = n, y = k$ in range, but away from integers it is meromorphic and branch-sensitive through q^z . It is used for inversion in the upper or lower parameter and for the $q \uparrow 1$ bridge to the ordinary gamma quotient.

Continuous-linear naturality.

LEAN-PROVED

Continuous-linear naturality states that a convergent random or deterministic series commutes with a continuous linear map: $T(\sum x_n) = \sum T(x_n)$. It transports weighted-uniform constructions between scalar and vector spaces, sends laws through affine projections, and justifies termwise transform or coordinate calculations under the declared summability mode.

Contour deformation.

CLASSICAL

Contour deformation replaces an integration path in the complex plane by a homologous path on which the integral is easier to estimate, using Cauchy's theorem and accounting for any crossed singularities by residues. For an unbounded contour one must also control connecting arcs and convergence at infinity. In Bromwich saddle analysis, the original vertical line is localized or deformed toward a steepest-descent path through the saddle. The repository separates local central estimates from tail and minor-arc bounds so that the deformation is quantitatively justified.

Contour integral.

CLASSICAL

A contour integral $\int_\gamma f(z) dz$ integrates a complex function along a parameterized curve γ . Holomorphy allows path deformation, while poles contribute residues. Fourier inversion, Laplace inversion, Mellin inversion, and coefficient extraction all admit contour formulations. In the Fabius endpoint proof, the contour integral carries a large exponential phase determined by the negative-Laplace transform; its central portion yields the saddle expansion and its remote portions require uniform decay estimates.

Contraction constant.

CLASSICAL

A contraction constant is a number $q < 1$ satisfying $d(Tx, Ty) \leq qd(x, y)$. It governs the geometric rate q^n of fixed-point iteration and the size of certified error bounds. In a Fabius fixed-point construction, the exact value depends on the chosen operator and function norm. Recording it turns uniqueness into an effective approximation theorem.

Contraction mapping.

CLASSICAL

A contraction mapping on a metric space satisfies $d(Tx, Ty) \leq qd(x, y)$ for a constant $q < 1$. Contractions are automatically continuous and, on complete spaces, have unique fixed points. Their iterates converge geometrically, giving explicit numerical error bounds. The Fabius defining operator is arranged to be a contraction on a space of bounded continuous functions with fixed tails and symmetry. This construction avoids assuming beforehand that the desired smooth function or probability law exists.

Convexity and concavity.

CLASSICAL

A function is convex on an interval when $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$ for $0 \leq t \leq 1$; it is concave when the inequality is reversed. For twice differentiable functions, $f'' \geq 0$ implies convexity and $f'' \leq 0$ implies concavity. The bounded Fabius function is strictly convex on the left half of $[0, 1]$, strictly concave on the right half, and has a unique inflection point at $1/2$. This follows from the monotonicity and symmetry of its derivative, which is an affine copy of the up-function.

Convolution.

CLASSICAL

For integrable functions, $(f * g)(x) = \int_{\mathbb{R}} f(x-y)g(y) \, dy$; for measures, convolution is the law of the sum of independent random variables. Convolution is associative and commutative, preserves total mass, adds supports by Minkowski sum, and turns into multiplication under the Fourier transform. The Fabius probability model is an infinite convolution of scaled uniform laws. Finite convolutions are splines, while the limiting centered density is the up-function.

Convolution of measures.

CLASSICAL

For finite Borel measures μ, ν on $\mathbb{R}$, their convolution is $(\mu * \nu)(A) = \int \int \mathbf{1}_A(x+y) \mu(dx) \nu(dy)$. It is the pushforward of $\mu \otimes \nu$ under addition. Characteristic functions multiply, and supports add. The finite and infinite convolution measures in the Fabius probability construction are defined at this measure level before smooth densities are identified.

Convolution power.

CLASSICAL

The k -fold convolution power f^{*k} is defined recursively by $f^{*1} = f$ and $f^{*(k+1)} = f^{*k} * f$. It represents the density of a sum of k independent identically distributed variables when f is a density. Refinement identities for the up-function lead to exact self-convolution and scaled convolution formulas. Fourier transformation converts a convolution power into an ordinary power of the transform, often making normalization and support calculations immediate.

Cornish–Fisher expansion.

FRONTIER

A Cornish–Fisher expansion reverses an Edgeworth CDF expansion to approximate quantiles:

$$Q(u) = \Phi^{-1}(u) + \sum_{j \geq 1} \epsilon^j q_j(\Phi^{-1}(u)).$$

The coefficient polynomials are obtained by series reversion and Hermite identities. Uniformity is normally restricted to u away from 0 and 1, where the Gaussian density stays bounded away from zero.

Correction term.

CLASSICAL

A correction term is a subordinate contribution added to a leading approximation to reach a sharper error scale. Corrections can be algebraic powers, logarithms, periodic functions, or exponentially small terms. Their relative size must be judged in the variable actually governing the limit. In the Fabius endpoint expansion, the centered one-periodic fluctuation is part of the main exponent rather than a dispensable numerical decoration; subsequent saddle corrections occur in inverse powers of $\lambda(x)$.

Cosine series.

CLASSICAL

A cosine series is a Fourier series involving only $\cos(n\pi x/L)$, appropriate for even functions on a symmetric interval. If f is even, its sine coefficients vanish. The up-function admits a cosine expansion whose coefficients are samples of its Fourier transform; the sinc product forces all nonzero even-frequency coefficients in a particular normalization to vanish, leaving an odd cosine series. Smoothness gives rapid coefficient decay and hence strong convergence.

Coupling.

CLASSICAL

A coupling of probability measures μ and ν is a joint law with those marginals. Constructive couplings compare random variables on one sample space and turn pointwise differences into metric bounds. The natural head–tail coupling of a geometric uniform series with its truncations controls Wasserstein distance and supports inverse-distribution error estimates.

Covariance kernel.

PROJECT / MIXED

For a centered family X_t , the covariance kernel is $K(s, t) = \mathbb{E}[X_s X_t]$. It is positive definite. Normalizing the geometric-uniform family yields a rational kernel in the parameters, from which Gram determinants and projection errors can be computed exactly.

Critical value.

CLASSICAL

A critical value is $f(x_0)$ where $f'(x_0) = 0$ or, in a complex setting, where the derivative of the map vanishes. Local inverse branches can meet or ramify there. Critical values, not merely zeros and poles of the forward function, determine the branch atlas of inverse q -special functions.

Cumulant.

CLASSICAL

Cumulants κ_n are the coefficients of the logarithm of a moment-generating or characteristic function: $\log \mathbb{E}(e^{zX}) = \sum_{n \geq 1} \kappa_n z^n / n!$ near zero. The first cumulant is the mean, the second is the variance, and higher cumulants measure non-Gaussian structure. Cumulants add for independent sums, making them natural for the infinite random-series representation. Endpoint saddle modules use tilted cumulants and their recurrences to control local phase derivatives and sharp dyadic asymptotics.

Cumulant-generating function.

CLASSICAL

The cumulant-generating function is $K_X(z) = \log M_X(z)$, where $M_X(z) = \mathbb{E}(e^{zX})$, on a domain where a consistent logarithm is defined. Its derivatives at zero are cumulants. For products of moment-generating functions arising from independent summands, taking logarithms converts the product into a sum, revealing dyadic harmonic structure. The negative-real specialization for the Fabius law is denoted by a logarithmic Laplace function whose exact decomposition contains polynomial-logarithmic terms and a one-periodic remainder.

Cumulant scaling law.

LEAN-PROVED

If $Y = aX + b$, then $\kappa_1(Y) = a\kappa_1(X) + b$ and $\kappa_n(Y) = a^n \kappa_n(X)$ for $n \geq 2$. Independent sums add cumulants. For geometric uniform series this turns each cumulant into a geometric sum, producing rational functions of q and Bernoulli-number factors.

Cumulative distribution function.

CLASSICAL

The cumulative distribution function of a real random variable X is $F(x) = \mathbb{P}(X \leq x)$. It is nondecreasing, right-continuous, tends to zero at $-\infty$, and tends to one at $+\infty$. If the law has a continuous density f , then $F'(x) = f(x)$. The bounded Fabius function is a smooth CDF supported in the probabilistic sense on $[0, 1]$; its derivative is a scaled translate of Rvachev's up-function.

Cyclotomic inverse branch.

FRONTIER

Near a root of unity ζ , a q -polynomial or meromorphic q -function has a local expansion $Y - Y_0 = c(q - \zeta)^m + \dots$. If $m = 1$, the inverse is analytic; if $m > 1$, it has m Puiseux branches with leading power $(Y - Y_0)^{1/m}$. Cyclotomic factorization determines zeros, while derivative cancellation determines additional critical ramification.

Cyclotomic polynomial.

CLASSICAL

The m -th cyclotomic polynomial $\Phi_m(q) \in \mathbb{Z}[q]$ is the monic polynomial whose roots are the primitive m -th roots of unity. It satisfies

$$q^n - 1 = \prod_{d|n} \Phi_d(q).$$

Cyclotomic factorization converts products of $1 - q^j$ into exact zero-order and divisibility data. This is the arithmetic backbone of root-of-unity analysis for Gaussian coefficients and q -Catalan polynomials.

Cyclotomic valuation.

LEAN-PROVED

For a nonzero polynomial $P(q)$, the cyclotomic valuation $v_{\Phi_m}(P)$ is the exponent of Φ_m in its factorization. For products of $1 - q^j$, it is obtained by counting multiples of m . Subtracting numerator and denominator counts gives the exponent in a Gaussian coefficient and decides whether a primitive m -th root is a zero, a removable quotient singularity, or a nonzero value.

4 D **D -finite function.**

FRONTIER

A formal power series is D -finite when it satisfies a nontrivial linear differential equation with polynomial coefficients. Over characteristic zero, its coefficient sequence is P -recursive. Smooth nonanalytic compactly supported functions are not represented by one convergent power series on the whole support, so local or transform formulations must be distinguished.

Decay estimate.

CLASSICAL

A decay estimate bounds a function by a quantity tending to zero in a specified limit, often uniformly in auxiliary parameters. Polynomial, exponential, superpolynomial, and stretched-exponential decay have different analytic consequences. The Fourier transform of a compactly supported C^∞ function decays faster than every power on the real axis, while the Fabius endpoint value decays on a quadratic logarithmic scale. Quantitative decay estimates justify Fourier inversion, contour truncation, termwise operations, and numerical stopping rules.

Declaration arity.

CONVENTION

The arity of a definition or notation command is the number and order of its explicit arguments. Parameters suppressed by a local section are not part of the printed command but remain part of the mathematical context. Arity auditing prevents, for example, a base q , order ν , or evaluation variable from being silently exchanged when formulas are migrated between documents.

Declaration inventory.

CONVENTION

A declaration inventory is an explicit list or counted summary of the public definitions and theorems in a module or thematic tranche. It lets a prose document state exactly which formal objects support a section. The current README frequently records inventories as “definitions + theorems” and adds non-claims to prevent a finite or formal result from being read as a stronger analytic theorem.

Delay differential equation.

CLASSICAL

A delay differential equation relates a derivative at one point to function values at transformed or earlier points, rather than only to values at the same point. The Fabius relation $F'(x) = 2F(2x)$ is more precisely an advanced dilation equation on the left half when read in x , but after logarithmic change of variables it becomes a delay equation because multiplication of x by two becomes translation. Such equations encode self-similarity and can generate log-periodic phenomena.

Delta series.

CLASSICAL

A delta series is a formal power series $f(t)$ with zero constant term and nonzero linear coefficient. It has a compositional inverse and serves as the substitution component of a Sheffer sequence or exponential Riordan array. Normalized reversion coefficients are formed from its higher coefficients relative to the linear term.

Denominator bound.

CLASSICAL

A denominator bound gives an explicit integer divisible by the reduced denominators of a family of rational quantities. It may be universal over all dyadic numerators at a fixed level or tailored to one sequence. In the Fabius arithmetic, exact finite formulas prove rationality at dyadic points, while valuation formulas identify powers of two and least-common-multiple products control odd prime factors. A denominator bound certifies exact storage and comparison even when it is not minimal.

Denominator clearing.

PROJECT / MIXED

Denominator clearing multiplies a rational identity by a common nonzero factor to obtain an integral or polynomial identity. It enables valuation, divisibility, parity, and semiring arguments. The chosen factor should be recorded: an oversized denominator may obscure sharp arithmetic even though it preserves correctness.

Dense analytic locus.

LEAN-PROVED

A function has dense analytic locus when every nonempty open subinterval contains a point at which it is real analytic. Elementary and algebraic branches satisfy such a property under the repository's closure theorems. Nowhere analyticity of the Fabius function and its inverse therefore yields strong non-elementarity and non-algebraicity conclusions.

Derivative scaling.

CLASSICAL

Derivative scaling is the law obtained by repeatedly differentiating a dilation equation. From $\mathcal{F}'(x) = 2\mathcal{F}(2x)$, induction and the chain rule give $\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}}\mathcal{F}(2^m x)$. The triangular exponent comes from multiplying $2^{1+2+\dots+m}$. This identity yields exact dyadic derivatives, sharp uniform derivative bounds, terminating Taylor series at dyadic points, and high-order flatness estimates.

Derivative-value distribution.

LEAN-PROVED

The derivative-value distribution treats $F'(X)$, $\text{up}'(X)$, or a related derivative evaluated at a random argument as a random variable. The current formal development derives distributional symmetries and level-set identities from reflection and monotonicity. This viewpoint converts shape information into pushforward measures and moments, rather than considering derivative values only pointwise.

Determinant–Bell identity.

FRONTIER

A determinant–Bell identity expresses complete Bell polynomials, reciprocal-series coefficients, or cumulants as structured determinants. It packages a triangular recurrence into one closed algebraic object. Such identities are useful for exact arithmetic and symbolic verification but do not automatically improve numerical conditioning.

Determinate moment problem.

CLASSICAL

A moment problem is determinate when at most one positive measure has the given moments. Compact support implies determinacy by polynomial density. Carleman's condition provides a broader sufficient criterion on unbounded support. In the Fabius/Rvachev setting, compact support means that exact moment identities determine the probability law, not merely its formal generating series.

Differential entropy.

FRONTIER

For a density f ,

$$h(f) = - \int f(x) \log f(x) \, dx,$$

when the positive and negative parts are defined, with $0 \log 0 = 0$. Unlike discrete entropy it can be negative and changes by $\log |a|$ under the scaling aX .

Differential equation.

CLASSICAL

A differential equation imposes a relation among a function and its derivatives. Ordinary differential equations compare quantities at the same independent variable, while functional-differential equations may include dilated or translated arguments. The Fabius and up equations are of the latter type. Their global solution theory requires normalization and support or tail conditions; without these, the equation alone need not select the intended function. Repeated differentiation reveals the exact self-similar hierarchy of derivatives.

Differentially algebraic function.

FRONTIER

A function is differentially algebraic when it satisfies a polynomial differential equation involving the function and finitely many derivatives. Every D -finite function is differentially algebraic, but not conversely. A non-elementarity theorem does not imply differential transcendence; these are distinct levels of expressive complexity.

Differentially transcendental function.

FRONTIER

A function is differentially transcendental over a base differential field if it satisfies no nonzero algebraic differential equation over that field. Proving this is much stronger than showing non-elementarity or non- D -finiteness. Fabius-related differential-transcendence statements are frontier questions unless backed by a precise theorem.

Differentiation under the integral sign.

CLASSICAL

Differentiation under the integral sign exchanges $\frac{d}{dx} \int f(x, t) \, dt$ with $\int \partial_x f(x, t) \, dt$. It is justified by hypotheses such as continuity of the derivative and an integrable dominating function uniform in x . The method derives smoothness and differential identities for convolution and transform representations. In the Fabius setting, compact support and rapid decay often provide the needed domination, but endpoint or complex-contour integrals require separate estimates.

Digit character.

FRONTIER

A digit character assigns a root of unity to a digit statistic, for example $\omega^{sq(n)}$. The binary choice $\omega = -1$ gives the Thue–Morse sign. Character orthogonality decomposes digit-restricted sums and suggests multibase analogues of Prouhet cancellation, rarefaction, and geometric interpolation identities.

Digit multiplicity profile.

FRONTIER

A digit multiplicity profile records how often each digit occurs in a base expansion or across a finite block. Complete blocks have uniform positional profiles, while constrained or carry-conditioned sets do not. Such profiles feed weighted digit characters, rarefaction matrices, and valuation statistics.

Digit power sum.

FRONTIER

A digit power sum is a finite sum of powers of digit values or digit-weighted positions, such as $\sum_j d_j(n)^r$ or $\sum_j j^r d_j(n)$. These statistics refine the ordinary digit sum and can parameterize weighted automatic sequences, carry distributions, and higher derivatives of digit generating functions.

Digitwise factorization.

CLASSICAL

Digitwise factorization is an identity in which a sum indexed by binary words factors into independent contributions from each bit position. The basic Thue–Morse polynomial satisfies $\sum_{r=0}^{2^k-1} \varepsilon_r z^r = \prod_{j=0}^{k-1} (1 - z^{2^j})$. Choosing a bit either contributes 1 or contributes $-z^{2^j}$, so expanding the product enumerates all binary integers and records the parity of their digit sum. This factorization is the engine behind Prouhet cancellation and several dyadic formulas.

Dirac comb.

CLASSICAL

The Dirac comb of spacing $h > 0$ is the tempered distribution $\text{III}_h = \sum_{k \in \mathbb{Z}} \delta_{kh}$. Under a fixed Fourier convention it transforms to a reciprocal-spacing comb with a normalization factor. Multiplying by a function samples it, while convolving periodizes it. Poisson summation is the equality of these two Fourier descriptions.

Dirac measure.

CLASSICAL

The Dirac measure δ_a places unit mass at a : $\int f \, d\delta_a = f(a)$. Convolution with it translates, $f * \delta_a = f(\cdot - a)$, and its 2π -Fourier transform is $e^{-2\pi i a \xi}$. Finite atomic measures encode spline difference operators, sampling combs, and distributional derivatives of piecewise polynomials throughout the repository.

Dirichlet eta function.

CLASSICAL

The Dirichlet eta function is $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ for $\Re s > 0$, with analytic continuation to an entire function. It satisfies $\eta(s) = (1 - 2^{1-s})\zeta(s)$. Alternating sums involving powers or reciprocal powers can therefore be expressed in zeta terms with a dyadic correction factor. Eta-type combinations appear naturally when Thue–Morse or parity decompositions separate even and odd indices.

Discrete antiderivative.

CLASSICAL

For a sequence a_n , a discrete antiderivative is a sequence A_n with $\Delta A_n = A_{n+1} - A_n = a_n$. Choosing $A_0 = 0$ gives the prefix sum. Iterating produces binomial-kernel convolutions and raises polynomial degree, just as continuous integration does. Boundary constants are part of the definition.

Discrete B-spline.

CLASSICAL

A discrete B-spline is a finite sequence or grid function obtained by repeated discrete convolution of interval indicators, serving as the lattice analogue of a continuous B-spline. Iterated prefix sums are an equivalent construction. After normalization and rescaling, such sequences approximate compactly supported spline densities. The repository identifies corrected Thue–Morse grid arrays with histogram or discrete-spline data, allowing a combinatorial recurrence to be connected to the analytic up-function through a discrete limit.

Discrete convolution.

CLASSICAL

For sequences a, b , their discrete convolution is $(a * b)_n = \sum_k a_k b_{n-k}$, with the sum restricted to indices where both terms are defined. It models the distribution of sums of integer-valued random variables and is the coefficient rule for products of generating functions. Iterated convolution of finite box sequences produces discrete B-splines. In Thue–Morse limit arguments, discrete convolution and prefix summation reveal cancellation and smooth the signed data before scaling.

Discrete convolution-power kernel.

CLASSICAL

The r -fold convolution power of the constant-one sequence has coefficient $\binom{n+r-1}{r-1}$. Consequently repeated prefix summation is convolution with a binomial kernel. This observation turns nested sums into a single finite sum and links discrete antiderivatives to weak compositions.

Discrete Fourier transform.

CLASSICAL

For N samples $a_0, \dots, a_{N-1}$, the discrete Fourier transform is $\hat{a}_k = \sum_{n=0}^{N-1} a_n e^{-2\pi i n k / N}$. It diagonalizes cyclic convolution and converts periodic digit or valuation data into spectral modes. Normalization may be placed in the forward, inverse, or both transforms and is always declared.

Discrete limit.

CLASSICAL

A discrete limit theorem shows that normalized finite arrays, sums, or grid functions converge to a continuous object as the mesh tends to zero or the degree tends to infinity. Convergence may be pointwise, uniform, in L^p , or weak. The Fabius directory contains several discrete-limit constructions: spline grids converging to the up-function, complex-shift q -binomial rows converging to the signed global extension, and Thue–Morse histograms converging after exact normalization. Finite identities and stability bounds are kept separate from the limiting argument.

Distribution law.

CLASSICAL

The distribution law of a random variable is the probability measure it induces on its value space. Two random variables can be defined on different probability spaces yet have the same law. Laws are compared through cumulative distribution functions, moments when determinate, or characteristic functions. The Fabius function is most naturally defined as the CDF of an infinite weighted sum of

independent uniform variables; this law-based viewpoint makes positivity, symmetry, and convolution identities immediate.

Division-free identity.

PROJECT / MIXED

A division-free identity is written using multiplication and polynomial operations after all denominators have been cleared. It remains meaningful at specializations where a quotient formula would display $0/0$, and it is well-suited to integral, modular, and semiring proofs. Current Gaussian adjacent-ratio, moment-numerator, and cyclotomic arguments commonly prove a division-free statement first and derive quotient forms under nonvanishing hypotheses.

Division-free recurrence.

PROJECT / MIXED

A division-free recurrence computes each new term using additions and multiplications of previously known integers, with any apparent rational factors already shown integral. Such a recurrence is valuable for exact arithmetic because it avoids repeated fraction reduction and exposes positivity or divisibility. The repository gives division-free recurrences for normalized moment numerators and denominator sequences by clearing triangular factors such as factorials and Mersenne products. The proof must verify that every displayed quotient is an integer before the recurrence can be called division-free.

Dominated convergence theorem.

CLASSICAL

The dominated convergence theorem states that if measurable functions $f_n \rightarrow f$ almost everywhere and $|f_n| \leq g$ for an integrable g , then $\int f_n \rightarrow \int f$. It is a principal tool for interchanging limits with integrals. In the Fabius theory it supports limits of finite convolutions, differentiation or expansion of transform integrals, and passage from finite random sums to the infinite law. Uniform or local domination must be checked explicitly when a parameter varies.

Double factorial.

CLASSICAL

The double factorial is $n!! = n(n-2)(n-4)\cdots$, terminating at 1 or 2, with $(-1)!! = 0!! = 1$. Thus $(2m-1)!! = (2m)!/(2^m m!)$. Odd double factorials are Gaussian moments and appear in saddle-point integrals and in the normalization of Reshetnikov's integer sequence. Their product structure isolates powers of two from odd factors, which is useful in valuation and integrality arguments.

Dyadic argument.

PROJECT / MIXED

A dyadic argument is a real input of the form $a/2^n$, usually with integers $a, n \geq 0$. Reducing a to odd numerator identifies the exact dyadic level. The Fabius differential scaling maps dyadic points to integers after finitely many doublings, causing all sufficiently high derivatives to vanish there. Consequently the formal Taylor series terminates and exact rational evaluation becomes possible by finite recurrences or Thue–Morse sums.

Dyadic common denominator.

PROJECT / MIXED

A dyadic common denominator is an explicit integer D_n that clears the denominators of all Fabius values on a grid of level n , or of an associated family of normalized moments. Its construction combines factorials, powers of two, and least common multiples of odd factors. It is stronger than proving each value rational separately because it supports uniform exact tables and modular arguments. The repository develops both recurrence and closed-product controls for such denominators.

Dyadic derivative cell.

LEAN-PROVED

A dyadic derivative cell is an open interval between adjacent dyadic grid points on which a sufficiently high derivative of a finite spline approximant, or the exact derivative relation inherited from the Fabius scaling law, has a simple polynomial or signed form. Cell language separates interior formulas from knot behavior and records the level at which a rational point is reduced.

Dyadic derivative sign law.

LEAN-PROVED

At a reduced dyadic point, a distinguished highest nonzero derivative of the Fabius function has sign determined by a Thue–Morse or binary-weight parity. Lower and higher derivative behavior follows from the scaling law and cell structure. This exact sign law links analytic jets to automatic sequences and supplies noncancellation data in nowhere-analyticity arguments.

Dyadic direction profile.

FRONTIER

A dyadic direction profile records how many projected segment weights occur at each dyadic scale. It determines the integer zero multiplicities, cumulants, and comb precision of the corresponding one-dimensional product. Recovering the profile from spectral data is an inverse problem in discrete convexity.

Dyadic factorized-product extrapolant.

FRONTIER

A dyadic factorized-product extrapolant has the form $\widehat{\Phi}_{N,r} = \Phi_N \widehat{T}_{N,r}$, where $\widehat{T}_{N,r}$ approximates the omitted tail. Because the finite factor is retained exactly, included integer zeros, their multiplicities, and the real lobe sign are preserved. The hat denotes extrapolation of a transform, not a second Fourier transform.

Dyadic grid.

PROJECT / MIXED

The dyadic grid of level n is $2^{-n}\mathbb{Z}$, or its intersection with a bounded interval. It is nested under refinement: every level- n point also belongs to level $n + 1$. Functional equations with dilation by two interact especially well with this grid. Fabius values and derivatives on it admit finite exact formulas, while spline approximants and numerical verification suites use the same grid for certified comparison.

Dyadic harmonic sum.

PROJECT / MIXED

A dyadic harmonic sum has the form $H(s) = \sum_{j \in \mathbb{Z} \text{ or } j \geq 0} h(s/2^j)$. Mellin transformation turns scaling by 2^{-j} into a geometric series factor $(1 - 2^{-z})^{-1}$. Its imaginary lattice of poles produces

one-periodic functions of $\log_2 s$ or a related phase. The logarithm of the negative-Laplace Fabius product is such a sum.

Dyadic name.

PROJECT / MIXED

A dyadic name for a real number is a sequence of dyadic rationals $a_p/2^p$ converging with a prescribed fast error, commonly $|x - a_p/2^p| \leq 2^{-p}$. A signed-dyadic name may encode the numerator as a pair of naturals (a_p^+, a_p^-) . Names are representations used in computable analysis: an algorithm operates on approximations with guarantees rather than on an inaccessible exact real. The Fabius computability proof constructs such names uniformly from finite spline values.

Dyadic rational.

PROJECT / MIXED

A dyadic rational is a rational number whose reduced denominator is a power of two, equivalently an integer multiple of 2^{-n} . Dyadic rationals form a dense subring of $\mathbb{R}$ and are exactly the finite binary fractions. They are privileged for the Fabius function because the dilation equation repeatedly doubles the argument; after finitely many steps a dyadic point reaches an integer, producing terminating derivative and evaluation formulas. Every bounded Fabius value at a dyadic argument is rational.

Dyadic refinement invariance.

PROJECT / MIXED

Dyadic refinement invariance is the property that representing the same dyadic rational on a finer grid does not change the value produced by a formula or algorithm. For example, $a/2^n = (2a)/2^{n+1}$ must yield identical outputs. This is nontrivial when formulas contain level-dependent moments, factorials, or sign sums. The repository proves compatibility directly, ensuring that exact evaluators depend on the rational number itself rather than on a noncanonical numerator-denominator pair.

Dyadic spine.

LEAN-PROVED

The dyadic spine is the dense set of dyadic points and their iterated preimages used to propagate exact derivative information through scaling and composition. At these points the Fabius derivatives have finite-order termination or explicitly signed nonzero values. Nowhere-analyticity proofs use the spine to force incompatible jets arbitrarily close to any proposed analytic neighborhood.

Dyadic Taylor evaluator.

PROJECT / MIXED

The dyadic Taylor evaluator is a terminating exact algorithm for $F(a/2^n)$. It uses a Taylor identity whose derivatives at a dyadic center are known through the global scaling law and vanish above a finite order. A binary recursion removes a highest set bit or folds the numerator into a smaller block, so the number of recursive calls is tied to the binary weight rather than to the denominator size. All arithmetic is rational and the termination proof is explicit.

Dyadic valuation.

PROJECT / MIXED

The dyadic, or 2-adic, valuation $v_2(r)$ of a nonzero rational r is the exponent of two in its prime factorization: write $r = 2^k u/v$ with odd integers u, v , then $v_2(r) = k$. It extends the integer valuation and satisfies $v_2(rs) = v_2(r) + v_2(s)$ and $v_2(r + s) \geq \min(v_2(r), v_2(s))$. The repository proves an exact formula for $v_2(F(2^{-n}))$, thereby determining the power of two in the reduced denominator.

Dyadic-value denominator LCM.

LEAN-PROVED

At level n , the dyadic-value denominator LCM is the least common multiple of reduced denominators of the selected exact Fabius values on the odd grid. It is a positive integer with explicit boundary conventions. The quantity summarizes global denominator complexity at a level without claiming that every individual denominator attains it.

Dyadic zero classification.

LEAN-PROVED

For the up transform, a nonzero integer frequency m is a zero in every factor whose scale divides out the odd part of m . The number of vanishing factors gives an explicit multiplicity controlled by $v_2(m)$. Formal modules distinguish analytic zero order of an entire function from a 2-adic valuation of an integer.

5 E**Edgeworth expansion.**

FRONTIER

An Edgeworth expansion corrects a Gaussian density or CDF by Hermite polynomials whose coefficients are standardized cumulants. A rigorous theorem states the normalization, small parameter, number of terms, uniformity region, and remainder norm. The formal correction polynomial by itself need not be a nonnegative density.

Effective flatness.

CLASSICAL

Effective flatness is a quantitative version of infinite-order vanishing at an endpoint. Instead of merely asserting $F^{(m)}(0) = 0$ for every m , it gives explicit bounds such as $F(x) \leq 2^{\binom{n+1}{2}} x^n / n!$ when $2^n x \leq 1$. Such inequalities provide computable rates, justify endpoint truncations, and show decay faster than every fixed power. The repository derives both mean-value and iterated-integral forms from sharp derivative scaling.

Effective modulus.

CLASSICAL

An effective modulus converts a qualitative continuity or convergence statement into a recursive rate. For uniform continuity, a function $d(n)$ may guarantee that $|x - y| < 1/d(n)$ implies $|f(x) - f(y)| < 1/n$. For convergence, a modulus gives an index after which an error tolerance is met. The bounded Fabius function has an explicit modulus derived from its least Lipschitz constant, while the spline approximation has a direct precision-to-level rule. These data are essential in computable analysis.

Effective uniform continuity.

CLASSICAL

A function is effectively uniformly continuous when it has a recursively computable modulus valid over its whole domain. In the convention used by the repository, a recursive positive integer $d(n)$ satisfies $|x - y| < 1/d(n) \Rightarrow |f(x) - f(y)| < 1/n$ for every positive n . This strengthens ordinary uniform continuity by exposing a usable rate. For the constant-tail bounded Fabius function, the global Lipschitz bound yields the primitive-recursive choice $d(n) = 2n$.

Elementary-function class.

LEAN-PROVED

The elementary-function class is generated from rational functions, exponentials, logarithms, powers, trigonometric functions, and permitted compositions under a declared real or complex branch discipline. The formal definition used in the project is designed to prove a dense analytic-locus theorem. Since the Fabius function is nowhere analytic, it cannot belong to that class on any nontrivial open interval.

Elementary logarithmic asymptotic.

CLASSICAL

An elementary logarithmic asymptotic rewrites a special-function coordinate, such as the lower Lambert variable, in nested elementary logarithms. For small x , $W_{-1}(-cx)$ can be expanded using $\log x$, $\log |\log x|$, and inverse powers of $|\log x|$. Substitution makes the scale more familiar but also lengthens the formula and can obscure the periodic phase. The repository derives this form only after establishing the exact Lambert-coordinate expansion, so error propagation remains controlled.

Elementary symmetric polynomial.

CLASSICAL

The elementary symmetric polynomial $e_m(x_1, \dots, x_r)$ sums squarefree monomials over m -element subsets. Its generating function is $\prod_i (1 + x_i t)$. Finite q -binomial expansions are evaluations on geometric progressions.

Empty sum, empty product, and zero-power conventions.

CONVENTION

A sum over an empty finite index set is 0, a product over one is 1, and a zeroth convolution or operator power is the identity. In the natural-number polynomial identities used by the project, $x^0 = 1$, including the formal order-zero case $0^0 = 1$. These neutral-element conventions are not cosmetic: they make boundary formulas at $n = 0$, empty compositions, finite q -products, and triangular recurrences genuinely total.

Endpoint asymptotic.

CLASSICAL

An endpoint asymptotic describes a function as its argument approaches the boundary of its support or domain, here $x \downarrow 0$ for the bounded Fabius function. Infinite flatness makes ordinary Taylor expansion useless: every derivative at the endpoint vanishes. The correct analysis instead uses Laplace transforms, a saddle whose location diverges, a lower-Lambert coordinate, and a log-periodic correction. Symmetry transports the result to the opposite endpoint.

Endpoint convention.

CONVENTION

An endpoint convention specifies values at the boundary of an interval or support. It matters for pointwise refinement equations, half-weighted indicators, totalized inverses, and clamped CDFs even when it does not change a Lebesgue integral. The repository states whether an interval is open or closed, whether a density vanishes at support endpoints, and whether an inverse uses left, right, or canonical endpoint values.

Endpoint derivative jet.

LEAN-PROVED

The endpoint derivative jet is the sequence $(f^{(n)}(a))_{n \geq 0}$ at a support or interval endpoint. For F at zero and for up at ± 1 , the nonconstant derivatives vanish to all orders. Formal theorems package these values because they control extension by zero, repeated integration, inverse asymptotics, and nonanalyticity.

Endpoint flatness.

CLASSICAL

A smooth function is flat at an endpoint if all derivatives of positive order vanish there. For a compactly supported C^∞ function extended by zero, flatness at the support boundary is necessary for global smoothness. Flatness implies $f(x) = o(x^N)$ for every fixed N under suitable one-sided conditions, but it does not imply local analyticity. The Fabius and up-functions are infinitely flat at their transition endpoints, and the repository supplies explicit bounds quantifying that fact.

Entire function.

CLASSICAL

An entire function is holomorphic on all of $\mathbb{C}$. Its Taylor series at any point has infinite radius of convergence, though its growth can vary greatly. The Fourier transform of the compactly supported up-function is entire and is represented by a locally uniformly convergent infinite product of sinc factors. Its zeros, growth, and real-axis decay encode partitions of unity, sampling identities, and inversion formulas. Entirety follows from the product or from compact support together with standard Fourier-Laplace theory.

Entire function of exponential type.

CLASSICAL

An entire function F is of exponential type τ if roughly $|F(z)| \leq C_\varepsilon \exp((\tau + \varepsilon)|z|)$ for every $\varepsilon > 0$. Fourier transforms of functions supported in a compact interval have exponential type controlled by the support radius. For the up-function supported on $[-1, 1]$, this growth is compatible with rapid decay on the real axis. The distinction between complex-plane growth and real-axis decay is important: an entire function can grow exponentially off the real axis while decreasing superpolynomially along it.

Entire-product zero ledger.

CONVENTION

A zero ledger lists every zero of an entire product, its multiplicity, possible cancellations, and the factor responsible. It prevents a factorization valid only away from zeros from being used as a global quotient. The current q-product and Fourier-product modules pair product identities with separate zero-order and meromorphic-pole statements.

Entropy power.

FRONTIER

For an absolutely continuous real random variable with differential entropy $h(X)$, its entropy power is $N(X) = (2\pi e)^{-1} e^{2h(X)}$. It equals the variance for a Gaussian and satisfies the entropy-power inequality under independence. Compact support guarantees neither finite Fisher information nor smooth parameter dependence at every finite spline stage.

Equality in distribution.

CLASSICAL

Random variables X and Y are equal in distribution, written $X \stackrel{d}{=} Y$, when their pushforward probability measures coincide. They need not be equal on a common sample space. Independent-copy fixed-point equations for geometric uniform laws and the Fabius distribution are naturally distributional; after uniqueness and density theorems they induce pointwise identities for the canonical CDF and density.

Equality of measures.

CLASSICAL

Measures μ and ν are equal when they assign the same value to every measurable set. It is often enough to prove equality on a generating π -system or by equality of integrals against an appropriate test class. A density identity holding almost everywhere yields equality of the associated absolutely continuous measures, while pointwise endpoint discrepancies in the density remain immaterial to that measure equality.

Equicontinuity.

CLASSICAL

A family $\mathcal{F}$ of functions is equicontinuous if one neighborhood size works simultaneously for every member of the family at each point; uniform equicontinuity uses one size over the whole domain. Equicontinuity, boundedness, and compactness interact through the Arzela–Ascoli theorem. In approximation arguments for finite Fabius splines or convolutions, a common Lipschitz bound gives equicontinuity and helps upgrade pointwise or weak convergence to uniform convergence on compact sets.

Euler–Maclaurin formula.

CLASSICAL

The Euler–Maclaurin formula connects sums and integrals by endpoint derivatives and Bernoulli numbers. In one form, $\sum_{n=a}^b f(n) = \int_a^b f(t) dt + (f(a) + f(b))/2 + \sum_{k \geq 1} B_{2k}(f^{(2k-1)}(b) - f^{(2k-1)}(a))/(2k)! + R$. It is a principal tool for extracting constants, zeta values, and asymptotic corrections from harmonic sums. The Fabius Mellin and saddle analysis uses Euler–Maclaurin-type evaluations with explicit remainder control rather than as a purely formal expansion.

Euler–Mascheroni constant.

CLASSICAL

The Euler–Mascheroni constant is $\gamma = \lim_{n \rightarrow \infty} (\sum_{k=1}^n 1/k - \log n)$. It is the constant term in the Laurent expansion $\zeta(s) = 1/(s-1) + \gamma + \dots$ and satisfies $\Gamma'(1) = -\gamma$. Finite-part Mellin calculations and expansions of Gamma-zeta products naturally produce γ . In an endpoint asymptotic, it belongs to the nonoscillatory constant after the periodic component has been centered.

Euler pentagonal theorem.

PROJECT / MIXED

Euler’s pentagonal theorem is

$$(q; q)_{\infty} = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2}, \quad |q| < 1.$$

It is a specialization of the Jacobi triple product and gives the sparse coefficient pattern of the Euler product. The two generalized pentagonal exponents arise from positive and negative n .

Euler's first q -exponential.

LEAN-PROVED

For $|q| < 1$,

$$e_q(z) = \sum_{n \geq 0} \frac{z^n}{(q; q)_n} = \frac{1}{(z; q)_\infty}, \quad |z| < 1$$

in the unscaled convention. A classical-limit convention often replaces z by $(1 - q)z$. The radius and scaling must be stated because several functions called e_q coexist in the literature.

Euler's second q -exponential.

LEAN-PROVED

Euler's companion is

$$E_q(z) = \sum_{n \geq 0} \frac{q^{n(n-1)/2} z^n}{(q; q)_n} = (-z; q)_\infty.$$

It is entire in z for $|q| < 1$, and $e_q(z)E_q(-z) = 1$ on the common domain. The two functions correspond to the two signs in the finite q -binomial theorem.

Eulerian number.

CLASSICAL

The Eulerian number $\langle n \rangle_k$ counts permutations of n with k descents under the chosen indexing convention. It forms the coefficients of the Eulerian polynomial and transforms powers into rational generating functions of the form $\sum_{m \geq 0} m^n z^m$.

Eulerian polynomial.

CLASSICAL

The Eulerian polynomial packages Eulerian numbers, commonly as $A_n(t) = \sum_k \langle n \rangle_k t^k$. A shift in the descent index changes the displayed power of t , so the project prints the convention. These polynomials occur in finite-difference, polylogarithmic, and rational generating-function formulas.

Exact arithmetic.

CLASSICAL

Exact arithmetic manipulates integers, rationals, algebraic data, or symbolic products without rounding error. It contrasts with floating-point evaluation, where cancellation and underflow can erase tiny endpoint values. The dyadic Fabius algorithms use integer signs, binomial coefficients, moment numerators, factorials, and known denominators to return exact rational results. These values serve both as mathematical theorems and as high-precision reference data for validating asymptotic or numerical procedures.

Exact dyadic derivative.

PROJECT / MIXED

At a reduced interior dyadic point $x_0 = a/2^n$ with odd a , the Fabius derivative of order n has the exact magnitude $2^{\binom{n+1}{2}}$ and a sign determined by the Thue–Morse sequence; all higher derivatives vanish. This follows from global derivative scaling because $2^n x_0 = a$ is an integer where the signed extension takes a sign value. The result proves that the terminating Taylor polynomial has degree exactly n .

Exact rational moment table.

LEAN-PROVED

An exact rational moment table lists moments or energy partial sums as reduced fractions obtained symbolically from proved recurrences. It is not a floating-point reconstruction. Such tables provide regression tests, denominator data, and certified input for polynomial, Appell, Legendre, and asymptotic calculations.

Exact topological support.

LEAN-PROVED

An exact support theorem proves equality rather than only containment. For the up-function the topological support is precisely $[-1, 1]$; for its density law the support is the same closed interval. Endpoint membership comes from every neighborhood containing positive interior values even though the function itself vanishes at the endpoints.

Expansive matrix.

CLASSICAL

A square matrix is expansive when every eigenvalue has modulus greater than one. Its inverse contracts, so powers A^{-n} generate a shrinking multivariate scale. Expansiveness is the matrix analogue of a scalar refinement dilation factor exceeding one.

Expectation.

CLASSICAL

The expectation $\mathbb{E}[X]$ of an integrable random variable is its probability-weighted mean, defined as a Lebesgue integral. More generally, $\mathbb{E}[g(X)]$ integrates g against the law of X . Moments, characteristic functions, and moment-generating functions are expectations of X^n , e^{itX} , and e^{zX} . In the Fabius probability representation, linearity of expectation and independence translate the infinite weighted sum into exact moment recurrences and transform products.

Explicit non-claim.

CONVENTION

An explicit non-claim states a plausible strengthening that a theorem or module does not establish. Examples include lack of a convergence theorem for a formal identity, lack of a target-specific minimality result, or absence of complex branch continuation. Non-claims are part of rigorous documentation: they bound the semantic reach of names that might otherwise invite overinterpretation.

Explicit uniform error.

CLASSICAL

An explicit uniform error bound has the form $\sup_{x \in D} |f_n(x) - f(x)| \leq E_n$ with a fully specified $E_n \rightarrow 0$. It is stronger than an existence statement and directly yields a stopping rule. The centered finite-spline approximation to the bounded Fabius function satisfies a dyadic error bound independent of x . This estimate is used both for certified numerical evaluation and for constructing a computable real function with a recursive precision modulus.

Exponential formula.

CLASSICAL

The combinatorial exponential formula states that labeled assemblies of connected components have an exponential generating function equal to the exponential of the connected-component series. Algebraically it is the Bell polynomial identity. Moments versus cumulants and arbitrary partitions versus connected blocks are its probabilistic manifestation.

Exponential generating function.

CLASSICAL

For a sequence (a_n) , the exponential generating function is $A(z) = \sum_{n \geq 0} a_n z^n / n!$. The factorial denominator makes multiplication encode binomial convolution: the coefficient of $z^n / n!$ in $A(z)B(z)$ is $\sum_k \binom{n}{k} a_k b_{n-k}$. Moment-generating functions are canonical examples. Fabius moment and half-moment recurrences become compact product or functional equations when expressed in this form.

Exponential Riordan array.

LEAN-PROVED

For $g(0) \neq 0$ and $f(z) = f_1 z + \dots$ with $f_1 \neq 0$, the exponential Riordan array has entries

$$R_{n,k} = \frac{n!}{k!} [z^n] g(z) f(z)^k.$$

It sends an exponential generating function A to $g(z)A(f(z))$. Composition and inversion of such arrays encode Sheffer basis changes and series reversion.

Exponentially tilted measure.

PROJECT / MIXED

For a random variable with finite MGF,

$$\frac{d\mathbb{P}_t}{d\mathbb{P}} = \exp(tX - \Lambda(t))$$

defines the exponentially tilted law. Under it, the mean is $\Lambda'(t)$ and variance $\Lambda''(t)$. Saddlepoint and large-deviation arguments choose the tilt that makes the rare value typical.

Extended nonnegative reals.

CLASSICAL

The extended nonnegative reals are $[0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$. They support monotone limits and nonnegative integrals without first proving finiteness. Addition and order extend in the expected way, while subtraction and multiplication involving $0 \cdot \infty$ require explicit conventions or avoidance. Lean measure-theoretic statements often live first in this type before finite moment or norm hypotheses convert them to ordinary real equalities.

Extension by zero.

CONVENTION

Extension by zero takes a function defined on a subset and declares it to vanish outside that subset. For a function on a closed interval to extend as a C^m function, its derivatives through order m must match zero at the boundary; C^∞ extension therefore requires flatness. Rvachev's up-function is naturally a compactly supported global function, while finite spline pieces are often first defined on their support and then extended by zero. Endpoint normalization determines whether the extension has the claimed regularity.

Exterior constant tails.

LEAN-PROVED

The bounded Fabius function is identically 0 on the nonpositive half-line and identically 1 on $[1, \infty)$. These constant tails make it a total CDF-style function on $\mathbb{R}$, simplify convolution and inverse statements, and distinguish it from the signed global extension. Derivatives of every positive order vanish on both exterior regions.

6 F

Faa di Bruno formula.

CLASSICAL

The Faa di Bruno formula is the higher-order chain rule for a composite $f \circ g$. One form is $\frac{d^n}{dx^n} f(g(x)) = \sum_{k=1}^n f^{(k)}(g(x)) B_{n,k}(g'(x), \dots, g^{(n-k+1)}(x))$, where $B_{n,k}$ are partial Bell polynomials. It organizes the derivatives of exponentials, logarithms, and inverse functions. In the all-orders saddle and inverse-Lambert algebra, the formula explains why coefficients are finite polynomials in lower-order phase jets, even when the Lean proof unfolds them through recurrences.

Faà di Bruno partition propagation.

FRONTIER

Faà di Bruno partition propagation tracks the first potentially nonzero derivative of a composition by summing over set partitions or Bell-polynomial indices. When lower derivatives vanish, most partitions disappear and a small surviving family determines the new jet. This mechanism drives the dyadic-spine proofs for compositional iterates and also appears in formal series reversion.

Fabius CDF–density bridge.

LEAN-PROVED

The bounded Fabius function is a cumulative distribution function, and its derivative is an affine rescaling of the up-density:

$$F'(x) = 2 \text{ up}(2x - 1).$$

The formula is pointwise for the canonical smooth representatives and both sides vanish outside $[0, 1]$. It transports support, positivity, smoothness, moments, and transform identities between the CDF and density pictures.

Fabius distribution.

PROJECT / MIXED

The Fabius distribution is the compactly supported probability law whose cumulative distribution function is the bounded Fabius function. It can be realized by an infinite sum of independent uniform random variables with geometrically decreasing weights, after the normalization used in the repository. Its law is symmetric about $1/2$, has a smooth density on $(0, 1)$, and is infinitely flat at the support endpoints. Centering and rescaling the density yields Rvachev's up-function.

Fabius fractional shift.

FRONTIER

A Fabius fractional shift is a proposed or derived continuation of the integer derivative/primitive ladder to a real or complex order using fractional integration, Mellin coordinates, or operator semigroups. It must specify terminal, branch, normalization, and function space. Different choices need not agree, so the term denotes a framework rather than one canonical globally formalized object.

Fabius function.

PROJECT / MIXED

The bounded Fabius function is the unique C^∞ map $F : \mathbb{R} \rightarrow [0, 1]$ with $F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$, reflection law $F(1 - x) = 1 - F(x)$, and differential equation $F'(x) = 2F(2x)$ on $[0, 1/2]$. It is a strictly increasing distribution function on $[0, 1]$, strictly convex then strictly concave, and nowhere analytic on its nonconstant interval. Its dyadic values are rational, and its endpoint decay contains a nonconstant log-periodic correction.

Fabius recurrence sequence.

PROJECT / MIXED

A Fabius recurrence sequence is one of the exact rational or integer sequences introduced to encode moments, inverse-power values, dyadic numerators, or normalized endpoint values. Such sequences are usually triangular: the n -th term is determined from earlier terms by binomial coefficients, factorials, and factors $2^j - 1$. The phrase is repository-specific rather than universally standardized, so the defining recurrence and normalization must accompany it. Different source papers use related symbols for moments, half moments, numerators, and Reshetnikov integers.

Fabius–Rvachev fold map.

LEAN-PROVED

The fold map sends a bounded CDF-style function to an even compactly supported profile by

$$u(x) = F(1 - |x|).$$

For the canonical objects this is Rvachev’s up-function. The absolute value folds the two sides of the unit interval onto the origin, transfers endpoint flatness to both support endpoints, and turns complement reflection of F into even symmetry and unimodality of u .

Fabius self-map dynamics.

FRONTIER

On $[0, 1]$, the bounded Fabius function is an increasing self-map with fixed endpoints and a symmetry-determined midpoint value. Iteration raises questions about fixed points, convergence rates, conjugacies, Carleman linearization, and the propagation of nonanalyticity. The frontier literature treats these dynamics while keeping them distinct from pointwise powers and from the signed global extension.

Factorial normalization.

CONVENTION

Factorial normalization records whether a coefficient family is encoded by $\sum a_n z^n$ or $\sum a_n z^n / n!$. The two conventions change convolution, Bell-polynomial, determinant, and reversion formulas. A notation such as f_n is not transported between ordinary and exponential series without the corresponding $n!$ factors.

Falling factorial.

CLASSICAL

The falling factorial is $x^{\underline{n}} = x(x-1) \cdots (x-n+1)$, with $x^{\underline{0}} = 1$. It is the natural basis for forward differences and counts injections when x is a nonnegative integer. Signed Stirling numbers of the first kind convert it to monomials.

Fast signed-dyadic name.

PROJECT / MIXED

A fast signed-dyadic name represents a real number at precision p by a pair $(c^+, c^-) \in \mathbb{N}^2$, interpreted as $(c^+ - c^-)/2^p$, with error at most 2^{-p} . The adjective fast refers to the guaranteed exponential error rate in the precision index, not necessarily to computational complexity. This redundant coding avoids negative integers inside primitive-recursive constructions and is extensionally equivalent to ordinary signed dyadic approximation. It is the representation used in the repository’s computable-analysis layer.

Filter convergence.

CLASSICAL

Filter convergence expresses limits without choosing a particular sequence. The notation $f \rightarrow L$ along a filter includes ordinary limits, one-sided limits, limits at infinity, and uniform convergence on sets. In Lean this is the common language behind `Tendsto`, `TendstoUniformly`, and `TendstoUniformlyOn`. The project uses it to keep endpoint, parameter, series, and approximation limits typed and to prevent an unstated limiting regime from entering a theorem.

Finite difference.

CLASSICAL

The forward finite-difference operator is $(\Delta_h f)(x) = f(x + h) - f(x)$; its n -fold iterate is an alternating binomial sum. It annihilates polynomials of degree below n and extracts the leading coefficient of a degree- n polynomial up to $n!h^n$. Gaussian q -differences replace equally spaced nodes by geometric nodes and ordinary binomial weights by q -binomial weights. The Fabius dyadic formulas combine such leading-coefficient extraction with Thue–Morse cancellation.

Finite-difference table.

CLASSICAL

A finite-difference table arranges a sequence and its successive forward differences in triangular form. Constant d -th differences characterize polynomials of degree at most d . In exact Fabius and repeated Thue–Morse computations, tables reveal diagonal polynomiality, integrality, and the first level where cancellation fails.

Finite field.

CLASSICAL

A finite field $\mathbb{F}_q$ exists exactly when q is a prime power and is unique up to isomorphism. Vector subspaces over $\mathbb{F}_q$ are counted by Gaussian binomial coefficients. The base q here is a field cardinality, not an arbitrary complex nome, although the resulting counting polynomial can later be evaluated elsewhere.

Finite-part integral.

CLASSICAL

A finite-part integral assigns a canonical finite value to an integral with a controlled endpoint divergence by subtracting the singular asymptotic terms before taking the limit. For example, if $f(t) = a_{-1}/t + O(1)$, the finite part of $\int_0^1 f(t) dt$ removes $a_{-1} \log(1/\varepsilon)$ from the cutoff integral. It differs from a Cauchy principal value, which relies on symmetric cancellation. Mellin and Bose-kernel calculations in the Fabius asymptotic use Hadamard finite parts to isolate constants and periodic terms.

Finite q -binomial theorem.

LEAN-PROVED

The finite theorem is

$$(z; q)_n = \sum_{k=0}^n (-1)^k q^{k(k-1)/2} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k.$$

Equivalently, $\prod_{j=0}^{n-1} (1 + zq^j)$ has coefficients $q^{k(k-1)/2} \begin{bmatrix} n \\ k \end{bmatrix}_q$. The sign convention depends on whether the product uses $1 - zq^j$ or $1 + zq^j$; the glossary records both through this single normalized identity.

Finite q -Pochhammer product.

LEAN-PROVED

For $n \in \mathbb{N}$,

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_0 = 1.$$

The empty-product convention is part of the definition. As a polynomial in a , it has degree n ; as a polynomial in q , its degree depends on the specialization of a . The formal development proves shift, splitting, reversal, and scalar-extension laws and uses this product as the basic atom of Gaussian coefficients and basic hypergeometric terms.

Finite shifted-up synthesis.

LEAN-PROVED

A finite shifted-up synthesis represents a polynomial on a prescribed compact interval by a finite linear combination of sufficiently wide translates and dilates of the up-function. On that interval every selected atom lies in one polynomial cell of its global differential structure, reducing the problem to a finite Vandermonde or finite-difference system. The equality is local to the declared interval, not global on $\mathbb{R}$.

Finite sinc product.

LEAN-PROVED

A finite sinc product

$$\Phi_N(\xi) = \prod_{j=0}^N \text{sinc}_\pi(\xi/2^j)$$

is the Fourier transform of a finite convolution of centered interval densities. Its inverse transform is a compactly supported spline with explicit knots and finite smoothness. As N grows, the products converge normally on compact sets to the entire up transform.

Finite spline approximation.

CLASSICAL

A finite spline approximation replaces the limiting Fabius law by the distribution of a finite partial random sum, whose density is a compactly supported piecewise polynomial. Every approximant can be evaluated using finite rational arithmetic on a dyadic grid. Centering the support removes a half-step bias, and coupling the omitted tail gives an explicit uniform error. These approximants are simultaneously probabilistic truncations, convolution splines, and constructive witnesses for computability.

First-defect monomial.

LEAN-PROVED

For a mesh M , the first-defect monomial has degree $v_2(M) + 1$. A formal theorem supplies a phase at which its shifted Rvachev reconstruction fails. This proves sharpness of the universal degree bound rather than merely failure of a proof technique.

First nonzero shell.

FRONTIER

In a sum organized by weight or defect, the first nonzero shell is the lowest layer that survives all parity, moment, or partition cancellations. Identifying it gives the leading asymptotic or the highest nonzero derivative. Subsequent shells supply systematic corrections.

First saddle correction.

PROJECT / MIXED

The first saddle correction is the coefficient of $1/\lambda$ after the leading Gaussian saddle approximation. It combines the quartic phase term, the square of the cubic phase term, derivatives of the amplitude, and any first-order periodic dependence of the transform. In logarithmic form it becomes $A_1(\lambda(x))/\lambda(x)$. The repository derives it explicitly and proves an error one order smaller.

Fisher information.

FRONTIER

For a positive differentiable density f ,

$$J(f) = \int \frac{(f'(x))^2}{f(x)} dx$$

when finite. It measures translation sensitivity and enters de Bruijn-type entropy expansions. Endpoint flatness can make the quotient delicate even when both numerator and density vanish; finiteness requires proof.

Fixed point.

CLASSICAL

A fixed point of a map T is an element x satisfying $T(x) = x$. Functional equations often become fixed-point equations on spaces of functions. A unique fixed point can be obtained from contraction, order, compactness, or monotone-iteration arguments. The Fabius distribution is constructed as the fixed point of an averaging/refinement operator; the operator identity is equivalent to the probabilistic self-similarity of the underlying random series and ultimately to the differential equation.

Fixed-scale up synthesis.

LEAN-PROVED

In fixed-scale synthesis all atoms share one dilation and differ only by translation. A finite consecutive set of centers spans polynomials up to a degree determined by the scale and active cells. Keeping one scale simplifies implementation and exposes Toeplitz or finite-difference structure in the coefficient matrix.

Flat function.

CLASSICAL

A smooth function is flat at a point x_0 if $f^{(n)}(x_0) = 0$ for every $n \geq 0$, or, in some conventions, for every positive n after the value is separated. The model e^{-1/x^2} extended by zero is flat but not analytic. The Fabius function is flat at its transition endpoints relative to its constant tails, and the up-function is flat at ± 1 . Effective flatness estimates quantify the otherwise qualitative vanishing of every derivative.

Flat jet.

CLASSICAL

A function has a flat jet at x_0 when every derivative at x_0 vanishes. Its Taylor series is then identically zero, although the function need not vanish in any neighborhood. The Fabius function has flat endpoint jets, and the up-function is flat at both support endpoints; these are canonical examples separating C^∞ smoothness from analyticity.

Focused import.

CONVENTION

A focused import exposes a small thematic slice of the library rather than the complete root facade. It is useful for faster builds, dependency audits, and frontier-facing algebraic experiments. A focused module name describes what has been formalized there, but does not license conclusions absent from its public declarations—for example analytic convergence of a merely formal germ or approximation bounds for a finite identity.

Formal asymptotic series.

CLASSICAL

A formal asymptotic series is an ordered coefficient expression manipulated algebraically without asserting that the infinite sum converges. Its analytic meaning is supplied by a remainder theorem: every finite truncation approximates the target to a declared order in a stated limit. The repository keeps formal coefficient identities separate from realized Fabius, Lambert, gamma, and q -function asymptotics.

Formal germ.

PROJECT / MIXED

A formal germ is a power series considered algebraically near a chosen base point, without an assertion that it converges to an analytic function. It can satisfy composition, inversion, or algebraic equations coefficient by coefficient. The quarter-Catalan and dyadic-rescaling constructions in the current tree are formal germs; their uniqueness in a power-series ring is distinct from existence of a global analytic branch.

Formal logarithm.

CLASSICAL

For a formal power series $A(z) = 1 + U(z)$ with zero constant term in U , its formal logarithm is $\log A = \sum_{r \geq 1} (-1)^{r-1} U^r / r$. Every coefficient depends on only finitely many coefficients of U , so no analytic convergence is required. In the saddle expansion, the normalized mass series is converted into logarithmic correction coefficients by this operation. Ordered compositions give an explicit coefficient formula for the formal logarithm.

Formal normal form.

PROJECT / MIXED

A formal normal form is a canonical arrangement of terms in a polynomial, power series, transseries, or recurrence after algebraic equivalence has been resolved. Typical choices order monomials, collect equal exponents, and separate leading from residual blocks. Normal forms make coefficient extraction and series reversal deterministic; they do not assert analytic convergence unless a separate theorem supplies it.

Formal power series.

CLASSICAL

A formal power series $\sum_{n \geq 0} a_n z^n$ is an algebraic object whose coefficients are manipulated without assigning a numerical value to z or requiring convergence. Addition is coefficientwise and multiplication is the Cauchy product. Formal differentiation, composition under suitable constant-term conditions, exponentiation, and logarithms are defined coefficientwise. The repository uses formal series to certify moment recurrences, saddle jets, and all-orders coefficient identities before separate analytic estimates justify any asymptotic interpretation.

Formal versus analytic identity.

CONVENTION

A formal identity is proved by coefficient algebra in a series ring; an analytic identity additionally requires convergence and equality of the resulting functions on a domain. The same printed series may support both interpretations under different hypotheses. The repository repeatedly marks this boundary in Catalan–Gaussian filters, complex-order products, inverse series, and transseries so that convergence is never inferred from formal manipulation alone.

Formal versus realized transseries.

CONVENTION

A formal transseries identity is an algebraic equality of ordered supports and coefficients. A realized asymptotic identity additionally names a function, domain, branch, sector, and remainder bound. The repository marks this boundary explicitly: formal inversion may guide a conjecture, but it is not an analytic theorem until realization and error transfer are proved.

Formalization crosswalk.

CONVENTION

A formalization crosswalk maps a mathematical notion or displayed formula to its Lean name, module, hypotheses, and exact conclusion. It prevents prose terminology from being mistaken for a theorem of greater generality. This glossary supplies a selected crosswalk for major objects; the root README and module docstrings remain authoritative when exact declaration names or source checkpoints matter.

Fourier coefficient.

CLASSICAL

For a one-periodic integrable function P , its k -th complex Fourier coefficient is $\widehat{P}(k) = \int_0^1 P(u) e^{-2\pi i k u} du$. The coefficient $k = 0$ is the mean; nonzero coefficients measure oscillation. In the corrected Fabius endpoint theory, the centered periodic fluctuation has coefficients proportional to $-\Gamma(-\chi_k) \zeta(1 - \chi_k) / \log 2$, where $\chi_k = 2\pi i k / \log 2$. Their nonvanishing proves that the fluctuation is genuinely nonconstant.

Fourier energy.

LEAN-PROVED

Fourier energy is an L^2 norm such as $\int_{\mathbb{R}} |\widehat{f}(\xi)|^2 d\xi$. Plancherel identifies it with the spatial L^2 energy of f . For up, the positive-half-line sinc product integral, the square integral of up, and a rational Legendre coefficient series are formally connected.

Fourier inversion.

CLASSICAL

Fourier inversion reconstructs a function from its Fourier transform under suitable integrability or regularity assumptions. With the convention $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \xi} dx$, one has $f(x) = \int \widehat{f}(\xi) e^{2\pi i x \xi} d\xi$ when both sides are appropriately integrable, with standard variants at discontinuities. The up-function's rapid transform decay makes inversion straightforward and supports cosine expansions, derivative formulas, and sampling identities.

Fourier-Legendre coefficient.

CLASSICAL

The Fourier-Legendre coefficient of $f \in L^2[-1, 1]$ is the scalar projection onto a Legendre polynomial: $a_n = (2n + 1) \int_{-1}^1 f(x) P_n(x) dx / 2$ in the standard normalization. These coefficients minimize mean-square error among polynomial approximations when truncated. The repository derives exact coefficients for the up-function, exploiting moments and symmetry, and proves more than L^2 convergence: the resulting series converges absolutely and uniformly.

Fourier-Legendre series.

CLASSICAL

A Fourier-Legendre series expands a function on $[-1, 1]$ in the orthogonal basis (P_n) of Legendre polynomials. It is analogous to a trigonometric Fourier series but adapted to polynomial approximation with unit weight. For even functions only even degrees occur. The up-function's series has exact arithmetic coefficients, least-squares optimality of partial sums, and absolute uniform convergence established through derivative and coefficient estimates. It should not be confused with the ordinary Fourier transform of the same function.

Fourier mode.

CLASSICAL

A Fourier mode is a basic exponential $u \mapsto e^{2\pi i k u}$ or its sine-cosine components at integer frequency k . Periodic functions decompose into such modes. Under a dyadic dilation equation, complex exponents with imaginary parts $2\pi k / \log 2$ recur because multiplication by two advances the logarithmic phase by one period. The nonzero modes of the Fabius periodic correction are expressed by Gamma-zeta coefficients and cannot be replaced by a constant.

Fourier series.

CLASSICAL

A Fourier series represents a periodic function as $\sum_{k \in \mathbb{Z}} c_k e^{2\pi i k x}$, or equivalently by sine and cosine terms. Convergence may be in L^2 , pointwise, uniform, or absolute, depending on regularity. Smooth periodic functions have rapidly decaying coefficients. The repository uses Fourier series both for the endpoint's one-periodic correction and for periodicizations of the up-function; the two applications have different variables and should not be conflated.

Fourier shell.

FRONTIER

A Fourier shell groups frequencies or product factors with the same dyadic valuation, radial size, or newly introduced zero set. Shell decomposition separates coarse zeros from fine tail corrections and is useful for factorized approximation, spectral counting, and Thue-Morse sign analysis.

Fourier transform.

CLASSICAL

The Fourier transform converts translation into multiplication by a phase, convolution into multiplication, and differentiation into multiplication by frequency. Under the repository's convention, $\hat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i x z} dx$. For the up-function it equals the entire product $\prod_{n \geq 0} \text{sinc}(\pi z / 2^n)$. Zeros of the first factors yield partition and sampling identities, while compact support and smoothness determine complex growth and real-axis decay.

Fourier-transform convention.

CONVENTION

A Fourier-transform convention specifies the signs and factors of 2π in the transform and inverse transform. The reference article uses $\widehat{f}(z) = \int f(x)e^{-2\pi i x z} dx$, for which the centered interval density has a $\text{sinc}(\pi z)$ transform. Sampling frequencies, derivative multipliers, and Poisson-summation constants all depend on this choice. Stating the convention prevents normalization errors.

Fourier transform of a distribution.

CLASSICAL

For a tempered distribution T , the Fourier transform is defined by duality against Schwartz test functions. Dirac masses, combs, polynomial factors, and derivatives then transform by exact algebraic rules even when no ordinary integral exists. Distributional formulas must be distinguished from pointwise identities between functions.

Fractional CDF layer-cake formula.

LEAN-PROVED

Fractional layer-cake identities weight level sets by powers such as $t^{\alpha-1}$ or $(x-a)^{\alpha-1}$. They connect fractional moments, Riemann–Liouville integrals, and CDF tails through Beta and Gamma constants. In the Fabius corpus they provide a systematic extension of integer moment relations to real orders under positivity and integrability assumptions.

Fractional integration semigroup law.

PROJECT / MIXED

Under appropriate integrability hypotheses, $I^\alpha I^\beta = I^{\alpha+\beta}$ for positive orders. The proof is a Beta-integral computation justified by Tonelli or Fubini. This law makes the fractional order a continuous analogue of an iteration count and underlies operator interpolation between integer primitive levels.

Fractional iterate.

FRONTIER

A fractional iterate $f^{\circ t}$ extends integer composition so that $f^{\circ(s+t)} = f^{\circ s} \circ f^{\circ t}$ and $f^{\circ 1} = f$. Constructing one requires a chosen conjugacy or semigroup and is generally nonunique. The notation is never interpreted as a pointwise power $f(x)^t$.

Fractional order lowering.

PROJECT / MIXED

Order lowering applies a suitable derivative or resolvent identity to reduce a generalized integral order. It is valid only after endpoint terms and regularity are controlled. The repository's real Stieltjes-resolvent hierarchies and fractional Fabius formulas state the admissible range explicitly rather than treating order as a purely formal index.

Fractional order raising.

PROJECT / MIXED

Order raising composes a fractional integral of order α with an ordinary or fractional integral to obtain order $\alpha + \beta$. At the kernel level it is the Beta convolution of positive-part powers. Fabius primitive ladders use this rule to interpolate between discrete integration levels and to transfer support and scaling relations.

Fractional part.

CONVENTION

The fractional part of a real number is $\{x\} = x - \lfloor x \rfloor \in [0, 1)$. Any one-periodic function $P(x)$ can be viewed as a function of $\{x\}$. In dyadic asymptotics, the relevant phase is often the fractional part of a logarithmic or Lambert coordinate; changing x by a dyadic scale shifts that coordinate by approximately an integer. The periodic correction records the residual dependence on this phase.

Frame bound.

FRONTIER

A family (ϕ_k) in a Hilbert space has frame bounds $A, B > 0$ when

$$A\|f\|^2 \leq \sum_k |\langle f, \phi_k \rangle|^2 \leq B\|f\|^2.$$

Frames permit redundancy. Shifted compactly supported kernels may reproduce polynomials while failing a global frame inequality, so approximation order and frame stability are distinct properties.

Free cumulant.

FRONTIER

Free cumulants r_n are related to moments by summing over noncrossing partitions:

$$m_n = \sum_{\pi \in NC(n)} \prod_{B \in \pi} r_{|B|}.$$

They linearize free additive convolution. They are not classical cumulants, whose partition formula uses all set partitions.

Free infinite divisibility.

FRONTIER

A probability law is freely infinitely divisible when, for every n , it is the free additive convolution of n identical laws. Conditional positivity of the free-cumulant sequence gives algebraic tests. Failure of a finite Hankel necessary condition proves nondivisibility; passing all tested blocks is not a proof of divisibility.

Fubini theorem.

CLASSICAL

Fubini's theorem permits the order of integration in a product space to be exchanged when the integrand is absolutely integrable. Tonelli's theorem covers nonnegative measurable integrands without prior integrability. These results justify factorization of expectations for independent variables, convolution formulas, and rearrangement of transform integrals. Signed Thue–Morse expressions require absolute bounds before Fubini can be invoked.

Full logarithmic expansion.

CLASSICAL

The full logarithmic expansion is the all-orders expansion of $\log F(x)$, rather than of $F(x)$ itself. In the repository it has the form $\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} A_j(\lambda(x))/\lambda(x)^j + \mathcal{O}(\lambda(x)^{-N})$, where the A_j are smooth one-periodic functions. Taking the logarithm separates multiplicative saddle corrections into additive coefficients and makes the uniqueness and size of the periodic term transparent.

Full moment sequence c_n .

CLASSICAL

The sequence c_n in the repository denotes a normalized family of full centered moments of the up-function, with exact normalization fixed by the displayed integral and generating function in the source. Symmetry means only even powers contribute to the principal sequence. The triangular recurrence for c_n feeds the dyadic closed formula, Fourier-Legendre coefficients, and moment numerator arithmetic. Because symbols vary between papers, c_n should always be read with the repository's normalization.

Functional-differential equation.

CLASSICAL

A functional-differential equation combines derivatives with values at transformed arguments, for example $f'(x) = 2f(2x)$ or an equation involving translates $f(ax + b)$. Such equations are common in refinement theory, delay equations, and self-similar probability laws. They generally require boundary, support, or normalization data for uniqueness. The Fabius, up, and signed global forms satisfy related but not identical global equations, so the domain and chosen normalization are part of the statement.

Functional equation.

CLASSICAL

A functional equation constrains a function through values at different arguments, such as reflection, translation, dilation, or convolution identities. The equations $F(1-x) = 1-F(x)$, $\mathcal{F}'(x) = 2\mathcal{F}(2x)$, and the up-function refinement law are central examples. Functional equations often generate recurrences on special grids and transform products after Fourier or Laplace transformation. Local validity does not automatically imply a global identity.

Fundamental theorem of calculus.

CLASSICAL

The fundamental theorem of calculus links differentiation and integration: a continuous function g has an antiderivative $G(x) = \int_a^x g(t) dt$, and a continuously differentiable F satisfies $F(b) - F(a) = \int_a^b F'(t) dt$. Iterating this identity with vanishing endpoint derivatives gives factorially improved flatness bounds. It also converts the up-density into the Fabius cumulative distribution and supports moment integration by parts.

7 G**Gamma-derivative tower.**

LEAN-PROVED

A gamma-derivative tower is a recursively organized family of derivatives of a product or quotient built from gamma and reciprocal-gamma factors. Bell polynomials in polygamma values express the n -th derivative. The current Thue–Morse gamma modules track zeros, poles, jets, and exact ratios rather than differentiate a meromorphic expression informally.

Gamma function.

CLASSICAL

The Gamma function extends the factorial by $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ for $\Re s > 0$ and meromorphic continuation elsewhere. It satisfies $\Gamma(s+1) = s\Gamma(s)$ and has simple poles at the nonpositive integers. Gamma factors arise in Mellin transforms and Gaussian saddle integrals. In the Fabius periodic correction, $\Gamma(-\chi_k)$ combines with a zeta value at a purely imaginary dyadic frequency.

Gamma ratio.

PROJECT / MIXED

A gamma ratio has the form $\Gamma(z+a)/\Gamma(z+b)$. For large z in a sector it admits a power expansion beginning with z^{a-b} , with coefficients expressed by Bernoulli polynomials. Ratios often cancel the dominant Stirling exponential and are therefore more stable than separate gamma evaluations.

Gamma-tower ratio.

LEAN-PROVED

A gamma-tower ratio compares consecutive levels of a recursively defined Thue–Morse gamma product. Binary block factorization reduces it to a smaller set of shifted gamma factors and makes induction possible. The ratio is stated as a meromorphic identity together with explicit nonvanishing conditions or a cross-multiplied entire form.

Gamma-zeta coefficient.

CLASSICAL

A Gamma-zeta coefficient is a Fourier coefficient expressed as a product of values of the Gamma and Riemann zeta functions. For the centered Fabius periodic correction, the nonzero coefficient at frequency k has the form $-\Gamma(-\chi_k)\zeta(1-\chi_k)/\log 2$, with $\chi_k = 2\pi i k / \log 2$. Mellin inversion explains this form: poles repeated along a vertical arithmetic progression become Fourier modes in a logarithmic variable. Exponential Gamma decay makes the periodic fluctuation extremely smooth and numerically small.

Gärtner–Ellis theorem.

FRONTIER

The Gärtner–Ellis theorem derives an LDP from convergence of scaled log-moment-generating functions under regularity such as essential smoothness or, in compact settings, appropriate exposed-point control. Merely computing a pointwise log-MGF limit does not automatically supply the full lower bound.

Gauss–Lobatto quadrature.

FRONTIER

A Gauss–Lobatto rule includes both endpoints of an interval and chooses the remaining nodes for maximal polynomial precision. With n total nodes its classical precision is $2n-3$. For Fabius/Rvachev weighted rules, endpoint flatness changes numerical behavior but not the algebraic moment conditions.

Gauss–Radau quadrature.

FRONTIER

A Gauss–Radau rule prescribes one endpoint and chooses the remaining nodes to achieve degree $2n-2$ exactness with n nodes. It can be obtained from a modified orthogonality measure or a rank-one Jacobi-matrix perturbation. This differs from a dyadic-comb cubature, whose nodes are fixed arithmetically rather than optimized.

Gaussian-binomial bounds.

LEAN-PROVED

For real $0 \leq q < 1$, positivity gives

$$1 \leq \left[\begin{matrix} n \\ k \end{matrix} \right]_q \leq \frac{1}{(q; q)_\infty},$$

with sharper finite bounds obtained from the product representation. For $q > 1$, palindromy transfers estimates to q^{-1} and contributes the factor $q^{k(n-k)}$. Such bounds provide domination for fixed-column

limits, Toeplitz rows, and q -interpolation kernels.

Gaussian binomial coefficient.

CLASSICAL

The Gaussian, or q -binomial, coefficient is defined by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}, \quad 0 \leq k \leq n.$$

It is a polynomial in q with nonnegative integer coefficients and specializes to $\binom{n}{k}$ as $q \rightarrow 1$. At $q = 1/2$, normalized Gaussian rows act as finite-difference weights on geometric nodes. The Fabius formulas use them to extract leading coefficients, prove translation invariance, and construct complex discrete limits.

Gaussian binomial palindromy.

LEAN-PROVED

The Gaussian polynomial satisfies

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}.$$

Thus its coefficient sequence is palindromic. This is polynomial reciprocity, not merely equality of two rational expressions for $q \neq 0$; after multiplication by the displayed power it extends through $q = 0$. The identity also controls large- q asymptotics and reciprocal-base inverse branches.

Gaussian binomial polynomial.

LEAN-PROVED

For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}$$

is a polynomial in q with nonnegative integer coefficients, despite the displayed quotient. Its degree is $k(n-k)$, its constant and leading coefficients are one, and it is symmetric under $k \leftrightarrow n-k$. The formal development proves polynomial structure before using rational quotient presentations at exceptional bases.

Gaussian coefficients at $q = -1$.

LEAN-PROVED

Evaluation at $q = -1$ is governed by parity: zeros caused by even cyclotomic factors may cancel between numerator and denominator. The surviving value reduces to an ordinary binomial coefficient with halved indices in the compatible parity cases and vanishes in the remaining odd case. Current modules also track the first derivative at -1 , which requires the polynomial realization rather than substitution into the quotient formula.

Gaussian contraction.

CLASSICAL

A Gaussian contraction is the pairing operation used to integrate products against a centered Gaussian law. Wick's rule expresses an even Gaussian moment as a sum over pairings and makes every odd moment vanish. In saddle-point expansions the same combinatorics contracts powers of the local integration variable after the quadratic phase has been normalized; it organizes higher correction coefficients and connects them with Bell polynomials and cumulants.

Gaussian finite difference.

PROJECT / MIXED

A Gaussian finite difference is a q -analogue of an ordinary finite difference, with geometrically spaced nodes and Gaussian binomial weights. Its finite q -binomial theorem implies annihilation of node polynomials below a prescribed degree and an exact response at the leading degree. In the Fabius dyadic evaluation, the base $q = 1/2$ matches the powers of two in the dilation equation, while a second Thue–Morse cancellation acts inside binary blocks.

Gaussian integral.

CLASSICAL

The Gaussian integral is $\int_{-\infty}^{\infty} e^{-t^2/2} dt = \sqrt{2\pi}$. Its even moments are $\int t^{2m} e^{-t^2/2} dt = \sqrt{2\pi}(2m - 1)!!$, while odd moments vanish. These identities produce the universal coefficients in saddle mass expansions. The reference Gaussian weight is obtained by scaling according to the second derivative of the Fabius phase at the saddle.

Gaussian Pascal recurrence.

LEAN-PROVED

Two equivalent q -Pascal recurrences are

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q.$$

Boundary values are 1 at $k = 0, n$ and 0 outside the natural range. The two forms correspond to adjoining a terminal or initial geometric node and are used in induction, summation, and interpolation proofs.

Gaussian polynomial.

CLASSICAL

Gaussian polynomial is another name for a Gaussian q -binomial coefficient, emphasizing that $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is genuinely a polynomial in q . More generally, the term may refer to finite products and combinations built from these coefficients. Repository modules on Gaussian polynomial contraction establish identities and bounds for rows at $q = 1/2$, including normalization and total-variation estimates required by the Toeplitz limit.

Gaussian quadrature.

PROJECT / MIXED

An n -node Gaussian quadrature rule for a positive measure integrates all polynomials of degree at most $2n - 1$ exactly. Its nodes are the zeros of the degree- n orthogonal polynomial and its weights are positive. Exact Rvachev moments determine such rules algebraically; numerical node certification is a separate analytic or interval-arithmetic task.

Gaussian rational.

CLASSICAL

A Gaussian rational is a complex number $a + bi$ with $a, b \in \mathbb{Q}$, that is, an element of $\mathbb{Q}(i)$. It is the fraction field of the Gaussian integers after allowing rational denominators. Exact complex q -binomial approximants can be evaluated in this field when the shift parameter is Gaussian rational. Scalar-extension lemmas then interpret rational polynomial identities over $\mathbb{C}$ without redoing the arithmetic proof.

Generalized quantile function.

CLASSICAL

For a CDF F , the left-continuous generalized inverse is $Q(u) = \inf\{x : F(x) \geq u\}$. It exists even when F has jumps or flat parts. For the continuous strictly increasing Fabius CDF on $[0, 1]$, this quantile agrees with the ordinary inverse. The generalized formulation is useful when transferring approximation bounds from CDFs to inverse distributions.

Generating function.

CLASSICAL

A generating function encodes a sequence as coefficients of a formal or analytic series. Ordinary generating functions use $\sum a_n z^n$; exponential generating functions use $\sum a_n z^n / n!$. Products encode convolutions, derivatives encode weighted index shifts, and functional equations encode recurrences. Fabius moment sequences, Thue–Morse signs, Bernoulli coefficients, and saddle corrections all have useful generating functions, but their convergence domains and normalizations differ.

Genocchi number.

CLASSICAL

Genocchi numbers are defined by a generating function such as

$$\frac{2t}{e^t + 1} = \sum_{n \geq 0} G_n \frac{t^n}{n!}.$$

They are closely related to Bernoulli numbers and vanish in one parity beyond the initial terms. Sign and index conventions vary and are therefore stated explicitly in the corpus.

Genocchi polynomial.

CLASSICAL

Genocchi polynomials insert e^{xt} into the Genocchi generating function. They form an Appell sequence and satisfy translation and reflection identities analogous to Bernoulli and Euler polynomials. They occur in the broader combinatorial-coefficient and inversion materials.

Geometric Bernoulli polynomial.

FRONTIER

For a normalized geometric-uniform law with moment-generating function $M_q(t)$, the geometric Bernoulli polynomials are defined by

$$\frac{e^{xt}}{M_q(t)} = \sum_{n \geq 0} B_{n,q}^{\text{geom}}(x) \frac{t^n}{n!}.$$

They are monic Appell polynomials that invert averaging by the law. At limiting parameters they connect to classical Bernoulli-type families after the stated normalization.

Geometric interpolation nodes.

CLASSICAL

Geometric interpolation nodes form a progression such as $-1, -2, \dots, -2^n$, with constant ratio rather than constant spacing. Lagrange denominators on these nodes factor into q -Pochhammer products, so the leading-coefficient functional is expressed by Gaussian binomial weights with $q = 1/2$. This is the interpolation mechanism behind the finite Fabius q -binomial formulas. A common translation of all nodes changes lower-degree terms but leaves the extracted leading coefficient unchanged.

Geometric Lagrange interpolation.

PROJECT / MIXED

For distinct geometric nodes $x_j = q^j$, the Lagrange basis is

$$L_j(x) = \prod_{m \neq j} \frac{x - q^m}{q^j - q^m}.$$

Factoring the denominator yields powers of q , signs, and finite q -Pochhammer products. At $q = 1/2$, these weights explain the Gaussian coefficients and Mersenne factors in exact Fabius evaluation.

Geometric Newton interpolation.

LEAN-PROVED

Newton interpolation on nodes $1, q, q^2, \dots$ uses divided differences adapted to a geometric comb. The node polynomial is a rescaled q -Pochhammer product, and the interpolation coefficients are expressible through Gaussian factors. It is the q -analogue of equally spaced Newton differences and supplies exact formulas for dyadic and reciprocal-power tables.

Geometric uniform CDF.

LEAN-PROVED

The geometric uniform CDF is $F_q(x) = \mathbb{P}(X_q \leq x)$ for the normalized geometric uniform law. The current formalization proves monotonicity, measurability, continuity, reflection, endpoint values, integral representations, and smoothness in the positive-contraction regime. Exterior formulas use the support normalization and should not be extrapolated to signed or complex q .

Geometric uniform density.

LEAN-PROVED

The geometric uniform density is the continuous compactly supported derivative of the geometric uniform CDF when $0 < q < 1$. It is nonnegative, reflective about the support midpoint, integrates to one, and realizes the law as a measure with density. At $q = 1/2$, an affine rescaling yields the Rvachev up-density.

Geometric uniform law.

LEAN-PROVED

The geometric uniform law is the distribution of a normalized geometric uniform random series. Its support, reflection symmetry, transform product, moments, cumulants, CDF, and smooth density are developed uniformly in the ratio q under explicit hypotheses such as $0 < q < 1$. The Fabius law is a distinguished $q = 1/2$ member after affine normalization.

Geometric uniform random series.

LEAN-PROVED

For $|q| < 1$, a geometric uniform random series is a sum

$$X_q = \sum_{j \geq 0} a q^j U_j$$

of independent uniform variables with a normalization a chosen to fix the support or total weight. Almost-sure convergence follows from bounded geometric tails. At the dyadic specialization the law is affinely related to the Fabius distribution and the centered density to Rvachev's up-function.

Germ.

CLASSICAL

A germ at a point is an equivalence class of functions that agree on some neighborhood of that point. Germs let local properties be discussed without fixing a particular neighborhood. A smooth germ records all nearby values, whereas a formal Taylor series records only derivatives at the point; the latter does not determine a smooth germ. The terminating Taylor series of the Fabius function at a dyadic point is therefore compatible with a nonpolynomial, nowhere-analytic germ.

Global binary series.

CLASSICAL

The global binary reduction formula evaluates the signed global Fabius extension by peeling an argument through its binary scales and adding finite Taylor blocks determined by exact dyadic data. For a fixed dyadic input the process terminates; for more general encodings the accompanying theorem specifies the limiting or summability mode. Its principal value is structural: it converts the global functional equation into digitwise arithmetic while preserving signs, endpoint conventions, and the distinction between the bounded and global functions.

Global differential equation.

CLASSICAL

The global differential equation is $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ for every real x , satisfied by the signed global extension $\mathcal{F}$. The bounded CDF does not satisfy this equation on all of $\mathbb{R}$ because of its constant tails, and the compactly supported up-function satisfies a folded variant. Keeping this distinction explicit prevents invalid differentiation or scaling outside the natural interval. Repeated differentiation of the global equation yields exact derivative scaling.

Global Fabius object.

LEAN-PROVED

The repository's `globalFabius` or equivalent exported object denotes the signed extension of the bounded function to all real arguments. It agrees with F on the unit interval but follows the unclamped global differential law elsewhere. Names containing “global” are therefore semantically significant: substituting the bounded function outside $[0, 1]$ can change both values and signs.

Global first-jet reflection law.

LEAN-PROVED

A first-jet reflection law couples both function values and first derivatives at reflected points. For the bounded Fabius normalization, complement reflection gives $F(x) + F(1 - x) = 1$, and differentiation gives the corresponding equality of derivatives after accounting for the chain-rule sign. The current development packages global first-jet consequences so that shape and inverse arguments need not rederive them locally.

Global signed extension.

PROJECT / MIXED

The global signed extension, written $\mathcal{F}$ here and implemented by the repository's global Fabius object, is the real-valued extension that agrees with the bounded Fabius function on $[0, 1]$ and obeys $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$. Outside the unit interval it oscillates rather than remaining clamped; for example the documented global evaluator gives $\mathcal{F}(3) = -1$. It is constructed through exact translation and reduction identities, so it must not be confused with the bounded CDF-style function or with the even compactly supported up-function.

Global uniform spline approximation.

LEAN-PROVED

The global uniform spline approximation extends the finite spline model to all real arguments in the normalization appropriate to the bounded or signed global function. The formal error theorem gives a uniform dyadic bound and a filter-convergence theorem to the global Fabius object. This analytic approximation is stronger than the coarse estimate required merely to certify computability.

Gram determinant.

CLASSICAL

The Gram determinant is $\det G$. It equals the squared volume of the parallelotope spanned by the vectors and vanishes exactly under linear dependence. Ratios of consecutive Hankel–Gram determinants give squared norms and recurrence coefficients of monic orthogonal polynomials.

Gram matrix.

LEAN-PROVED

For vectors or functions $v_0, \dots, v_{N-1}$ in an inner-product space, the Gram matrix is $G_{ij} = \langle v_i, v_j \rangle$. It is Hermitian positive semidefinite and is positive definite exactly when the family is linearly independent. For monomials under a moment inner product, the Gram matrix is the Hankel moment matrix.

Gregory coefficient.

FRONTIER

Gregory coefficients are the coefficients in $z/\log(1+z)$ or a sign-shifted equivalent convention. They appear in numerical integration, reciprocal logarithms, and series reversion. Because the literature alternates signs and starting indices, the defining generating function is part of the notation.

Grid-based transseries.

FRONTIER

A grid-based transseries has support contained in a finitely generated multiplicative grid of monomials. It can often be computed like a multivariate formal power series while retaining an asymptotic order. Not every well-based transseries is grid based, but explicit Lambert and polynomial–logarithmic examples usually are.

Grid histogram.

PROJECT / MIXED

A grid histogram is a piecewise-constant or discrete array representation of mass on a regular mesh. Its values may be cumulative counts, normalized bin heights, or samples of a spline. In the Thue–Morse part of the development, a corrected finite grid is identified exactly with a normalized histogram produced by iterated discrete smoothing. This identity bridges combinatorial sign sums and weak or uniform convergence to the continuous Fabius-related density.

Grzegorzczak computability.

CLASSICAL

Grzegorzczak computability for real functions is an early constructive framework based on computable real sequences and effective uniform continuity. A function must transform fast rational names effectively and possess a recursive continuity modulus. This avoids treating arbitrary real inputs as finite data. The repository proves the bounded Fabius function computable in this sense by combining primitive-recursive evaluation of centered finite splines with a uniform approximation theorem and the modulus $d(n) = 2n$.

8 H

Hadamard finite part.

CLASSICAL

The Hadamard finite part regularizes an integral or divergent expression by subtracting all explicitly known singular terms in its cutoff expansion and retaining the constant term. It handles power and logarithmic divergences and is local to the singular endpoint. Unlike analytic continuation, it is defined directly from asymptotic subtraction, although the two methods often agree when both apply. The Fabius Mellin/Bose analysis uses finite parts at the Mellin origin to isolate the constant and periodic components of a dyadic harmonic sum.

Hahn series.

PROJECT / MIXED

For an ordered abelian exponent group Γ , a Hahn series is

$$\sum_{\gamma \in S} a_{\gamma} t^{\gamma}$$

with well-ordered support $S \subseteq \Gamma$ in the chosen direction. Laurent and Puiseux series are special cases. Hahn power-log series let the coefficients a_{γ} be logarithmic polynomials and support real exponent scales used in generalized reversion.

Half moment.

PROJECT / MIXED

A half moment is a moment integrated over one half of a symmetric support, with normalization chosen in the source. For the up-function or bounded Fabius law, the sequence d_n packages integrals over a half interval and is linked by symmetry to the full even moments c_k . A key bridge identifies normalized half moments with endpoint values $F(2^{-n})$. This converts analytic integrals into exact rational arithmetic and makes parity information about numerator sequences yield dyadic valuation theorems.

Half-moment numerator.

PROJECT / MIXED

A half-moment numerator is an integer sequence obtained by clearing the canonical factorial and power-of-two denominators from the rational half moments. The repository uses a dedicated symbol, often G_n or a related normalization depending on the source layer. Its recurrence is triangular and division-free after the correct scaling. Parity and positivity properties of these numerators are the arithmetic input for exact endpoint valuations and Reshetnikov integer formulas.

Half-moment numerator G_n .

PROJECT / MIXED

The symbol G_n is used in the arithmetic layer for an integer normalization of the half moments, with the exact scaling fixed in the repository. Its recurrence is derived from the rational half-moment recurrence after clearing factorial, dyadic, and Mersenne denominators. Parity and divisibility results for G_n are structural inputs to denominator theorems. It should not be confused with the transform sometimes also denoted $G(z)$ in analytic sections.

Half-moment sequence d_n .

PROJECT / MIXED

The sequence d_n is the repository's normalized half-moment sequence. It is related to the full moments by a finite even-index binomial transform and to endpoint values by an exact inverse-power bridge of the form $F(2^{-n})$ equals a specified factorial and power-of-two normalization times d_n . Arithmetic properties of d_n , especially the oddness of a normalized numerator, yield the exact 2-adic valuation and Reshetnikov integers.

Half-weighted interval indicator.

CONVENTION

A half-weighted interval indicator equals 1 in the interior of an interval, $1/2$ at its two endpoints, and 0 outside. It is useful when adjacent closed cells are summed, because shared endpoints then receive total weight one. It is a separate object from the ordinary indicator and appears in exact periodization, spline, and sampling formulas where endpoint values are pointwise data rather than negligible measure-zero choices.

Hamming weight.

CLASSICAL

The Hamming weight of a binary word or nonnegative integer is the number of one bits. For integers it is the same as the binary digit sum $\text{wt}_2(n)$. Its parity determines the Thue–Morse sign and its exact value counts how many active scales occur in a binary recursion. Algorithms that remove one set bit per step therefore have complexity proportional to Hamming weight rather than to the numerical size of the numerator.

Hankel moment matrix.

PROJECT / MIXED

The $N \times N$ Hankel moment matrix is

$$H_N = (m_{i+j})_{0 \leq i, j < N}.$$

Its entries are constant on anti-diagonals. Positivity of $\int p(x)^2 d\mu(x)$ implies positive semidefiniteness; strict positivity through degree $N - 1$ makes H_N positive definite. Its determinants control existence and normalization of monic orthogonal polynomials.

Harmonic number.

CLASSICAL

The n -th harmonic number is $H_n = \sum_{k=1}^n 1/k$, with $H_0 = 0$. Harmonic numbers arise in derivatives of Gamma functions, coefficients of logarithms of factorial products, and finite-part expansions. Generalized harmonic numbers use $\sum k^{-r}$. In saddle and Mellin coefficient formulas, they may appear when expanding Gamma ratios or zeta corrections near integer arguments.

HasSum and tsum.

CONVENTION

In Lean, `HasSum f a` records a proof that the series with terms $f(n)$ converges to a , whereas `tsum f` denotes the library's totalized infinite sum. A theorem named `*_hasSum` retains the summability witness and a theorem named `*_eq_tsum` states the resulting equality. The distinction is important in the repository because a formal expression may exist as a totalized value even when the intended analytic theorem still requires a summability hypothesis.

Hausdorff moment sequence.

PROJECT / MIXED

A sequence is a Hausdorff moment sequence when $m_n = \int_0^1 x^n d\mu(x)$ for a finite positive measure. Hausdorff's criterion is complete monotonicity of the finite differences: $(-1)^k \Delta^k m_n \geq 0$. The representing measure on $[0, 1]$ is unique. Bounded Fabius moments and normalized profile sequences fit this compact moment setting.

Head–tail decomposition.

LEAN-PROVED

A head–tail decomposition splits an infinite random series at level N into a finite partial sum and an independent remainder. The head has an explicit spline density, while the tail is confined to a short interval or is a scaled copy of the whole law. This yields uniform CDF sandwiches, self-similar recurrences, and computable approximation errors.

Heat trace.

CLASSICAL

For a nonnegative operator with discrete eigenvalues λ_j , the heat trace is $\Theta(t) = \sum_j e^{-t\lambda_j}$ when trace class. Its small- t asymptotics encode spectral growth, while Mellin transformation connects it to the spectral zeta function. Frontier work applies analogous constructions to dyadically structured zero spectra.

Heine transformation.

LEAN-PROVED

Heine's transformation rewrites a ${}_2\phi_1$ series as another ${}_2\phi_1$ multiplied by an infinite-product quotient. One standard form is

$${}_2\phi_1(a, b; c; q, z) = \frac{(b, az; q)_\infty}{(c, z; q)_\infty} {}_2\phi_1(c/b, z; az; q, b).$$

Its validity requires the stated convergence and nonvanishing conditions; other equivalent forms permute parameters and should not be conflated.

Herglotz function.

LEAN-PROVED

A Herglotz, or Pick–Nevanlinna, function maps the upper half-plane into itself. With the $z - x$ Cauchy-transform convention, a positive measure may instead produce a function with negative imaginary part, so a minus sign is sometimes inserted. The repository fixes this orientation before applying representation and boundary-value theorems.

Highest nonzero dyadic derivative.

LEAN-PROVED

At a fixed dyadic point, scaling and flatness can force all derivatives above a finite order to vanish while leaving one distinguished top derivative nonzero. The order and sign are computed from the reduced dyadic level and binary data. This terminating jet is a central local invariant in exact Taylor evaluation and dyadic-spine nonanalyticity proofs.

Highest-set-bit decomposition.

LEAN-PROVED

Every positive integer has a unique decomposition $n = 2^m + r$ with $0 \leq r < 2^m$. Recursions based on the highest set bit process one nonzero binary digit at a time and expose complete aligned blocks. Their depth is bounded by bit length and often more closely by Hamming weight.

Highest-set-bit recursion.

CLASSICAL

A highest-set-bit recursion decomposes a positive integer a as $2^m + r$ with $0 \leq r < 2^m$, processes the leading binary block, and recurses on a smaller integer. For Fabius dyadic evaluation, reflection and Taylor reduction turn this decomposition into an exact arithmetic algorithm. Each recursive call removes at least one set bit, providing a transparent termination measure and an operation count controlled by the binary representation.

Hilbert-space projection.

LEAN-PROVED

For a closed subspace V of a Hilbert space, every x has a unique orthogonal projection $P_V x$ minimizing $\|x - v\|$. The residual is orthogonal to V , and Pythagoras splits the squared norm. Finite spans of standardized geometric-uniform random variables give exact least-squares predictors in the current corpus.

Hilbert transform.

CLASSICAL

The Hilbert transform is the principal-value singular integral

$$Hf(x) = \frac{1}{\pi} \text{p. v.} \int_{\mathbb{R}} \frac{f(t)}{x - t} dt.$$

It is the real boundary part conjugate to a Cauchy transform and multiplies Fourier modes by $-i \operatorname{sgn} \xi$ under a standard convention. Compact support does not remove the principal-value issue inside the support.

Historical notation scope.

HISTORICAL

A historical notation scope is a clearly marked region in which a quoted or transcribed source retains its original symbols. It preserves bibliographic fidelity without changing the active project's notation. The scope ends with the quotation or transcription; surrounding commentary returns to canonical commands. A historical marker is therefore provenance metadata, not a blanket waiver from notation consistency.

Hockey-stick identity.

CLASSICAL

The hockey-stick identity is $\sum_{j=r}^n \binom{j}{r} = \binom{n+1}{r+1}$. Its name comes from the shape of the selected entries in Pascal's triangle. Repeated prefix summation and discrete B-spline formulas often reduce to this identity, as do sums over ordered compositions with a final free part. The repository uses it to collapse nested finite sums arising from Thue–Morse cancellation and spline integration.

Holomorphic function.

CLASSICAL

A holomorphic function is complex differentiable at every point of an open domain. Complex differentiability implies analytic power-series expansion and strong rigidity properties such as the identity theorem and Cauchy's integral formula. Entire functions are holomorphic on all of $\mathbb{C}$. The complex moment-generating function and Fourier transform of the up-function are holomorphic objects; contour and Mellin arguments rely on their domains, singularities, and growth estimates.

Holonomic function.

FRONTIER

In one variable, "holonomic" is commonly synonymous with D -finite. In several variables it refers to a system with minimal-dimensional characteristic variety. Closure under sums, products, derivatives, and definite integration makes holonomic methods powerful, but composition and functional inversion have restricted closure properties.

Homothety.

CLASSICAL

A homothety is a dilation about a fixed center, $x \mapsto c + a(x - c)$. Homothetic copies of a compactly supported function are rescaled and shifted versions that preserve shape while changing support. Rvachev's atomic-function theory builds approximation systems from homothetic copies of the up-function. The Fabius dilation equation is centered at zero and uses ratio two, while folding and translating connect it to support-centered copies.

Horner scheme.

CLASSICAL

Horner's scheme evaluates a polynomial $a_0 + a_1x + \cdots + a_nx^n$ as $a_0 + x(a_1 + x(\cdots + xa_n))$. It uses n multiplications and n additions and reduces intermediate expression growth. Exact dyadic formulas involving moment polynomials can be implemented in Horner form to avoid repeated exponentiation. In rational arithmetic it also helps control numerator and denominator sizes, although bit complexity still depends on coefficient growth.

Hypergeometric term.

CLASSICAL

A sequence term a_n is hypergeometric when the ratio a_{n+1}/a_n is a rational function of n . Factorials, binomial coefficients, and double factorials are basic examples. Their sums can often be manipulated by telescoping or symbolic recurrence methods. The Fabius arithmetic formulas combine hypergeometric prefactors with recursively defined moment numerators; the q -analogues replace rational dependence on n by rational dependence on q^n .

Hypothesis ledger.

CONVENTION

A hypothesis ledger lists the assumptions attached to a family of formulas: domains, positivity, nonvanishing denominators, branch choices, convergence regions, and endpoint exclusions. The current q -series and inverse materials use ledgers because superficially similar identities can require different parameter regimes. A glossary definition summarizes the object; the ledger determines when a particular theorem may be applied.

9 I

Identity theorem.

CLASSICAL

The identity theorem says that two holomorphic functions on a connected domain that agree on a set with an accumulation point agree everywhere. A real-analytic analogue holds on intervals. It is used conceptually in nowhere-analytic arguments: if a smooth Fabius germ were analytic and agreed with a terminating Taylor polynomial on a neighborhood, overlap and continuation would force polynomial behavior incompatible with the global scaling and shape.

Immutable recovery point.

CONVENTION

An immutable recovery point is a commit or archived artifact retained so that an earlier source state can be reconstructed exactly. It differs from an active document, which should receive current notation and corrections. Recovery points preserve scholarly provenance and allow regressions to be diagnosed without forcing obsolete material back into the live corpus.

Improper integral.

CLASSICAL

An improper integral is defined as a limit when the interval is unbounded or the integrand is singular at an endpoint. Convergence requires the cutoff integrals to approach a finite limit. Absolute convergence is stronger and permits more transformations. Laplace, Mellin, and Fourier integrals in the Fabius theory are often improper; when a singular Mellin term prevents ordinary convergence, a finite-part integral rather than an improper integral is used.

Inclusion–exclusion principle.

CLASSICAL

Inclusion–exclusion computes the size or measure of a union by alternating over intersections. Algebraically it is a Boolean-cube Möbius inversion. Finite-difference and Thue–Morse subset sums share the same alternating-sign architecture, although their summands represent function values rather than event intersections.

Independent-copy equation.

LEAN-PROVED

An independent-copy equation is a distributional fixed point such as

$$X \stackrel{d}{=} aU + qX',$$

where X' is independent of U and has the same law as X . It encodes the first term and rescaled tail of a geometric random series. Taking transforms, moments, or CDFs produces the corresponding analytic functional equations, while uniqueness requires an appropriate class of probability laws.

Independent random variables.

CLASSICAL

Random variables are independent when probabilities of joint events factor, equivalently when expectations of suitable products factor. Independence implies that characteristic and moment-generating functions multiply under addition and that probability laws convolve. The Fabius distribution arises from an infinite sum of independent uniform variables at dyadically decreasing scales. This representation explains the infinite product transform and produces finite convolution splines by truncation.

Indicator function.

CONVENTION

The indicator of a set A is $\mathbf{1}_A(x) = 1$ for $x \in A$ and 0 otherwise. Normalized interval indicators are uniform probability densities. Their repeated convolutions are splines, and their Fourier transforms are phase factors times sinc functions. The finite approximants to the up-function begin with scaled interval indicators, so elementary support and mass calculations propagate through convolution.

Infinite convolution.

CLASSICAL

An infinite convolution is a limit of finite convolutions of probability measures or functions. For probability laws it represents the distribution of an infinite sum of independent random variables, provided the partial sums converge. Weak convergence is often the natural initial topology, while additional smoothness and tail control may yield densities and uniform convergence. The Fabius law is the infinite convolution of dyadically scaled uniform laws; its Fourier transform is the corresponding infinite product of sinc factors.

Infinite product.

CLASSICAL

An infinite product $\prod_{n \geq 0} (1 + u_n)$ converges to a nonzero limit when the partial products converge and, under standard conditions, when $\sum |u_n| < \infty$. Locally uniform convergence of holomorphic factors produces a holomorphic limit unless zeros accumulate pathologically. The up-function transform is an entire sinc product. Its zero structure, logarithm, and negative-real specialization drive the Fourier and endpoint theories.

Infinite q -binomial theorem.

LEAN-PROVED

For $|q| < 1$ and $|z| < 1$,

$$\frac{(az; q)_\infty}{(z; q)_\infty} = \sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n.$$

The domain stated here guarantees absolute convergence; analytic continuation may enlarge it only after poles and zeros are controlled. Euler's two q -exponential series are special cases.

Infinite q -Pochhammer product.

LEAN-PROVED

When $|q| < 1$,

$$(a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

The product is an analytic function of a , locally uniformly on $\mathbb{C}$; its zeros are exactly $a = q^{-j}$ unless repeated or cancelled in a quotient. The repository separates the product-valued definition, normal-convergence theorems, zero-order statements, and logarithmic-derivative formulas so that division by the product is never justified merely by a formal factorization.

Infinite random series.

CLASSICAL

An infinite random series is a sum $\sum_{n \geq 0} X_n$ of random variables defined as a limit of partial sums in a chosen mode, such as almost surely or in L^2 . If X_n are uniformly bounded by an absolutely summable deterministic sequence, convergence is immediate for every outcome. The Fabius variable

is a weighted series of independent uniforms with geometric weights, so it has compact support and an entire moment-generating function.

Infinite weighted sum.

CLASSICAL

An infinite weighted sum has the form $\sum a_n x_n$, with convergence governed by the weights and the size of the terms. In probability, independent bounded variables with $\sum |a_n| < \infty$ give almost-sure absolute convergence. Dyadic weights make the tail comparable to the next scale and create exact self-similarity. This is the probabilistic source of the Fabius dilation equation and of the finite spline truncations.

Inflection point.

CLASSICAL

An inflection point is a point where the local convexity changes from convex to concave or conversely. A zero of the second derivative alone is not sufficient unless a sign change or equivalent convexity statement is proved. The bounded Fabius function has a unique inflection point at $x = 1/2$: its derivative is strictly increasing on the left half and strictly decreasing on the right half. Symmetry forces the location, while strict unimodality of the up-density gives uniqueness.

Information density.

FRONTIER

When the joint law is absolutely continuous with respect to the product of marginals, the information density is the log Radon–Nikodym derivative

$$\iota_{X;Y}(x, y) = \log \frac{dP_{XY}}{d(P_X P_Y)}(x, y).$$

Its expectation is mutual information and its variance is an information dispersion quantity.

Injective, surjective, and bijective.

CLASSICAL

A map is injective if equal outputs imply equal inputs, surjective if every target value is attained, and bijective if both hold. Strict monotonicity gives injectivity on an interval; continuity and endpoint values often give surjectivity onto the interval between them. The Fabius restriction $F : [0, 1] \rightarrow [0, 1]$ is bijective, so it has a well-defined inverse. Globally, the constant tails destroy injectivity, which is why inverse statements always specify the restricted domain.

Integer composition.

CLASSICAL

A composition of a positive integer n is an ordered tuple of positive integers $(m_1, \dots, m_r)$ with sum n . Unlike a partition, order matters. Weak compositions allow zero parts. Compositions index coefficients of powers and logarithms of formal series: the coefficient of z^n in $U(z)^r$ is a sum over r -part compositions of n . The all-orders Fabius saddle coefficients use such sums explicitly.

Integer numerator normalization.

LEAN-PROVED

An integer numerator normalization chooses a prescribed denominator D_n and defines $N_n = D_n m_n$ for a rational moment or value m_n . The normalization is useful when N_n obeys a division-free recurrence, has controlled parity, or factors naturally. Different valid choices produce different numerator sequences and are not interchangeable.

Integer partition.

CLASSICAL

An integer partition of n is a multiset of positive integers whose sum is n , with order ignored. Partitions classify multiplicity patterns in Bell-polynomial and Faà di Bruno expansions. They are distinct from partitions of unity, which are families of functions summing to one, and from ordered compositions. In all-orders coefficient formulas one may pass between partition-indexed multiplicities and composition-indexed ordered products depending on which is easier to formalize.

Integer-valued polynomial.

CLASSICAL

An integer-valued polynomial is a rational-coefficient polynomial taking integer values at every integer. Such polynomials are exactly the finite integer combinations of $\binom{x}{k}$. Repeated-sum diagonals often belong to this ring even when their monomial-basis coefficients have denominators.

Integrability.

CLASSICAL

Integrability means that a function has a well-defined integral in the relevant sense, usually Lebesgue or Riemann. Absolute integrability, $\int |f| < \infty$, ensures stable Fourier transforms and interchange of sums and integrals. Compactly supported continuous functions are absolutely integrable, as are the up-function and its derivatives. Mellin kernels may fail to be integrable at zero until singular terms are subtracted.

Integral transform.

CLASSICAL

An integral transform maps f to a new function $Tf(s) = \int K(s, x)f(x) \, dx$ using a kernel K . Fourier, Laplace, Mellin, and moment transforms emphasize different structures: translation, exponential decay, scaling, and polynomial moments. The Fabius theory moves among all four. Each transform has its own convergence region and inversion theorem, so identities must be transferred with explicit hypotheses.

Integrality.

CLASSICAL

An integrality theorem proves that a rationally defined expression is in fact an integer. Such results often require divisibility, valuation, or recurrence arguments that are invisible in the original quotient. The repository proves positivity and oddness of normalized Reshetnikov integers and constructs division-free numerator recurrences. Integrality is stronger than rationality and enables modular or parity conclusions that feed back into exact denominator formulas.

Interior of the support.

CLASSICAL

The interior of the support is the largest open set contained in the support. For the up-function, the support is $[-1, 1]$ and its interior is $(-1, 1)$, where the function is strictly positive. Boundary points belong to the topological support even though the value and every derivative vanish there. Distinguishing support from its interior is important in statements about positivity, analyticity, and zero sets.

Interpolating polynomial.

CLASSICAL

Given distinct nodes $x_0, \dots, x_n$ and values y_j , the interpolating polynomial is the unique polynomial of degree at most n satisfying $p(x_j) = y_j$. Lagrange's formula writes it as a sum of basis polynomials. On geometric nodes, its leading coefficient is a Gaussian-weighted functional of the data. The Fabius q -binomial identities identify this leading coefficient after Thue–Morse cancellation with an exact dyadic value.

Interval partition.

FRONTIER

An interval partition has every block equal to a consecutive interval in the natural order. Such partitions index Boolean moment–cumulant relations. They are fewer than noncrossing partitions and depend on the linear order.

Invariant measure.

PROJECT / MIXED

A measure μ is invariant under T when $T_{\#}\mu = \mu$, equivalently $\int f \circ T \, d\mu = \int f \, d\mu$ for bounded measurable f . An invariant density is a fixed point of the corresponding transfer operator. A functional equation for a Fabius-related function does not by itself define an invariant dynamical measure.

Inverse Barnes- G asymptotic.

FRONTIER

The inverse of Barnes G at large positive values is obtained by reverting its quadratic-logarithmic expansion. The leading coordinate is comparable to $\sqrt{\log y / \log \log y}$, after which nested logarithmic corrections are determined recursively. Complex inverse branches require additional monodromy and critical-value analysis.

Inverse-branch closure.

LEAN-PROVED

Inverse-branch closure is the class generated from elementary functions by allowing inverses on intervals or domains where a function is injective and an analytic branch exists. The repository uses an explicitly typed version to state that the inverse Fabius function is not obtained even after adjoining such inverse branches to the elementary class. Branch domains and nonvanishing derivatives are indispensable hypotheses.

Inverse-dyadic value.

PROJECT / MIXED

An inverse-dyadic value is a number $x \in [0, 1]$ satisfying $F(x) = a/2^n$ for a dyadic ordinate. Existence and uniqueness follow from the bijection $F : [0, 1] \rightarrow [0, 1]$. Closed forms are generally harder than direct dyadic evaluation because the argument need not itself be dyadic. The repository studies compositions and exact identities at selected inverse-dyadic values, using symmetry and functional equations rather than assuming a simple rational form.

Inverse-Fabius self-map dynamics.

FRONTIER

The inverse Fabius function is likewise an increasing self-map of the unit interval. Its iterates reverse the time direction of the bounded Fabius dynamics wherever ordinary inversion applies. Endpoint flatness of F becomes extreme expansion for the inverse, so local derivative formulas, fractional iteration proposals, and numerical conditioning require separate endpoint analysis.

Inverse function.

CLASSICAL

If $f : I \rightarrow J$ is bijective, its inverse $f^{-1} : J \rightarrow I$ satisfies $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$. For a continuous strictly monotone map between intervals, the inverse is continuous; if $f' \neq 0$, it is differentiable locally with derivative $1/f'$. The Fabius inverse is defined on $[0, 1]$ using the strictly increasing bounded function. Near the endpoints its behavior is governed by inverting the corrected flat asymptotic.

Inverse-function condition number.

PROJECT / MIXED

For a differentiable scalar map $y = f(x)$ with $f'(x) \neq 0$, the local absolute sensitivity of the inverse is $|1/f'(x)|$. A relative condition number also scales by $|y/x|$ when meaningful. Near a critical point or a flat Fabius endpoint, inverse evaluation becomes ill conditioned even though the inverse remains continuous.

Inverse-gamma asymptotic.

FRONTIER

An inverse-gamma asymptotic solves $\Gamma(x) = y$ or $\log \Gamma(x) = \log y$ for large y . Stirling's formula gives a first Lambert-type approximation; polynomial-logarithmic reversion supplies successive corrections. Since gamma is not globally injective on $\mathbb{C}$, the branch and real monotonicity region are part of the result.

Inverse-Lambert expansion.

PROJECT / MIXED

An inverse-Lambert expansion rewrites the solution of an equation such as $\lambda 2^{-\lambda} = x$ in elementary logarithmic terms. Exactly, $\lambda = -W_{-1}(-x \log 2)/\log 2$. Iterating the relation $\lambda \log 2 = |\log x| + \log \lambda$ gives nested logarithms and a full asymptotic series. The repository formalizes the coefficient algebra and remainder bounds needed to transfer the sharp saddle expansion from λ to powers of $1/|\log x|$.

Inverse-logarithmic error.

CLASSICAL

An inverse-logarithmic error is a remainder bounded by a negative power of a growing logarithm, for example $\mathcal{O}(1/|\log x|)$ as $x \downarrow 0$. Such errors decay very slowly and are therefore sensitive to bounded oscillatory terms: a missing nonconstant periodic contribution is not swallowed by them. The corrected Fabius asymptotic attains an inverse-logarithmic error only after evaluating the periodic fluctuation at the exact lower-Lambert phase.

Inverse-Mellin residue expansion.

PROJECT / MIXED

Shifting a Mellin inversion contour and summing crossed residues yields an asymptotic expansion. The remaining contour supplies the error term and requires growth bounds. Repeated poles create logarithmic factors; arithmetic pole lattices create periodic fluctuations. Formal residue lists alone do not justify the contour shift.

Inverse modulus from a gap bound.

LEAN-PROVED

An inverse modulus converts a positive lower bound on $F(y) - F(x)$ for separated rational or dyadic inputs into a modulus of continuity for F^{-1} . The generic formal interface assumes a computable sequence of positive rational gap bounds; constructing a particular sharp sequence is a separate task. This separation makes the logical content of inverse computability explicit.

Inverse-power table.

CLASSICAL

An inverse-power table records exact or formally generated coefficients in expansions involving $x^{-1}, x^{-2}, \dots$, usually after a large coordinate has been selected. In saddle and Lambert analyses it stores the algebra needed to divide, compose, exponentiate, or reverse truncated series without repeatedly expanding from first principles. The table is an indexed coefficient family, not the reciprocal of a single scalar. The former headword “inverse-power value” is retained only as a historical alias.

Inversion formula.

CLASSICAL

An inversion formula reconstructs an original object from a transform or cumulative operation. Fourier, Laplace, and Mellin transforms each have inversion integrals; finite differences invert prefix summation; binomial inversion undoes a binomial transform. In the Fabius directory, inversion formulas recover the up-density from its sinc product and the endpoint function from a Bromwich contour. Each inversion requires its own convergence and regularity hypotheses.

IsFabius characterization.

PROJECT / MIXED

IsFabius is the Lean predicate collecting the axioms that characterize the bounded Fabius function: constant tails, reflection symmetry, differentiability or smoothness, and the left-half differential equation in the exact repository formulation. Mathematically, it defines a property of candidate functions rather than a new function. Existence supplies one witness and uniqueness proves that any two witnesses agree, allowing later theorems to be stated abstractly for every function satisfying the characterization.

Iterated derivative.

CLASSICAL

The iterated derivative $f^{(m)}$ is obtained by differentiating m times. For a self-similar differential equation, each iteration introduces both a coefficient and a chain-rule scaling. The signed Fabius extension satisfies the closed formula $\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x)$. This turns an apparently analytic hierarchy into exact arithmetic at dyadic points and yields the sharp derivative suprema.

Iterated prefix kernel.

LEAN-PROVED

The r -fold prefix-sum operator has kernel

$$S^r a(n) = \sum_{j=0}^{n-1} \binom{n-1-j+r-1}{r-1} a(j)$$

for $r \geq 1$, with the chosen exclusive-prefix convention. Alternative inclusive conventions shift the binomial factor. This kernel makes repeated Thue–Morse summation accessible to coefficient extraction and polynomial diagonal analysis.

Iterated prefix sum.

CLASSICAL

For a sequence a_n , its prefix sum is $A_n = \sum_{k \leq n} a_k$; iterating this operation produces higher discrete antiderivatives. Repeated prefix sums of an indicator sequence yield binomial coefficients and discrete B-splines. When applied to Thue–Morse signs, low-order sums vanish over complete dyadic blocks by Prouhet cancellation, and the first nonzero iterated sum has a structured polynomial form. This connects binary combinatorics to spline limits.

10 J**Jackson fundamental theorem of calculus.**

LEAN-PROVED

Under convergence and endpoint hypotheses,

$$D_q \left(\int_0^x f(t) d_q t \right) = f(x), \quad \int_0^a D_q f(t) d_q t = f(a) - f(0).$$

The proof is a telescoping geometric series. This is a q -calculus theorem, not a direct consequence of the ordinary fundamental theorem, and the behavior at the accumulation point must be controlled.

Jackson integration by parts.

LEAN-PROVED

The integration-by-parts identity is

$$\int_0^a f(t) D_q g(t) d_q t = f(a)g(a) - f(0)g(0) - \int_0^a g(qt) D_q f(t) d_q t.$$

Absolute convergence sufficient for rearrangement is part of the hypothesis. The shift $g(qt)$ encodes the nonlocal product rule.

Jackson product rule.

LEAN-PROVED

The Jackson derivative obeys

$$D_q(fg)(x) = g(x)D_q f(x) + f(qx)D_q g(x)$$

and equivalently $D_q(fg) = f D_q g + g(q \cdot) D_q f$. The shifted factor is not optional. Iteration produces Gaussian-binomial Leibniz formulas whose powers of q record how shifts pass through derivatives.

Jackson q -derivative.

LEAN-PROVED

For $q \neq 1$ and $x \neq 0$,

$$D_q f(x) = \frac{f(x) - f(qx)}{(1 - q)x}.$$

At $x = 0$ a value is supplied only under an explicit limiting hypothesis. The operator satisfies $D_q x^n = [n]_q x^{n-1}$ and a shifted product rule. It approaches the ordinary derivative as $q \rightarrow 1$ under suitable regularity, but it is a finite-difference operator at fixed q .

Jackson q -integral.

LEAN-PROVED

For $0 < q < 1$ and $a > 0$,

$$\int_0^a f(t) d_q t = (1-q)a \sum_{n \geq 0} q^n f(aq^n),$$

when the series converges absolutely. It samples on a geometric comb and is therefore naturally paired with D_q . The endpoint 0 is approached through the accumulating nodes rather than through a continuum of points.

Jacobi continued fraction.

PROJECT / MIXED

The Stieltjes or Cauchy transform of a moment functional has a Jacobi continued fraction,

$$\frac{1}{z - \alpha_0 - \frac{\beta_1}{z - \alpha_1 - \frac{\beta_2}{\ddots}}}.$$

Its coefficients are the three-term recurrence data. Finite truncations are rational functions whose poles are Gaussian nodes and whose residues are quadrature weights.

Jacobi cubic identity.

LEAN-PROVED

Jacobi's cubic identity is a theta/product relation obtained by a three-dissection of the triple product. It organizes coefficients by residues modulo three and yields cubic theta series. The current formalization treats it as an exact q -series theorem with absolute-convergence hypotheses, not as a heuristic rearrangement of bilateral sums.

Jacobi matrix.

PROJECT / MIXED

The Jacobi matrix is the tridiagonal matrix representing multiplication by x in an orthonormal polynomial basis. Its diagonal entries are the recurrence coefficients α_n , and its positive off-diagonal entries are square roots of β_n . Finite principal blocks have eigenvalues equal to Gaussian quadrature nodes.

Jacobi theta function.

PROJECT / MIXED

A Jacobi theta function is a doubly infinite q -series with quasi-periodicity and an equivalent infinite-product representation. One multiplicative normalization is

$$\theta(z; q) = (z; q)_\infty (q/z; q)_\infty (q; q)_\infty.$$

Different classical ϑ_j notations include powers of q , signs, and additive variables; every use must state its normalization.

Jacobi triple product.

LEAN-PROVED

For $|q| < 1$ and $z \neq 0$, a standard form is

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{n(n-1)/2} z^n = (z; q)_\infty (q/z; q)_\infty (q; q)_\infty.$$

It identifies a bilateral series with a theta product. Sign and exponent shifts vary among conventions; the displayed normalization is the one from which quasi-periodicity follows directly.

Jet.

CLASSICAL

The m -jet of a smooth function at a point is the finite list of derivatives through order m , often encoded as its Taylor polynomial modulo terms of degree $m + 1$. Jets capture the local data needed for finite-order approximation without asserting convergence of the full Taylor series. Saddle expansions use phase and amplitude jets at a moving critical point; formal recurrences combine these jets into all-orders coefficients. At a dyadic Fabius point the infinite jet is eventually zero, yet the function is not locally polynomial.

11 K**K-fold convolution.**

CLASSICAL

A K -fold convolution combines K functions or measures by repeated convolution. For probability laws it is the distribution of a sum of K independent variables; for compact supports, the support is the Minkowski sum of the individual supports. K -fold identities involving up-functions and Thue–Morse coefficients generalize the basic self-convolution relation. Transform methods reduce them to products or powers and make normalization transparent.

K-fold Thue–Morse sum.

PROJECT / MIXED

A K -fold Thue–Morse sum is a multiple discrete convolution or signed sum built from the sequence $\varepsilon_n = (-1)^{\text{wt}_2(n)}$. Its generating function is a power of the finite Thue–Morse product, so high-order zeros at one translate into vanishing moments and Prouhet-type cancellation. In the repository such sums are related to convolution powers, discrete splines, and scaled limits of the Fabius/up functions.

K-function (generalized hyperfactorial).

FRONTIER

The K -function is a continuous extension of the hyperfactorial satisfying a functional equation related to powers z^z and Barnes G . Its normalization varies in the literature. Large-argument inversion combines a quadratic logarithmic leading term with Stirling-type corrections and must state the selected real or complex branch.

Kernel.

CLASSICAL

A kernel is a function that defines an integral, convolution, summation, or transformation operator. Examples include the Fourier exponential, the Laplace exponential, interval indicators, and Bose–Einstein kernels. Kernel properties such as positivity, normalization, support, symmetry, and decay

determine the operator's behavior. The Fabius development uses probability kernels for the fixed point, sinc factors as transform kernels, and regularized Mellin kernels for endpoint asymptotics.

Knot multiplicity.

CLASSICAL

The multiplicity of a spline knot is the number of repeated breakpoint factors at that location. Higher multiplicity reduces smoothness across the knot. For unequal-scale box splines and finite sinc prefixes, coincident signed subset sums create multiplicities that must be counted before differentiability or polynomial degree is inferred.

Kolmogorov distance.

CLASSICAL

For real distribution functions F and G , the Kolmogorov distance is

$$d_K(F, G) = \sup_{x \in \mathbb{R}} |F(x) - G(x)|.$$

It is the natural norm for uniform CDF approximation. A deterministic bound on the omitted tail of a monotone random sum yields a Kolmogorov sandwich after shifting the finite-sum CDF, while a direct density bound may give a sharper estimate.

Koopman eigenfunction.

FRONTIER

A nonzero observable f is a Koopman eigenfunction when $f \circ T = \lambda f$. Products of eigenfunctions multiply eigenvalues when the function space is an algebra. Boundary behavior and the choice of space can exclude formally plausible eigenfunctions, so spectral claims always name their ambient domain.

Koopman operator.

FRONTIER

For a measurable self-map $T : X \rightarrow X$, the Koopman operator is

$$(\mathcal{K}_T f)(x) = f(Tx)$$

on a declared function space. It is linear even when T is nonlinear. Its spectrum depends on the Banach or Hilbert space and invariant measure. Fabius Koopman models remain frontier objects unless those data and a spectral theorem are explicitly supplied.

Kummer's theorem.

CLASSICAL

For a prime p , Kummer's theorem states that $v_p\binom{n}{k}$ equals the number of carries when adding k and $n - k$ in base p . Equivalently, it is $(s_p(k) + s_p(n - k) - s_p(n))/(p - 1)$. The theorem converts valuation questions into digit combinatorics and complements Lucas's congruence theorem.

L2 norm.

CLASSICAL

For a square-integrable function on a measure space, the L^2 norm is $\|f\|_2 = (\int |f|^2)^{1/2}$. It comes from the inner product $\langle f, g \rangle = \int f \bar{g}$, so orthogonality and Pythagorean identities are available. Fourier-Legendre partial sums are orthogonal projections in $L^2[-1, 1]$ and therefore give the unique least-squares polynomial approximants. Uniform convergence proved later is stronger than convergence in this norm.

Lagrange–Good multivariate inversion.

FRONTIER

Good's multivariate inversion formula extracts coefficients of a solution to $w_i = z_i \phi_i(\mathbf{w})$ through a determinant correcting the Jacobian of the substitution. It generalizes Lagrange inversion and is effective for coupled exponentially small variables in oscillatory inverse problems. The full determinant factor is essential.

Lagrange interpolation.

CLASSICAL

Lagrange interpolation writes the unique degree-at-most- n polynomial through distinct data (x_j, y_j) as $p(x) = \sum_{j=0}^n y_j \ell_j(x)$, where $\ell_j(x) = \prod_{m \neq j} (x - x_m)/(x_j - x_m)$. Its leading coefficient is $\sum_j y_j / \prod_{m \neq j} (x_j - x_m)$. For geometric nodes -2^j , these denominators simplify to q -Pochhammer products and Gaussian binomial weights, revealing the interpolation structure of exact Fabius dyadic formulas.

Lagrange inversion formula.

PROJECT / MIXED

If $w = z\phi(w)$ with $\phi(0) \neq 0$, then

$$[z^n]H(w(z)) = \frac{1}{n} [t^{n-1}]H'(t)\phi(t)^n$$

for $n \geq 1$. In particular it gives coefficients of the inverse of a power series with nonzero linear term. Formal and analytic versions share the coefficient formula but have different convergence hypotheses.

Lagrange up-atom.

LEAN-PROVED

A Lagrange up-atom is a normalized translate/dilate of the up-function chosen so that, on a target interval, its active polynomial piece realizes one Lagrange cardinal basis polynomial. Finite sums then interpolate arbitrary data at the selected nodes. Outside the interval the atom remains a smooth compactly supported function rather than the global polynomial.

Lah number.

CLASSICAL

The unsigned Lah number $L(n, k) = \binom{n-1}{k-1} n! / k!$ counts partitions of an n -element set into k nonempty linearly ordered lists. Lah numbers convert rising and falling factorial bases. Signed variants insert $(-1)^{n-k}$.

Lambert polynomial.

PROJECT / MIXED

The Lambert polynomials $L_n(u)$ package the logarithmic coefficient blocks in the large- x expansion of Wright omega or Lambert W . In the current zero-based normalization $L_0(u) = -u$, and for $n \geq 1$ the degree is n with leading coefficient $1/n$. A differential-integral recurrence generates all orders.

Lambert W branch point.

PROJECT / MIXED

The map $w \mapsto we^w$ has derivative zero at $w = -1$, giving the critical value $-e^{-1}$. The branches W_0 and W_{-1} meet there. With $p = \sqrt{2(1 + ez)}$, both admit convergent local Puiseux series whose sign selects the branch.

Lambert W function.

PROJECT / MIXED

The Lambert W function is the multivalued inverse of $w \mapsto we^w$: it satisfies $W(z)e^{W(z)} = z$. On $(-e^{-1}, 0)$ there are two real branches, $W_0 \in (-1, 0)$ and $W_{-1} \in (-\infty, -1)$. The Fabius endpoint saddle diverges as $x \downarrow 0$, so the lower branch W_{-1} , not the principal branch, gives the relevant coordinate. Its nested-log expansion translates the exact saddle result into elementary logarithms.

Lambert W Puiseux coordinate.

PROJECT / MIXED

Near $z = -e^{-1}$, the signed square-root coordinate

$$p = \pm \sqrt{2(1 + ez)}$$

linearizes the leading branch separation: $W(z) = -1 + p - \frac{1}{3}p^2 + \dots$. The sign and square-root cut must be stated. Ordinary Taylor expansion in $z + e^{-1}$ cannot represent either inverse branch.

Laplace transform.

CLASSICAL

The Laplace transform of a function or measure on a half-line is $\mathcal{L}f(s) = \int_0^\infty e^{-sx} f(x) \, dx$, on the region where the integral converges. For a compactly supported probability law it extends to an entire moment-generating function after a sign change. Convolution becomes multiplication. The Fabius transform factors into elementary terms from the independent uniform summands, and its large negative-real behavior is the starting point for the corrected endpoint asymptotic.

Large-deviation principle.

FRONTIER

A sequence satisfies an LDP with speed a_n and rate I when probabilities of sets decay like $e^{-a_n \inf I}$, with separate open-set lower and closed-set upper bounds. The rate is often the Legendre–Fenchel transform of a limiting log-MGF. An LDP describes exponential tails, not central Gaussian fluctuations.

Laurent expansion.

CLASSICAL

A Laurent expansion represents a holomorphic function on an annulus as $\sum_{n=-\infty}^\infty a_n(z - z_0)^n$. Finitely many negative powers characterize a pole; the coefficient of $(z - z_0)^{-1}$ is the residue. Mellin transforms in the Fabius analysis have Laurent expansions at singular points such as the origin or $s = 1$. The finite part is the constant coefficient after the principal polar terms are removed.

Laurent–logarithmic series.

PROJECT / MIXED

A Laurent–logarithmic series has terms $t^n p_n(\log t)$, where $n \in \mathbb{Z}$ and each p_n is a polynomial. It models inverse problems in which logarithms enter at every algebraic order. Formal differentiation changes both the power and the logarithmic coefficient; analytic realization requires a logarithm branch.

Law as a pushforward measure.

CLASSICAL

The law of a random variable $X : \Omega \rightarrow S$ is the pushforward $X_{\#}\mathbb{P}$, defined by $\text{Law}(X)(B) = \mathbb{P}(X \in B)$. An almost-sure equality of random variables implies equality of laws, and a measurable map sends laws by another pushforward. The repository uses this identity to pass from random-series recursions to CDF and density statements.

Layer-cake representation.

CLASSICAL

For a nonnegative measurable function f ,

$$f(x) = \int_0^\infty \mathbf{1}_{\{f(x) > t\}} dt$$

pointwise, and Tonelli gives $\int f = \int_0^\infty \mu\{f > t\} dt$. Weighted variants yield moment and fractional-power formulas. The project uses layer-cake arguments for CDFs, densities, inverse functions, and level-set geometry.

Leading-coefficient functional.

CLASSICAL

On polynomials of degree at most n , the leading-coefficient functional sends $p(x) = a_n x^n + \dots$ to a_n . Evaluating p at $n + 1$ distinct nodes gives another coordinate system, so the functional becomes a weighted sum of node values. For geometric nodes the weights are normalized Gaussian binomial coefficients. The Fabius q -calculus shows that this functional, followed by Thue–Morse block cancellation and exact normalization, returns the desired dyadic value.

Least common multiple.

CLASSICAL

The least common multiple $\text{lcm}(a_1, \dots, a_m)$ is the smallest positive integer divisible by each a_j . Prime valuations satisfy $v_p(\text{lcm}(a_j)) = \max_j v_p(a_j)$. LCM products give economical common denominators for families of rational moment or dyadic values. In the Fabius arithmetic layer, they control odd prime factors separately from the exactly known power of two.

Least Lipschitz constant.

PROJECT / MIXED

The least Lipschitz constant of f is $\sup_{x \neq y} |f(x) - f(y)|/|x - y|$, possibly infinite. For a continuously differentiable function on an interval it equals $\sup |f'|$. The bounded Fabius function is globally Lipschitz with optimal constant 2: the derivative bound gives the upper estimate, and equality is attained at the midpoint because $F'(1/2) = 2$. Optimality matters for the sharp effective modulus of continuity.

Least-squares approximation.

CLASSICAL

A least-squares approximation minimizes the squared L^2 error $\int |f - p|^2$ over a finite-dimensional approximation space. In an orthogonal basis, the minimizer is obtained by truncating the corresponding Fourier coefficients. The Legendre partial sum of degree n is therefore the unique best polynomial approximation to the up-function in $L^2[-1, 1]$. This optimality is metric-specific and does not by itself imply best uniform approximation.

Least-squares prediction error.

LEAN-PROVED

The least-squares prediction error of Y from a subspace $V \subset L^2$ is

$$\mathbb{E}[(Y - P_V Y)^2].$$

For standardized geometric-uniform variables constructed from common digits, the current exact formula factors this error into squared hyperbolic-distance terms $\prod_j ((q - q_j)/(1 - qq_j))^2$. The common probability coupling is part of the statement.

Lebesgue integral.

CLASSICAL

The Lebesgue integral integrates measurable functions by measuring level sets and is stable under limit theorems such as monotone and dominated convergence. It naturally handles probability measures and almost-everywhere equivalence. Moments, expectations, convolution measures, and transform integrals in the Fabius development are most naturally Lebesgue integrals, even when continuity and compact support make them agree with classical Riemann integrals.

Legendre block.

LEAN-PROVED

A Legendre block is a finite polynomial assembled from a consecutive or parity-restricted range of Legendre modes. Blocks are used to isolate scale, parity, or a translate contribution before summing an infinite expansion. Exact rational coefficients may be computed from moments, while analytic estimates control the block norm.

Legendre coefficient.

CLASSICAL

A Legendre coefficient is the component of a function along the Legendre polynomial P_n under the inner product on $[-1, 1]$. With $\int_{-1}^1 P_n^2 = 2/(2n + 1)$, the coefficient is $(2n + 1) \int f P_n/2$. Symmetry annihilates odd coefficients for an even function. The Fabius directory expresses the up-function coefficients exactly through moments and proves bounds strong enough for absolute uniform convergence.

Legendre coefficient parity completion.

LEAN-PROVED

If a_r records only the even coefficient c_{2r} , its parity-completed sequence is $c_{2r} = a_r$, $c_{2r+1} = 0$. The completion is a new sequence, not a change of index convention. It allows full-series theorems to consume an efficiently stored even sequence without losing the parity proof.

Legendre derivative identity.

PROJECT / MIXED

A useful identity is

$$(2n+1)P_n(x) = P'_{n+1}(x) - P'_{n-1}(x), \quad n \geq 1.$$

It transfers derivatives from a smooth function to polynomial combinations under integration by parts. Repeated use, together with endpoint flatness, gives rapid decay of Legendre coefficients of the up-function.

Legendre energy tail.

LEAN-PROVED

With classical Legendre normalization, the squared L^2 error after degree N is

$$\sum_{n>N} \frac{2}{2n+1} |c_n|^2.$$

This is the Legendre energy tail. Exact rational coefficients give exact finite energy sums; decay estimates certify the infinite remainder. For even Rvachev data only even indices contribute.

Legendre–Fenchel transform.

CLASSICAL

For a function Λ , its convex conjugate is

$$\Lambda^*(x) = \sup_t \{tx - \Lambda(t)\}.$$

When Λ is a differentiable steep log-MGF, the maximizing t solves $\Lambda'(t) = x$. This transform supplies large-deviation rate functions for normalized geometric-uniform laws.

Legendre Gaunt coefficient.

LEAN-PROVED

The Legendre Gaunt coefficient is the triple product

$$G_{\ell mn} = \int_{-1}^1 P_\ell(x) P_m(x) P_n(x) \, dx.$$

It vanishes unless the triangle inequalities hold and $\ell + m + n$ is even. When admissible, it has an exact factorial formula equivalent to a squared Wigner $3j$ -symbol. The repository proves rational and real-cast forms.

Legendre linearization coefficient.

LEAN-PROVED

The product of two Legendre polynomials expands as

$$P_\ell(x) P_m(x) = \sum_n L_{\ell m}^n P_n(x), \quad L_{\ell m}^n = \frac{2n+1}{2} G_{\ell mn}.$$

Only triangle-admissible indices of the correct parity occur. These coefficients turn products and nonlinear energies into finite convolutions of Legendre coefficient sequences.

Legendre normalization.

CONVENTION

The classical Legendre polynomial P_n is normalized by $P_n(1) = 1$ and

$$\int_{-1}^1 P_n(x)P_m(x) \, dx = \frac{2}{2n+1}\delta_{nm}.$$

This is neither monic nor orthonormal. Every coefficient formula in the project includes the factor $(2n+1)/2$ appropriate to this normalization.

Legendre parity.

LEAN-PROVED

Legendre polynomials satisfy $P_n(-x) = (-1)^n P_n(x)$. Therefore an even function has zero odd Legendre coefficients and an odd function has zero even coefficients. The Rvachev up-function is even, so its full coefficient sequence is the parity completion of an even-indexed rational sequence.

Legendre partial sum.

LEAN-PROVED

For coefficients c_n , the degree- N partial sum is $S_N(x) = \sum_{n=0}^N c_n P_n(x)$. It is the L^2 -orthogonal projection onto polynomials of degree at most N when c_n are Fourier–Legendre coefficients. Uniform convergence requires more than Hilbert-space projection and is proved from absolute coefficient bounds.

Legendre polynomial.

CLASSICAL

The Legendre polynomial P_n is the degree- n polynomial normalized by $P_n(1) = 1$ and orthogonal to lower-degree polynomials on $[-1, 1]$ with unit weight. Rodrigues' formula is $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$, and it solves $((1-x^2)P_n')' + n(n+1)P_n = 0$. These properties support integration by parts, coefficient recurrences, and least-squares approximation of the up-function.

Legendre's factorial-valuation formula.

CLASSICAL

For a prime p and $n \in \mathbb{N}$, Legendre's formula is

$$v_p(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor = \frac{n - s_p(n)}{p-1},$$

where $s_p(n)$ is the base- p digit sum. The sum is finite. It converts factorial quotients into digit and carry data and underlies valuation proofs for binomial, multinomial, Gaussian-specialization, and denominator expressions in the arithmetic parts of the project.

Legendre superconvergence.

FRONTIER

Superconvergence is accuracy at selected points or for selected functionals that is higher than the global truncation order predicts. Symmetry, endpoint flatness, and cancellation of leading Legendre tails can produce it for Fabius/Rvachev expansions. A superconvergence claim must state the point set, norm or functional, and asymptotic scale.

Leibniz rule.

CLASSICAL

The Leibniz rule for the n -th derivative of a product is $(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$. It is the differential analogue of binomial convolution. Formalized moment, transform, and saddle calculations repeatedly use it to expand products with exact finite index ranges. For exponential products it combines naturally with Bell polynomials and cumulants.

Levy continuity theorem.

CLASSICAL

Levy's continuity theorem states that pointwise convergence of characteristic functions to a function continuous at zero is equivalent to weak convergence of the corresponding probability measures. It is a standard route from finite convolution products to an infinite probability law. In the Fabius construction, the explicit convergence of the sinc products can identify the weak limit and its characteristic function.

Lipschitz function.

CLASSICAL

A function is Lipschitz with constant L when $|f(x) - f(y)| \leq L|x - y|$ for all relevant x, y . Lipschitz continuity is stronger than uniform continuity and implies absolute continuity on intervals. A bounded derivative gives a Lipschitz bound by the mean value theorem. The Fabius function has least global constant 2, while related spline approximants inherit uniform Lipschitz control.

Local phase.

CLASSICAL

The local phase is the Taylor expansion of the exponent in a contour integral around its saddle after the linear term has vanished. In a scaled variable t , it begins with a negative quadratic Gaussian term, followed by cubic and higher corrections whose coefficients are phase jets. The Fabius saddle modules isolate the local phase from the global periodic factor, prove a central approximation, and bound the remainder on a radius where the expansion is uniformly valid.

Locally finite sum.

CLASSICAL

A sum of functions $\sum_j f_j(x)$ is locally finite if every point has a neighborhood meeting the support of only finitely many summands. Such a sum is unambiguously defined and inherits continuity or smoothness termwise without convergence estimates. Translate sums of compactly supported up-functions are often locally finite. The signed global Fabius extension also has presentations where only finitely many translated support pieces contribute at a fixed point, alongside other genuinely infinite absolutely convergent series.

Locally uniform convergence.

CLASSICAL

A sequence converges locally uniformly on a domain if it converges uniformly on every compact subset. For holomorphic functions, locally uniform convergence preserves holomorphy and permits termwise differentiation. Infinite products are also commonly controlled locally uniformly. The sinc product for the up-function's Fourier transform and the negative-Laplace logarithmic series are established on compact subsets of their natural domains before global growth or tail estimates are considered.

Log-gamma function.

PROJECT / MIXED

The log-gamma function is a branch of $\log \Gamma(z)$ on a domain avoiding gamma poles and chosen cuts. On the positive real axis it is the real logarithm of $\Gamma(x)$. Stirling expansions are naturally stated for $\log \Gamma$, while inverse gamma problems first invert this large polynomial–logarithmic scale and then exponentiate if needed.

Log-periodic asymptotic.

LEAN-PROVED

A log-periodic asymptotic contains a periodic function of a logarithmic or Lambert-type scale, such as $\Psi(\log_b(1/x))$. Multiplying x by a fixed base shifts the phase by an integer and leaves the oscillation invariant. In the Fabius endpoint law, the fluctuation is nonconstant and must remain in the exponential main term to obtain ratio asymptotics.

Log-periodic function.

CLASSICAL

A function of $x > 0$ is log-periodic with ratio $b > 1$ if it is periodic in $\log_b x$, equivalently $Q(bx) = Q(x)$. Such oscillations arise in discrete-scale-invariant problems because scaling by b advances the logarithmic phase by one. The Fabius endpoint correction is one-periodic in a lower-Lambert coordinate rather than exactly in $\log_2 x$, but it has the same dyadic origin. A log-periodic term can be tiny yet prevent convergence to a constant asymptotic ratio.

Log-squared asymptotic.

CLASSICAL

A log-squared asymptotic has a dominant term proportional to $(\log(1/x))^2$, often in the logarithm of a very flat function. For the Fabius endpoint, $\log F(x)$ has a leading negative quadratic logarithmic scale, modified by lower nested-log, constant, and periodic contributions. This explains decay faster than every power of x but slower than typical $\exp(-c/x^\alpha)$ scales. The exact coefficient is determined by the dyadic dilation factor $\log 2$.

Logarithmic delay equation.

CLASSICAL

A logarithmic change of variables converts multiplicative scaling into additive translation. If $x = e^{-t}$, then $f(2x)$ becomes a shifted function of $t - \log 2$. The resulting relation is a delay or advance equation with fixed step $\log 2$. This viewpoint explains why periodic functions of a logarithmic coordinate appear as homogeneous or residual solutions in the Fabius endpoint analysis.

Logarithmic height.

CONVENTION

Logarithmic height records the deepest iterated logarithm needed by a transmonomial, while exponential height records nested exponentials. The height is structural, not the power of a logarithm. Declaring it helps choose a closed transseries field before composition or inversion.

Logarithmic scale.

CLASSICAL

A logarithmic scale uses $L = \log(1/x)$, $\log L$, and inverse powers of these quantities to organize a limit as $x \downarrow 0$. It is appropriate when the decisive saddle grows only logarithmically. The Fabius expansion is most exact in $\lambda = -W_{-1}(-x \log 2)/\log 2$, but λ itself has a nested-log expansion.

Distinguishing the exact coordinate from its elementary logarithmic approximation prevents phase errors.

Logarithmic Stieltjes fixed point.

LEAN-PROVED

A logarithmic Stieltjes representation integrates $\log(z - x)$ or its derivative against a compactly supported law and satisfies a scale-recursive fixed-point equation. The repository proves a real logarithmic identity for $z > 1$. No unmentioned complex logarithm branch or cut continuation is part of that theorem.

Lower Lambert branch.

CONVENTION

The lower real branch W_{-1} of Lambert W is defined on $[-e^{-1}, 0)$, takes values in $(-\infty, -1]$, and satisfies $W_{-1}(z)e^{W_{-1}(z)} = z$. As $z \uparrow 0$, it tends to $-\infty$. This branch solves the large positive saddle equation in the Fabius endpoint problem. Using the principal branch would produce a bounded coordinate and the wrong asymptotic regime.

Lower Lambert coordinate.

PROJECT / MIXED

For small $x > 0$, the repository defines the lower Lambert coordinate by

$$\lambda(x) = -\frac{W_{-1}(-x \log 2)}{\log 2}, \quad \lambda(x)2^{-\lambda(x)} = x.$$

It is the exact real saddle parameter, tends to infinity as $x \downarrow 0$, and advances by approximately one under dyadic rescaling. The periodic correction and all inverse-power coefficients are most naturally evaluated at $\lambda(x)$, not at a truncated logarithmic surrogate.

Lower-Lambert phase.

LEAN-PROVED

The lower-Lambert phase is the large positive coordinate obtained from the W_{-1} solution of the endpoint saddle equation. It converts the implicit balance between a small argument and a logarithm into a monotone parameter tending to infinity. Fabius endpoint expansions are organized in inverse powers of this phase, with a separate period-one fluctuation.

Lower-order term.

CLASSICAL

A lower-order term is asymptotically smaller than the leading term in a specified scale. The phrase is meaningful only after the comparison variable and mode of comparison are fixed. A bounded periodic term is lower order than a quadratic logarithm but is not lower order than an $o(1)$ remainder in the exponent. The Fabius correction illustrates why informal claims that a term is negligible must be matched to the claimed error scale.

Lucas's theorem.

CLASSICAL

Let p be prime and write $n = \sum n_j p^j$ and $k = \sum k_j p^j$. Lucas's theorem states

$$\binom{n}{k} \equiv \prod_j \binom{n_j}{k_j} \pmod{p}.$$

It detects binomial coefficients nonzero modulo p digit by digit. The repository uses the binary specialization together with carry theorems and q -Lucas analogues to analyze Pascal rows, parity masks, cyclotomic specializations, and Thue–Morse structure.

13 M

Maclaurin series.

CLASSICAL

A Maclaurin series is a Taylor series centered at zero. When the function is analytic there,

$$f(z) = \sum_{n \geq 0} \frac{f^{(n)}(0)}{n!} z^n.$$

Moment-generating and Fourier transforms of compactly supported laws have entire Maclaurin series whose coefficients are moments. By contrast, the bounded Fabius function itself has the zero Taylor series at the flat endpoint 0 but is positive immediately to the right, a standard demonstration that smoothness does not imply analyticity.

Mahler equation.

PROJECT / MIXED

A Mahler equation relates a function at z to its values at $z^q, z^{q^2}, \dots$, for example $\sum_{j=0}^r p_j(z) f(z^{q^j}) = 0$. Generating functions of automatic and regular sequences often satisfy such equations. The reciprocal-Gamma and geometric-product parts of the current tree also contain dilation equations of Mahler type.

Mahler function.

PROJECT / MIXED

A Mahler function is an analytic or formal function satisfying a Mahler equation for an integer dilation $q \geq 2$. Its arithmetic values, transcendence, and singularities are constrained by the functional equation. The Fabius corpus invokes this viewpoint for digit sequences, q -products, and scale-recursive transforms, while distinguishing formal and analytic solutions.

Major arc.

CLASSICAL

In a contour or oscillatory integral, a major arc is the neighborhood of the dominant saddle or stationary point where the main contribution is extracted. Its width is chosen to make a local Taylor expansion accurate while retaining nearly all Gaussian mass. The complement is the minor arc or tail. Fabius saddle modules prove separate central estimates on the major arc and exponential or comparison bounds outside it, then balance the chosen radius to obtain a uniform remainder.

Mass expansion.

CLASSICAL

A mass expansion is the asymptotic series for the normalized integral remaining after the leading exponential value and Gaussian scale have been factored out. It typically begins $1 + a_1/\lambda + a_2/\lambda^2 + \dots$. The coefficients combine phase and amplitude jets and Gaussian moments. Taking the formal logarithm of this series gives the additive coefficients in $\log F(x)$. The repository provides both recurrences and explicit ordered-composition formulas for these saddle mass coefficients.

Mathematical interface theorem.

PROJECT / MIXED

An interface theorem translates between representations without exposing the internal construction of either side. Typical interfaces identify a rational recurrence with a real integral, a formal coefficient with an analytic transform derivative, or a distributional fixed point with a CDF equation. Such theorems stabilize downstream work: later modules can use the mathematical contract even if implementation details change.

Matrix dilation.

FRONTIER

For an expansive real matrix A , the dilation $D_A f(x) = f(Ax)$ scales Fourier variables by A^{-T} and volume by $|\det A|^{-1}$. A matrix refinement equation combines this dilation with lattice translates. Scalar dyadic theorems do not automatically extend because anisotropy and lattice geometry enter.

Mean value theorem.

CLASSICAL

The mean value theorem states that a continuous function on $[a, b]$, differentiable on (a, b) , satisfies $f(b) - f(a) = f'(c)(b - a)$ for some c . Consequently $|f(b) - f(a)| \leq \sup |f'| |b - a|$. Iterating such estimates with the exact derivative scaling yields one form of effective Fabius flatness. It also proves Lipschitz continuity and turns derivative maxima into sharp global constants.

Measure with density.

LEAN-PROVED

Given a measure μ and a nonnegative measurable function f , the measure $f\mu$, written $\mu.\text{withDensity}(f)$ in Lean, satisfies $\int g \, d(f\mu) = \int gf \, d\mu$ under the standard extended-integral conventions. The geometric-uniform modules identify their probability law with Lebesgue measure weighted by the explicit density.

Measure-zero set.

CLASSICAL

A set has Lebesgue measure zero if it can be covered by intervals of arbitrarily small total length. Statements holding outside such a set hold almost everywhere. Countable sets, including the dyadic rationals, have measure zero even when dense. In analytic-locus discussions this distinction matters: exceptional behavior on every dyadic point may be topologically dense yet measure-theoretically negligible. The Fabius nowhere-analytic result is stronger than an almost-everywhere statement.

Mellin finite part.

CLASSICAL

A Mellin finite part is the constant term obtained when a Mellin transform has a pole at the point of interest. One subtracts the polar Laurent terms or equivalently regularizes the original endpoint integral. In the Fabius negative-Laplace decomposition, the Mellin origin carries the polynomial-logarithmic main terms, while shifted imaginary poles generate the periodic correction. Correct extraction of the finite part fixes the endpoint constant and the centering of the periodic function.

Mellin inversion.

CLASSICAL

Mellin inversion reconstructs a function from

$$M(s) = \int_0^\infty f(x)x^{s-1} \, dx$$

by

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} M(s)x^{-s} ds,$$

under suitable hypotheses. Shifting the contour across poles produces asymptotic expansions. Periodically spaced imaginary poles yield Fourier series in $\log x$. This is the conceptual source of the Gamma–zeta periodic correction in the Fabius Laplace logarithm.

Mellin-pole asymptotics.

PROJECT / MIXED

In Mellin inversion, a pole of the transformed integrand contributes a power or power–logarithm term obtained from its residue. A vertical lattice of poles with equal real part contributes a periodic function of $\log x$. This is the mechanism by which dyadic harmonic sums produce log-periodic Fabius corrections.

Mellin strip.

CLASSICAL

The fundamental strip of a Mellin transform $\int_0^\infty f(x)x^{s-1} dx$ is the vertical strip in which the integral converges absolutely. Meromorphic continuation beyond the strip is a separate theorem. Inversion contours begin inside a valid strip before they are shifted across poles.

Mellin transform.

CLASSICAL

The Mellin transform is $M_f(s) = \int_0^\infty f(x)x^{s-1} dx$. It converts multiplicative scaling into multiplication by powers and is the Fourier transform in logarithmic coordinates. Harmonic sums over dyadic scales become products with $(1 - 2^{-s})^{-1}$, whose imaginary lattice of poles creates one-periodic functions. The Fabius endpoint analysis applies Mellin methods to a regularized Bose-type kernel derived from the negative-Laplace product.

Meromorphic continuation.

CLASSICAL

Meromorphic continuation extends a function beyond its initial domain while allowing isolated poles. Gamma and zeta functions are standard examples. Mellin transforms often converge only in a vertical strip but continue meromorphically through functional identities or subtraction of endpoint asymptotics. In the Fabius analysis, locating the poles and computing their residues is more important than merely knowing continuation exists, because the pole at zero and the imaginary dyadic lattice determine the main and periodic terms.

Mersenne factor.

CLASSICAL

A Mersenne factor is an integer of the form $2^m - 1$. Products of odd Mersenne factors arise when the half-base q -Pochhammer symbol $(1/2; 1/2)_n = \prod_{j=1}^n (1 - 2^{-j})$ is cleared of powers of two. These products appear in moment denominators, Gaussian binomial normalizations, and Reshetnikov integers. Their oddness makes them especially convenient for separating 2-adic valuations from odd divisibility.

Minimum mean-square error.

FRONTIER

The MMSE of estimating X from Y is

$$\text{mmse}(X|Y) = \mathbb{E}[(X - \mathbb{E}[X|Y])^2].$$

The conditional expectation is the unique L^2 optimizer. Restricting the estimator to a finite linear span gives a larger least-squares projection error, which must be distinguished from the unrestricted MMSE.

Minkowski sum of supports.

CLASSICAL

For subsets $A, B \subset \mathbb{R}$, their Minkowski sum is $A + B = \{a + b : a \in A, b \in B\}$. The support of a convolution satisfies $\text{supp}(f * g) \subseteq \text{supp } f + \text{supp } g$, with equality under positivity and mild nondegeneracy. Finite sums of scaled uniform variables therefore have support equal to the sum of their interval supports. This calculation fixes the centering and endpoint locations of the spline approximants.

Minor arc.

CLASSICAL

The minor arc is the portion of an oscillatory or contour integral outside the central saddle neighborhood. A successful saddle-point proof shows that its contribution is smaller than the target remainder, often by a strict loss in the real part of the phase or by transform decay. The Fabius development treats central, tail, and reference-weight estimates in separate modules, preventing a local Gaussian expansion from being mistaken for a complete contour proof.

Minor-arc bound.

LEAN-PROVED

A minor-arc bound shows that oscillatory or remote parts of a contour are smaller than the main saddle contribution, often exponentially so. It uses nonresonance, product decay, or a quantitative phase gap. The terminology is analogous to the circle method but here applies to the Bromwich/Fourier localization.

Möbius function of a subspace lattice.

CLASSICAL

If $U \subseteq Z$ and $d = \dim_{\mathbb{F}_q}(Z/U)$, then the subspace-lattice Möbius function is

$$\mu(U, Z) = (-1)^d q^{\binom{d}{2}}.$$

This is the q -analogue of the Boolean-lattice sign and is central to finite-field inversion formulas.

Möbius function of the partition lattice.

CLASSICAL

For partitions $\pi \leq \sigma$, the partition-lattice Möbius function is the incidence-algebra inverse of the constant-one zeta function. In particular, $\mu(\hat{0}, \hat{1}) = (-1)^{n-1} (n-1)!$. Möbius inversion converts moments to cumulants and connected combinatorial structures to arbitrary assemblies.

Mod-Gaussian convergence.

FRONTIER

Mod-Gaussian convergence factors a characteristic or moment-generating function into a diverging Gaussian term and a residual analytic limit. It refines the central limit theorem and can yield moderate deviations and local limits. The normalization and domain of locally uniform convergence are part of the definition.

Moderate-deviation regime.

FRONTIER

Moderate deviations lie between central-limit fluctuations and full large-deviation scale. The deviation size grows relative to the standard deviation but remains smaller than the macroscopic scale. Cumulant bounds or mod-Gaussian convergence can provide asymptotics there; neither a fixed-order Edgeworth expansion nor an LDP alone covers the entire intermediate range.

Module docstring.

CONVENTION

A module docstring is the opening documentation block that states a Lean file's purpose, conventions, imported prerequisites, and theorem boundaries. In a mathematically dense repository it functions as a local contract: it should identify normalization, endpoint behavior, parameter hypotheses, and known limitations before individual declarations are read. Documentation audits test the presence of these headers separately from theorem comments.

Modulus of continuity.

CLASSICAL

A modulus of continuity is a nondecreasing function $\omega(\delta) \rightarrow 0$ such that $|f(x) - f(y)| \leq \omega(|x - y|)$. A Lipschitz function has $\omega(\delta) = L\delta$. An effective modulus must be computable in a specified representation. For the Fabius function, the exact least Lipschitz constant gives both a sharp analytic modulus and a simple recursive modulus used in computability.

Moment.

CLASSICAL

The n -th moment of a random variable or finite measure is $\mathbb{E}[X^n]$ or $\int x^n \mu(dx)$, when finite. Moments encode location, scale, and shape; for compactly supported laws all moments exist and determine the law. The Fabius development uses full moments of the centered up-density and half moments on one side of the support. Their triangular recurrences and generating functions control exact dyadic values and Legendre coefficients.

Moment–cumulant transform.

LEAN-PROVED

Moments and cumulants are related by set partitions:

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}.$$

The inverse uses the Möbius function of the partition lattice. Complete exponential Bell polynomials encode both directions. Current moment, Appell, and standardized-law modules use these transforms with exact rational coefficients.

Moment functional.

LEAN-PROVED

A moment functional is a linear map $\mathcal{M}$ on polynomials determined by $\mathcal{M}(x^n) = m_n$. When $m_n = \int x^n d\mu(x)$, it is integration against a measure, but the algebraic definition does not require a measure. The Rvachev moment functional permits exact rational construction of orthogonal polynomials, Legendre coefficients, and Appell deconvolution before casting to real integrals.

Moment-generating function.

CLASSICAL

The moment-generating function is $M_X(t) = \mathbb{E}(e^{tX})$, defined near zero when the exponential moment exists. Its Taylor coefficients are moments. For bounded X , it is entire. Independence turns sums into products, while the logarithm gives cumulants. The Fabius random series therefore has a product MGF whose imaginary restriction is the sinc characteristic function and whose negative-real restriction drives endpoint analysis.

Moment Laurent expansion.

LEAN-PROVED

For compactly supported μ ,

$$G_\mu(z) = \sum_{n \geq 0} \frac{m_n}{z^{n+1}}$$

for $|z|$ larger than the support radius, where $m_n = \int x^n d\mu(x)$. The expansion is a convergent Laurent series, not merely formal. It transfers exact moment recurrences to the resolvent.

Moment numerator.

PROJECT / MIXED

A moment numerator is an integer sequence obtained by multiplying a rational moment by a canonical denominator built from factorials, powers of two, and Mersenne factors. In the repository the full-moment numerators and half-moment numerators are distinct sequences, often denoted F_n and G_n in the arithmetic paper's conventions. Their integer recurrences expose parity, positivity, and divisibility information hidden in the raw rational moments.

Moment numerator F_n .

PROJECT / MIXED

The arithmetic sequence F_n is a canonical integer numerator obtained from the full moments after clearing their prescribed denominator. It is distinct from the function $F(x)$, despite sharing the letter in source conventions. Its triangular recurrence and divisibility properties organize exact moment arithmetic. The glossary uses the phrase “moment numerator F_n ” whenever ambiguity with the bounded Fabius function is possible.

Moment recurrence.

CLASSICAL

A moment recurrence expresses a moment of order n in terms of lower moments, often by expanding a distributional fixed-point equation. If X has the same law as a scaled combination of X and an independent elementary variable, applying $\mathbb{E}[(\cdot)^n]$ and the binomial theorem yields a triangular relation. The Fabius full and half moment recurrences are rational and exact, and after normalization become division-free integer recurrences.

Moment sequence.

CLASSICAL

A moment sequence is a sequence $m_n = \int x^n \mu(dx)$ associated with a measure. Not every sequence is a moment sequence; positivity constraints and growth conditions are involved. Compact support makes the moment problem determinate. The repository's sequence c_n records even centered moments, while d_n records normalized half moments. Their arithmetic normalizations are used throughout the dyadic and Legendre layers.

Monic orthogonal polynomial.

LEAN-PROVED

The monic orthogonal polynomial π_n has degree n , leading coefficient one, and is orthogonal to every polynomial of degree $< n$ under the chosen moment functional. Positive definiteness makes it unique. Determinant formulas and Gram–Schmidt construct it entirely from $m_0, \dots, m_{2n}$.

Monodromy.

FRONTIER

Monodromy is the permutation of analytic branches produced by continuation around a singular value. For an algebraic critical point it cycles Puiseux branches; for logarithmic inverses it may add multiples of $2\pi i$. A branch label is therefore incomplete without a base point and continuation domain.

Monotone convergence theorem.

CLASSICAL

The monotone convergence theorem states that if $0 \leq f_n \uparrow f$ pointwise, then $\int f_n \uparrow \int f$, allowing the value $+\infty$. It is useful for increasing approximations by positive densities or distribution functions. Fabius finite probability constructions often allow stronger dominated or uniform convergence, but monotone convergence remains a foundational tool for nonnegative integral limits.

Monotone function.

CLASSICAL

A function is nondecreasing when $x < y$ implies $f(x) \leq f(y)$, and strictly increasing when the inequality is strict. Every cumulative distribution function is nondecreasing. The bounded Fabius function is strictly increasing on $[0, 1]$, because its density is positive in the interior, but it is constant on the exterior tails. The up-function is not globally monotone; it increases to the origin and then decreases.

Monotone order isomorphism.

LEAN-PROVED

A monotone order isomorphism is a bijection preserving and reflecting order, with a monotone inverse. The restriction of the canonical Fabius function to $[0, 1]$ has this property. Order-theoretic inverse lemmas yield continuity, endpoint values, and interval mapping without differentiability, and they remain valid in settings where analytic inverse-function theorems are inapplicable.

Multiple q -Pochhammer notation.

CONVENTION

A compact product such as

$$(a_1, \dots, a_r; q)_n = \prod_{\ell=1}^r (a_\ell; q)_n$$

is multiple q -Pochhammer notation. It abbreviates a product of independent factors and introduces no multivariate coupling. The base q and length n are shared; each upper parameter retains its own zero set.

Multiplicative periodic fluctuation.

CLASSICAL

A multiplicative periodic fluctuation is a bounded factor whose value depends periodically on a logarithmic phase, typically $\exp(\Psi(\lambda(x)))$. In logarithmic asymptotics it appears additively as Ψ . Because it need not approach a limit, it can prevent a ratio from converging even when all coarse logarithmic orders agree. The corrected Fabius endpoint formula contains exactly such a factor, with very small but explicitly nonzero Fourier modes.

Multiresolution analysis.

FRONTIER

A multiresolution analysis is a nested sequence of closed spaces V_j compatible with dilation and translation, whose union is dense and intersection is trivial. Orthogonal or biorthogonal detail spaces describe the increments. Finite nested interpolation spaces are a discrete analogue but do not constitute an infinite MRA until density and stability are proved.

Multiresolution detail operator.

LEAN-PROVED

For nested finite interpolation projectors P_{q-1}, P_q , the detail operator is $\Delta_q = P_q - P_{q-1}$, with $\Delta_0 = P_0$. Under compatible nesting these details are idempotent on the terminal coefficient space and mutually annihilating. Their ranks equal the increments in polynomial dimension.

Mutual information.

FRONTIER

Mutual information is

$$I(X; Y) = D_{\text{KL}}(\mathcal{L}(X, Y) \| \mathcal{L}(X) \otimes \mathcal{L}(Y)).$$

It measures dependence and is invariant under measurable bijections of either variable. Exact geometric-digit decompositions can turn it into a finite sum of scale contributions.

14 N

Namespace.

CONVENTION

A namespace groups related Lean declarations and prevents name collisions. The mathematical namespace `Fabius` contains functions, arithmetic sequences, transform objects, and approximation operators that would otherwise have generic names. Opening a namespace shortens notation locally but does not change a declaration's full identity; crosswalks use sufficiently complete names to remain unambiguous.

Negative-index Bell value.

FRONTIER

Negative-index Bell values in the corpus are defined by an explicit absolutely convergent Roman-type continuation, not by counting partitions of a negative set. A representative formula expresses B_{-k} through negative-index first-kind Stirling values or $e^{-1} \sum_{j \geq 1} (j^k j!)^{-1}$. The definition must precede any use of the familiar symbol B_n at negative n .

Negative-Laplace decomposition.

PROJECT / MIXED

The negative-Laplace decomposition analyzes $\log M(-s)$ for large $s > 0$, where M is the Fabius moment-generating function. An exact dyadic product is converted into a harmonic sum of the kernel $\kappa(u) = \log((1 - e^{-u})/u)$. Mellin analysis then separates explicit polynomial-logarithmic terms, a constant, a smooth one-periodic correction, and a decaying remainder. This decomposition is the global input for the Bromwich saddle method.

Negative-Laplace product.

PROJECT / MIXED

The negative-Laplace product is the exact factorization of the Fabius transform on the negative real axis, for example $G(-s) = \prod_{j \geq 1} (1 - e^{-s/2^j})/(s/2^j)$. Each factor is the Laplace transform of a uniform summand after normalization. The product is positive for $s > 0$, so its real logarithm is unambiguous. Its dyadic scale structure produces the Mellin denominator $1 - 2^{-z}$ and hence the periodic correction.

Newton identities.

CLASSICAL

Newton identities relate power sums to elementary or complete homogeneous symmetric polynomials. For example, $me_m = \sum_{j=1}^m (-1)^{j-1} e_{m-j} p_j$. They convert product data into logarithmic coefficients and underlie moment, cumulant, and q -product recurrences.

Newton iteration.

CLASSICAL

Newton iteration solves $h(x) = 0$ by $x_{n+1} = x_n - h(x_n)/h'(x_n)$. Near a simple root it converges quadratically under standard hypotheses. Stable endpoint evaluation of inverse Fabius values or saddle coordinates can use Newton's method with monotone bracketing and exact asymptotic initial guesses. The lower-Lambert equation $\lambda 2^{-\lambda} = x$ may also be solved numerically this way when a special-function implementation is unavailable.

Newton series.

CLASSICAL

A Newton series expands a function on integers as

$$f(x) = \sum_{k \geq 0} \binom{x}{k} \Delta^k f(0),$$

formally or under analytic growth conditions. Finite sums give ordinary polynomial interpolation on consecutive nodes. The expansion is the discrete counterpart of a Taylor series and interacts directly with prefix summation.

Noise floor.

FRONTIER

The numerical noise floor is the error level below which rounding, cancellation, underflow, or implementation effects dominate the signal being measured. The Fabius periodic correction has very small amplitude, so ordinary floating-point evaluation can obscure it. Exact rational endpoint values, arbitrary precision, and subtraction of the full nonoscillatory main term are required before interpreting residual plots. A flat residual at the noise floor does not establish absence of a mathematical term.

Nome.

CONVENTION

A nome is the base q used in theta, elliptic, and modular products, typically with $0 < |q| < 1$. Some literature uses $q = e^{\pi i \tau}$, while other sources use $e^{2\pi i \tau}$. This factor-of-two choice changes every theta exponent and modular transformation, so the project declares it before using theta notation.

Non-elementarity of the Fabius function.

LEAN-PROVED

The non-elementarity theorem states that the canonical bounded Fabius function cannot be represented by the project's elementary-function class on the open unit interval, and consequently not globally in the corresponding sense. The proof is qualitative: elementary expressions have a dense analytic locus, whereas the Fabius function is nowhere analytic. It does not depend on failure to find a particular closed form.

Non-elementarity of the inverse Fabius function.

LEAN-PROVED

The inverse non-elementarity theorem applies the same analytic-locus obstruction to the canonical inverse on $(0, 1)$. The result is strengthened to the class obtained by adjoining permitted inverse branches of elementary functions. Endpoint singularity alone would not prove this statement; the key input is nonanalyticity at every interior point.

Nonconstant periodic correction.

FRONTIER

A nonconstant periodic correction is a periodic term with at least one nonzero Fourier mode. In the Fabius endpoint asymptotic, every relevant Gamma–zeta coefficient is explicit, and nonvanishing proves that the correction cannot be absorbed into a constant. Its numerical amplitude is extremely small because Gamma values decay rapidly on vertical lines, but asymptotic correctness depends on logical size relative to the stated remainder, not on visual prominence.

Noncrossing partition.

FRONTIER

A partition of $\{1, \dots, n\}$ is noncrossing if no four indices $a < b < c < d$ have a, c in one block and b, d in another. Noncrossing partitions form a Catalan-counted lattice and index free moment–cumulant relations.

Nondegenerate saddle.

CLASSICAL

A saddle z_0 is nondegenerate when $\Phi''(z_0) \neq 0$. The local exponent is then quadratic to first order, producing a Gaussian factor and a half-power scale in the large parameter. Degenerate saddles require different normal forms such as Airy or higher catastrophe integrals.

Normal convergence.

CLASSICAL

A series of functions converges normally on a domain if it converges absolutely and uniformly on every compact subset, usually by the Weierstrass M-test. Normal convergence of holomorphic functions permits termwise differentiation and yields a holomorphic sum. Infinite transform products and Fourier series of the smooth periodic correction are controlled in this manner. The term is stronger than pointwise convergence and is especially useful for complex-analytic families.

Normal convergence of an infinite product.

PROJECT / MIXED

An infinite product $\prod(1 + u_n(z))$ converges normally on a domain when $\sum |u_n(z)|$ converges uniformly on every compact subset after finitely many exceptional factors. The limit is holomorphic, and zeros arise from factor zeros unless the limit collapses identically. This controls sinc, q -Pochhammer, theta, and reciprocal-Gamma products in the repository.

Normal convergence of q -products.

LEAN-PROVED

A family of analytic products converges normally on a domain when it converges uniformly on every compact subset. For $|q| < 1$, the estimate $\sum_j |aq^j| < \infty$ locally uniformly in a gives normal convergence of $(a; q)_\infty$. Normal convergence implies holomorphy of the limit and permits termwise differentiation after the usual nonvanishing or logarithmic localization hypotheses.

Normalized reversion coefficient.

FRONTIER

For a delta series $f(z) = f_1 z + f_2 z^2/2! + \dots$, a normalized reversion input is typically $f_{j+1}/((j+1)f_1)$. This isolates the linear scale before Bell-polynomial or partition formulas are applied. The exact normalization is printed because ordinary-series conventions use different factorial factors.

Normalized reversion input.

CONVENTION

A normalized reversion input is

$$a_j^{\text{rev}} = \frac{f_{j+1}}{(j+1)f_1}, \quad j \geq 1,$$

for an exponential-series forward map f . These quantities enter compact Bell-polynomial inverse formulas. They are neither ordinary Taylor coefficients nor the inverse coefficients themselves.

Normalized saddle mass.

PROJECT / MIXED

The normalized saddle mass is what remains of a saddle integral after factoring out the exponential at the saddle, the leading amplitude, and the Gaussian normalization. It tends to one and has a formal expansion $1 + \sum_{j \geq 1} a_j \lambda^{-j}$. This normalization isolates universal Gaussian moments from problem-specific phase jets. Its logarithm supplies the additive all-orders corrections A_j in the Fabius endpoint formula.

Normalized sinc functions.

CONVENTION

The radian sinc is $\text{sinc}_{\text{rad}} z = \sin z/z$, and the normalized sinc is $\text{sinc}_{\pi} x = \sin(\pi x)/(\pi x)$, both with removable value 1 at zero. Their zeros occur at $\pi\mathbb{Z} \setminus \{0\}$ and $\mathbb{Z} \setminus \{0\}$, respectively. The notation catalogue forbids a bare ambiguous sinc in normative formulas.

Normalized Volterra operator.

LEAN-PROVED

A normalized Volterra operator includes the Gamma or factorial factor that makes its order parameter additive, such as

$$(I^{\alpha} f)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} f(t) \, dt.$$

This normalization gives a clean semigroup law and matches ordinary repeated integration at positive integers.

Normative notation contract.

CONVENTION

The normative notation contract is the shared LaTeX command file together with the mathematical catalogue that defines each command's semantics. It fixes function normalizations, Fourier conventions, branch choices, and collision rules across active documents. Changing a command's arity or meaning requires a contract-version update and synchronized catalogue entries; historical sources are handled by explicit exceptions rather than silent reuse.

Notation collision.

CONVENTION

A notation collision occurs when the same glyph can denote different objects in nearby topics—for example F as the bounded Fabius function, a numerator sequence, or a generic map; P_n as a Legendre polynomial, a probability, or a Padé denominator; and μ as a measure, mean, or Möbius function. The project resolves collisions with semantic commands, superscripts, or mandatory local declarations rather than relying on context alone.

Nowhere analytic.

CLASSICAL

A smooth function is nowhere analytic on a set if it is not real analytic at any point of that set. This is stronger than having isolated nonanalytic points. The Fabius function is nowhere analytic on its nonconstant interval: dyadic points are dense, their formal Taylor series terminate, and if the function were analytic near any point, analytic continuation and overlap with such a dyadic neighborhood would force local polynomial behavior incompatible with strict shape and scaling. The exterior constant tails remain analytic away from the endpoints.

Nowhere-analytic Fabius iterate.

FRONTIER

A positive Fabius iterate $F^{\circ n}$ is asserted in the dedicated frontier development to be nonanalytic at every point of $[0, 1]$. The proof architecture follows dyadic points through the chain rule and Faà di Bruno partitions, locating a finite first nonzero derivative that cannot be reconciled with an analytic germ. This status is distinct from the separately formalized nonanalyticity of F itself.

Nowhere-analytic inverse-Fabius iterate.

FRONTIER

The corresponding frontier theorem treats $(F^{-1})^{on}$ for every positive n . Monotone conjugacy and dyadic-image structure transport singular jets through successive inverse branches. Care is required near endpoints, where ordinary derivative formulas may diverge; the result concerns real analyticity, not mere failure of a bounded first derivative.

Null set.

CLASSICAL

A null set is a set of measure zero. Altering an integrable function on a null set does not change its Lebesgue integral or its L^p equivalence class, but it can change pointwise properties such as continuity or exact functional equations. Formalizations that reason with continuous representatives cannot silently replace functions almost everywhere. The Fabius theory usually works with canonical continuous or smooth versions rather than equivalence classes.

Numerator sequence.

CLASSICAL

A numerator sequence records the integer numerators obtained after multiplying rational quantities by a prescribed denominator. The choice of denominator is part of the definition and can be designed to expose recurrence, parity, or divisibility. The Fabius directory contains several such sequences for moments, half moments, and dyadic endpoint values. Similar names in different papers need a notation crosswalk because their normalizations are not interchangeable.

Numerical residual.

FRONTIER

A numerical residual is the discrepancy obtained by substituting a computed approximation into an equation or by subtracting a predicted approximation from trusted data. A small residual is evidence, not by itself a proof, because conditioning and cancellation matter. Exact rational dyadic values provide unusually strong reference data for testing the corrected Fabius asymptotic. Comparing residuals across scales reveals the periodic oscillation and the order of the first omitted saddle term.

15 O**Odd cosine series.**

CLASSICAL

An odd cosine series is a cosine expansion containing only odd harmonics $\cos((2m+1)\pi x)$. It arises when even-frequency Fourier coefficients vanish. For the up-function, zeros of the sinc product force this pattern in the chosen periodization. Despite the adjective odd, the resulting function is even in x ; odd refers to the harmonic indices, not parity of the function.

Odd discrete Fourier transform.

FRONTIER

An odd DFT restricts or twists the discrete Fourier transform to odd residue classes, often modulo a power of two. It isolates primitive dyadic frequencies and avoids modes annihilated by even symmetry. The exact node and normalization convention is report-specific.

Odd part of an integer.

CLASSICAL

For nonzero $n \in \mathbb{Z}$, the odd part is $n/2^{v_2(n)}$, an odd integer up to sign. It retains the unit information after dyadic scale is removed. Odd-part recurrences are often more stable than raw numerator recurrences and make parity statements immediate.

One-periodic function.

CLASSICAL

A function P is one-periodic when $P(u+1) = P(u)$ for all u . It is determined by its values on any interval of length one and has Fourier frequencies in $\mathbb{Z}$. The endpoint correction and every all-orders coefficient A_j in the Fabius logarithmic expansion are one-periodic functions of the lower-Lambert coordinate. Centering sets the mean of the leading correction to zero.

Operation count.

CLASSICAL

An operation count estimates the number of arithmetic or logical steps performed by an algorithm as a function of input size. For integer inputs, bit length and Hamming weight are more informative than numerical magnitude. The fast dyadic Fabius evaluator makes one main recursive reduction per set bit and evaluates finite Taylor blocks with bounded-degree arithmetic. Exact bit complexity also depends on growth of numerators and denominators, so a scalar operation count is only one layer of analysis.

Optimal truncation.

PROJECT / MIXED

For a divergent asymptotic series, optimal truncation stops near the least term. The resulting error is often exponentially smaller than any fixed algebraic order. The optimal index depends on the large parameter and on coefficient growth; it is not obtained by summing the formal series to infinity.

Order of vanishing.

CLASSICAL

A function has a zero of order m at x_0 if its first $m-1$ derivatives vanish there and the m -th derivative does not. Infinite-order vanishing is flatness. Finite Thue–Morse generating polynomials have high-order zeros at $z=1$, encoding Prouhet cancellation of low powers. At an interior dyadic point, the Fabius Taylor polynomial has exact highest degree determined by the first nonzero scaled integer value, not by a zero of the function itself.

Ordered composition.

CLASSICAL

An ordered composition of n is a sequence of positive integers summing to n . It indexes terms produced by expanding powers of a series or taking a formal logarithm. The explicit Fabius saddle mass coefficients sum over compositions of $2j$, with additional subset choices selecting phase or amplitude contributions; the logarithmic coefficients sum over compositions of j . Ordered indexing avoids symmetry factors but produces more terms than partition notation.

Ordered-composition coefficient.

PROJECT / MIXED

An ordered-composition coefficient sums products over tuples of positive integers with prescribed total. It is the coefficient form of powers of a series and provides a direct alternative to Bell-polynomial

notation. Ordered compositions are useful in formalization because every index range is finite and elementary.

Ordinary Bell polynomial.

CLASSICAL

An ordinary Bell polynomial uses ordinary rather than exponential generating-function normalization. Factorials therefore move between its arguments and coefficients relative to the exponential Bell family. The project names the normalization explicitly whenever determinant or reversion formulas could otherwise differ by products of factorials.

Ordinary generating function.

CLASSICAL

The ordinary generating function of (a_n) is $A(z) = \sum_{n \geq 0} a_n z^n$. Its product corresponds to ordinary discrete convolution. The finite Thue–Morse generating polynomial factors digitwise, and iterated prefix sums correspond to division by powers of $1 - z$ in a formal setting. Ordinary and exponential generating functions occur side by side in the repository and encode different convolution laws.

Original–bounded Fabius bridge.

LEAN-PROVED

The original–bounded bridge proves that the compact-support predicate for up, together with the appropriate right-tail condition, is equivalent to the bounded IsFabius predicate after folding. It reconciles the historical differential-difference definition with the modern CDF-style normalization and makes uniqueness transferable in both directions.

Original compact-support characterization.

LEAN-PROVED

The original characterization identifies the normalized compactly supported solution of

$$u'(x) = 2(u(2x + 1) - u(2x - 1))$$

together with the appropriate smoothness, support, and value at the origin. The formal development proves existence and uniqueness and then connects this solution to the folded bounded Fabius function. The characterization is global; a local differential identity alone is not sufficient for uniqueness.

Original Fabius predicate IsOriginalFabius.

LEAN-PROVED

The original compact-support predicate characterizes the Rvachev-style function directly: smoothness, compact support, normalization, and the differential-difference law are packaged without first mentioning the bounded CDF. Current bridge theorems convert between this predicate for up and the bounded predicate IsFabius, with the required tail condition stated separately. It is a proposition about a candidate function, not an additional canonical function.

Orthogonal polynomial.

CLASSICAL

A sequence (p_n) is orthogonal with respect to a measure μ if $\int p_n p_m d\mu = 0$ for $m \neq n$, with $\deg p_n = n$. Orthogonal polynomials satisfy three-term recurrences and provide spectral bases for approximation. Legendre polynomials are orthogonal for unit weight on $[-1, 1]$ and are used to expand the up-function. Orthogonality makes coefficient extraction and least-squares optimality immediate.

Orthogonal projection.

CLASSICAL

The orthogonal projection of a Hilbert-space vector f onto a closed subspace V is the unique $p \in V$ with $f - p$ orthogonal to V . It minimizes $\|f - p\|$. The degree- n Fourier-Legendre partial sum is the orthogonal projection onto polynomials of degree at most n . The corresponding Pythagorean identity quantifies exactly how the squared error decreases when another Legendre coefficient is included.

Orthogonality.

CLASSICAL

Vectors or functions are orthogonal when their inner product is zero. In $L^2[-1, 1]$, $\langle f, g \rangle = \int f \bar{g}$. Orthogonality eliminates cross terms in norm calculations and yields the Pythagorean identity. The Legendre basis separates the polynomial components of the up-function, while Fourier exponentials separate periodic modes of the endpoint correction.

Orthonormal polynomial.

CLASSICAL

An orthonormal polynomial p_n satisfies $\langle p_n, p_m \rangle = \delta_{nm}$ and has a declared positive leading coefficient. It differs from the monic polynomial by division by its norm. Legendre polynomials in the repository are usually classically normalized by $P_n(1) = 1$, so a factor $\sqrt{(2n+1)/2}$ is required for orthonormality on $[-1, 1]$.

Oscillatory correction.

CLASSICAL

An oscillatory correction varies rather than tending monotonically to a constant, often through sine, cosine, or a periodic function of a slowly varying phase. Its mean may be zero while its pointwise effect persists indefinitely. The Fabius correction is smooth and one-periodic in $\lambda(x)$, so sequences approaching the endpoint through different phases have different normalized limits. The term is asymptotically indispensable despite its tiny amplitude.

16 P **P -recursive sequence.**

FRONTIER

A sequence (a_n) is P -recursive when

$$\sum_{j=0}^r p_j(n) a_{n+j} = 0$$

for polynomial coefficients p_j , eventually in n . Its ordinary generating function is D -finite in characteristic zero. Failure of a guessed recurrence at finite order is evidence, not a proof of non- P -recursiveness.

Padé approximant.

PROJECT / MIXED

A Padé approximant $[m/n]$ to a series is a rational function P_m/Q_n whose Taylor expansion agrees through the maximal prescribed order, with $Q_n(0) = 1$. For Stieltjes transforms, diagonal and near-diagonal Padé approximants inherit positivity and interlacing properties. The project uses moment determinants and continued fractions to organize certified rational approximations.

Paley–Wiener theorem.

CLASSICAL

Paley–Wiener theorems relate compact support to holomorphic growth of a Fourier transform. For a smooth function supported in $[-R, R]$, the transform extends to an entire function bounded by rapidly decaying powers on horizontal directions times $e^{2\pi R|\operatorname{Im} z|}$ under the 2π -frequency convention. Conversely, suitable entire functions of this exponential type arise from compactly supported data. This framework explains the simultaneous entireness and rapid real-axis decay of the up transform.

Parity.

CLASSICAL

Parity distinguishes even integers from odd integers. It is additive modulo two and governs the Thue–Morse sign through the parity of binary digit sum. Parity assertions about normalized Fabius moment numerators are stronger than nonvanishing and determine exact 2-adic valuations. In finite q -binomial sums, parity also controls sign patterns and even-odd term separation.

Parity-power series.

CLASSICAL

The parity-power series is a signed global Fabius representation whose summands are organized by binary parity data and powers arising from finite Taylor blocks. A corrected version in the repository includes the precise zeroth term and domain convention needed for global validity. It converges absolutely. The name is local to the development and should be understood through its displayed summands rather than as a standard class of power series.

Parseval identity.

CLASSICAL

Parseval’s identity equates the L^2 norm of a function with the sum of squared coefficients in a complete orthonormal basis. For Fourier series it is $\|f\|_2^2 = \sum |c_k|^2$; for Legendre series the norm factors reflect polynomial normalization. It justifies energy estimates and coefficient square summability. In the Fabius context it complements the stronger absolute-uniform convergence results for the up-function expansion.

Partial exponential Bell polynomial.

CLASSICAL

The partial exponential Bell polynomial $B_{n,k}(x_1, x_2, \dots)$ sums over set partitions of total size n with k blocks, weighting a block of size j by x_j . It is the coefficient engine in Faà di Bruno’s formula and in fixed-block moment–cumulant transforms.

Partition defect.

FRONTIER

A partition defect measures how far an index tuple falls short of a target weight, length, or admissibility condition. Organizing a recurrence by defect produces shells whose first nonzero layer often determines the leading coefficient. The term is used in frontier coefficient and derivative expansions rather than as a single universal invariant.

Partition lattice.

CLASSICAL

The partition lattice Π_n orders set partitions of $\{1, \dots, n\}$ by refinement. Its minimum is the discrete singleton partition and its maximum is the one-block partition. Intervals factor over blocks, and its Möbius function inverts the moment–cumulant relation.

Partition of unity.

CLASSICAL

A partition of unity is a family of nonnegative functions whose sum equals one, often locally finite and subordinate to a covering. Translates of the up-function satisfy exact partition identities because the Fourier transform vanishes at nonzero integers under an appropriate normalization. Scaled versions yield further partitions. These identities are analytically analogous to cardinal spline partitions and are useful in approximation and sampling.

Pascal row modulo a prime power.

FRONTIER

A Pascal row modulo p^r is the sequence $((\binom{n}{k} \bmod p^r)_{0 \leq k \leq n})$. Its zero pattern is controlled by carry counts and higher congruence data, not Lucas's theorem alone. Histograms and recursive block decompositions connect these rows to automatic sequences and Fabius denominator questions.

Peano kernel.

PROJECT / MIXED

If a linear functional annihilates polynomials below degree m , Peano's theorem may represent its error as

$$L(f) = \int K(t) f^{(m)}(t) dt.$$

The kernel K determines sign, sharp constants, and extremizers. Repeated integration and spline approximation in the project use positive Peano kernels to obtain certified convergence rather than only formal Taylor cancellation.

Periodic correction.

PROJECT / MIXED

A periodic correction is a periodic function added to an otherwise smooth asymptotic main term to account for discrete-scale effects. In the Fabius endpoint theory it arises from the nonzero imaginary poles of $(1 - 2^{-s})^{-1}$ in Mellin space. Its centered form has zero mean and explicit Gamma-zeta Fourier coefficients. Uniqueness means that no different continuous one-periodic function can satisfy the same sharp remainder estimate.

Periodic-correction Fourier coefficient.

LEAN-PROVED

For a one-periodic correction Ψ , its Fourier coefficients are

$$\widehat{\Psi}(k) = \int_0^1 \Psi(u) e^{-2\pi i k u} du.$$

The Fabius Mellin analysis expresses nonzero modes through explicit Gamma-zeta values. Centering sets the $k = 0$ coefficient to zero and moves its mean into the nonoscillatory constant.

Periodic correction Psi.

PROJECT / MIXED

The symbol Ψ denotes the centered smooth one-periodic function in the corrected Fabius endpoint expansion. Its mean over one period is zero, and its nonzero Fourier coefficients are explicit Gamma-zeta values. It is evaluated at the exact lower-Lambert coordinate $\lambda(x)$. The centering convention moves the zero Fourier mode into the endpoint constant, making Ψ unique under the stated sharp remainder.

Periodic mean.

CLASSICAL

The mean of a one-periodic integrable function P is $\int_0^1 P(u) \, du$. It is the zero-frequency Fourier coefficient. Subtracting it centers the function; adding it to the constant term keeps the full expression unchanged. Repository modules compute the mean of the raw Mellin periodic term so that the endpoint constant and centered oscillation have a canonical normalization.

Periodicity.

CONVENTION

A function has period $T > 0$ if $f(x + T) = f(x)$. The smallest positive period, when it exists, is the fundamental period. Periodicity can arise from translation symmetry or from multiplicative scaling after a logarithmic change of variables. The Fabius asymptotic uses period one in the lower-Lambert coordinate, corresponding to the base-two structure of the original equation.

Periodization.

CLASSICAL

The periodization of f with period h is $\sum_{k \in \mathbb{Z}} f(x + kh)$ when the sum is meaningful. Compact support makes it locally finite. Poisson summation expresses its Fourier coefficients as samples of $\hat{f}$. Integer zeros of the up transform make several periodizations exactly constant.

Perron–Frobenius transfer operator.

FRONTIER

A transfer operator $\mathcal{P}_T$ is dual to composition:

$$\int (\mathcal{P}_T f) g \, d\mu = \int f(g \circ T) \, d\mu.$$

When T is nonsingular it transports densities through inverse branches and Jacobian factors. It is not simply composition by T^{-1} , especially when T is noninjective.

Perturbation expansion.

CLASSICAL

A perturbation expansion expresses a solution or integral as a formal or asymptotic series in a small parameter. One expands the phase, amplitude, saddle location, or recurrence around a limiting model and solves coefficient equations successively. The inverse-saddle parameter $1/\lambda$ is the perturbation parameter in the Fabius all-orders analysis. Formal coefficient algebra and analytic remainder estimates are separate components of a valid perturbation proof.

Phase function.

CLASSICAL

The phase function is the exponent controlling an oscillatory or Laplace-type integral, commonly $\Phi(z)$ in $\int A(z) e^{\lambda \Phi(z)} \, dz$. Saddles satisfy $\Phi'(z) = 0$; the real part of Φ determines concentration and decay. In the Fabius Bromwich integral the phase incorporates the linear inversion term and the logarithm of the negative-Laplace product. Its exact saddle equation leads to the lower Lambert coordinate.

Physicists' Hermite polynomial.

CONVENTION

The physicists' Hermite polynomial is $H_n^{\text{phys}}(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$. It differs from the probabilists' normalization by $H_n^{\text{phys}}(x) = 2^{n/2} \text{He}_n(\sqrt{2}x)$. The two families must not share an undecorated H_n in probability asymptotics.

Piecewise polynomial.

CLASSICAL

A piecewise-polynomial function agrees with a polynomial on each interval of a finite or locally finite partition. Splines impose smoothness conditions at the knots. Finite convolutions of uniform densities are piecewise polynomial, and their degrees increase with the number of factors. The centered Fabius approximants are therefore exactly evaluable on each dyadic cell, which is crucial for primitive-recursive computation.

Plancherel theorem.

CLASSICAL

Plancherel's theorem extends the Fourier transform to a unitary map on L^2 , up to the selected normalization. It states equality of spatial and frequency energies and turns convolution into multiplication where defined. The theorem underlies exact up-function energy identities and L^2 approximation estimates.

Pochhammer symbol.

CLASSICAL

The ordinary Pochhammer symbol $(a)_n = a(a+1) \cdots (a+n-1)$ is the rising factorial, with $(a)_0 = 1$. It appears in hypergeometric series and Gamma ratios: $(a)_n = \Gamma(a+n)/\Gamma(a)$. The q -Pochhammer symbol is a different multiplicative object $(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j)$. The Fabius q -calculus mainly uses the latter, so the base parameter should always be displayed.

Poincaré asymptotic expansion.

CLASSICAL

For an asymptotic scale $\phi_0 \gg \phi_1 \gg \cdots$,

$$f(x) \sim \sum_{n \geq 0} a_n \phi_n(x)$$

means $f - \sum_{n < N} a_n \phi_n = o(\phi_{N-1})$ for every fixed N . The statement is termwise and finite at each order. It neither promises convergence nor specifies behavior when the truncation index grows with x .

Pointwise convergence.

CLASSICAL

A sequence f_n converges pointwise to f if $f_n(x) \rightarrow f(x)$ for each fixed x . The index after which an approximation is accurate may depend on x , unlike uniform convergence. Pointwise convergence alone does not preserve continuity or justify interchange with integration. The repository's stronger spline and Legendre results supply uniform convergence, while some discrete-limit formulas are first established pointwise and then strengthened through bounded rows or compactness.

Pointwise equality.

CLASSICAL

Two functions are pointwise equal when $f(x) = g(x)$ for every input in the declared domain. This is stronger than equality almost everywhere and differs from equality of distributions of random variables. Functional equations, support statements for continuous representatives, and exact dyadic evaluations in the Fabius development are pointwise unless a theorem explicitly uses a measure-theoretic equality.

Poisson summation.

CLASSICAL

Poisson summation relates samples of a function to samples of its Fourier transform: under suitable decay, $\sum_{n \in \mathbb{Z}} f(x + n) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{2\pi i k x}$. If all nonzero integer transform samples vanish, the periodized sum is constant. Applying this to the up-function's sinc product yields exact partitions of unity and sampling identities. Smooth compact support supplies the regularity needed for classical justification.

Pole.

CLASSICAL

A pole of order m at z_0 is an isolated singularity where $(z - z_0)^m f(z)$ extends holomorphically and nonvanishingly. The Laurent series has finitely many negative powers. Mellin contour shifts collect residues at poles; a real pole generates polynomial-logarithmic main terms, while imaginary translates generate periodic modes. The dyadic factor $(1 - 2^{-s})^{-1}$ creates the relevant arithmetic lattice of poles.

Polygamma function.

PROJECT / MIXED

The order- m polygamma function is $\psi^{(m)}(z) = \frac{d^{m+1}}{dz^{m+1}} \log \Gamma(z)$. For $m \geq 1$ and away from poles,

$$\psi^{(m)}(z) = (-1)^{m+1} m! \sum_{k \geq 0} (z + k)^{-m-1}.$$

These functions provide the derivative jets of gamma ratios and their asymptotic coefficients.

Polynomial approximation.

CLASSICAL

Polynomial approximation seeks polynomials close to a function in a chosen norm. Weierstrass' theorem gives uniform approximation for continuous functions on compact intervals, while orthogonal projections give least-squares approximants. The up-function's Fourier-Legendre series provides explicit polynomial approximants with exact coefficients and proven absolute uniform convergence. This is distinct from the finite spline approximation, whose pieces are polynomial but whose global approximant is not a single polynomial.

Polynomial cell.

LEAN-PROVED

A polynomial cell is a connected interval between consecutive knots on which a spline or finite sinc-prefix inverse transform is represented by one fixed polynomial. Cell boundaries depend on the dilation and translate parameters. Local synthesis arguments choose atoms so the target interval stays within specified cells, avoiding hidden changes of formula.

Polynomial commutator with integration.

LEAN-PROVED

For multiplication by x and an integration operator I , the commutator $xI - Ix$ is another integral operator with an extra kernel factor. Iterating this relation moves polynomials across Volterra or fractional integrals and yields finite binomial identities. Such commutators organize moment recurrences and polynomial reproduction formulas.

Polynomial deconvolution by the up-law.

LEAN-PROVED

Polynomial deconvolution assigns to a polynomial P the unique polynomial $D(P)$ satisfying

$$\int_{\mathbb{R}} D(P)(x - t) \operatorname{up}(t) \, dt = P(x).$$

Moment Appell polynomials make this map explicit. The formal operator is a linear, injective, degree-preserving map with computable rational coefficients for rational input.

Polynomial diagonal stabilization.

FRONTIER

Polynomial diagonal stabilization is the phenomenon that, for fixed offset from a natural boundary, entries of an iterated Thue–Morse summation array eventually agree with a polynomial in the iteration order. Degree, factorization, and denominator patterns reflect the binomial prefix kernel. A rigorous statement specifies the exact index region and whether the formula is an identity or an experimentally inferred continuation.

Polynomial identity over a ring.

CLASSICAL

A polynomial identity over a ring means equality in $R[x]$, hence equality of all coefficients. It is stronger than agreement at finitely many points unless an interpolation theorem applies, and it remains valid under every ring homomorphism. Gaussian, cyclotomic, q -Catalan, and quantum-plane results often live first in a polynomial or noncommutative ring before evaluation at a numeric base.

Polynomial–logarithmic transseries.

PROJECT / MIXED

A polynomial–logarithmic transseries at infinity is an expansion

$$X + \sum_{n \geq 0} X^{-n} p_n(\log X),$$

or a related Laurent form, with polynomial coefficient blocks. Lambert W , Wright’s omega function, inverse gamma-type functions, and logarithmic series reversions live in this class. Multiplication, division, composition, and reversion are coefficientwise finite under a triangular order.

Polynomial reproduction.

CLASSICAL

An approximation or summation scheme reproduces polynomials up to degree m if it returns every such polynomial exactly. Finite-difference annihilation and partition-of-unity identities are dual manifestations of reproduction. Gaussian rows on geometric nodes reproduce the leading-coefficient functional after annihilating lower degrees. In spline and Toeplitz limits, exact mass and low-moment conditions control the approximation order.

Polynomial reproduction order.

LEAN-PROVED

The reproduction order of a mesh-generator pair is the largest d for which every polynomial of degree at most d is reconstructed exactly from the prescribed translate sum. In the current Rvachev comb classification, a nonzero integer mesh M reproduces exactly through degree $v_2(M)$ in the universal shifted sense.

Polynomial sequence of binomial type.

CLASSICAL

A polynomial sequence $p_n(x)$ is of binomial type when $p_n(x+y) = \sum_k \binom{n}{k} p_k(x) p_{n-k}(y)$. Its exponential generating function is $e^{xf(t)}$ for a delta series f . Falling factorials and ordinary powers are basic examples.

Positive compositional iterate.

PROJECT / MIXED

For a self-map f , its positive compositional iterates are $f^{\circ n} = f \circ \dots \circ f$ for $n \geq 1$; $f^{\circ 0}$ is the identity. In the Fabius program the bounded function and its inverse are iterated on the invariant interval $[0, 1]$. Composition exponents must not be confused with pointwise powers.

Positive-definite kernel.

PROJECT / MIXED

A Hermitian kernel K is positive definite when

$$\sum_{i,j} \bar{c}_i c_j K(x_i, x_j) \geq 0$$

for every finite choice of points and scalars. Such a kernel determines a reproducing-kernel Hilbert space. Translation-invariant positive definiteness is characterized by Bochner's theorem through a positive spectral measure.

Positive-definite moment functional.

PROJECT / MIXED

A moment functional $\mathcal{M}$ is positive definite when $\mathcal{M}(p\bar{p}) > 0$ for every nonzero polynomial p . This ensures nonsingular Hankel matrices and a unique monic orthogonal polynomial of every degree. For the positive Rvachev density, the analytic integral supplies positivity; the rational moment calculus supplies the same Gram entries exactly.

Positive-half-line Fourier energy.

LEAN-PROVED

Because the up transform is even in modulus, its full real-axis squared mass is twice the integral over $(0, \infty)$. The formal Fabius-square-energy constant equals this positive-half-line mass under the fixed Fourier convention. A rescaled sinc-product integral gives an equivalent representation with its Jacobian displayed.

Positive-part kernel.

CLASSICAL

The positive-part power $(x-t)_+^\alpha = \max(x-t, 0)^\alpha$ packages a one-sided integration kernel on the whole line. It converts interval limits into support of the kernel and is convenient for translation, convolution, splines, and fractional integration. Endpoint exponents and the convention at zero must be stated when $\alpha \leq 0$.

Positive rational gap lower-bound sequence.

LEAN-PROVED

A positive rational gap sequence supplies computable numbers $g(n) > 0$ such that points separated by a prescribed mesh have Fabius values separated by at least $g(n)$. Positivity guarantees injective numerical decoding, while rationality allows exact verification. The current inverse computability module treats such a sequence as an input hypothesis rather than claiming a universal closed form for it.

Post-render source union.

CONVENTION

A post-render source union is material merged into a LaTeX source after the last retained PDF was built. The phrase warns that page counts describe the PDF, whereas the source contains additional sections or corrections. It is especially relevant to rapidly growing frontier, integration, representation, and q -series syntheses. Such material is active documentation even before a matching PDF is committed.

Power series.

CLASSICAL

A power series $\sum a_n(z - z_0)^n$ converges inside a disk determined by its radius of convergence and defines a holomorphic function there. Within the disk it may be differentiated and integrated termwise. A formal power series has the same algebra but no convergence claim. The Fabius function's Taylor series at dyadic points is finite yet fails to represent the function locally, whereas its transform power series is entire and analytically valid.

Power-sum symmetric polynomial.

CLASSICAL

The power sum is $p_m(x_1, \dots, x_r) = \sum_i x_i^m$. Power sums linearize disjoint unions of alphabets and are related to elementary and complete homogeneous functions by Newton identities. Cumulants of weighted independent sums are power sums of the weights times component cumulants.

Powerset expansion.

CLASSICAL

A powerset expansion sums over all subsets of a finite index set. Products of terms $1 + u_j$, iterated finite differences, and independent Bernoulli choices all expand this way. In formal proofs a finite-set formulation makes duplicate indices, empty cases, and sign exponents explicit.

Prefix polynomial.

PROJECT / MIXED

A prefix polynomial packages a finite signed prefix as $P_N(z) = \sum_{n < N} \varepsilon_n z^n$, or packages repeated prefix values as a polynomial in the endpoint N . Complete blocks factor by $\prod_{j < m} (1 - z^{2^j})$. Derivatives at $z = 1$ encode Prouhet moments and orders of cancellation.

Prefix-sum boundary convention.

CONVENTION

An exclusive prefix uses $\sum_{j < n} a_j$ and vanishes at $n = 0$; an inclusive prefix uses $\sum_{j \leq n} a_j$. Iteration shifts binomial kernels and diagonal formulas, so the convention must be fixed before comparing repeated-summation results. The current project states boundary values explicitly rather than relying on informal array indexing.

Primitive-recursive fold.

LEAN-PROVED

A primitive-recursive fold evaluates a finite list or bounded range by repeatedly applying an update operation whose iteration count is itself primitive recursive. The Fabius computability certificates use such folds to evaluate exact spline or dyadic data. A large unrestricted bound may make the certificate inefficient while still proving membership in the intended recursion class.

Primitive-recursive function.

CLASSICAL

Primitive-recursive functions are built from zero, successor, and projections using composition and primitive recursion. They form a robust class of total computable functions with explicit structural termination, though they do not include every computable function. The repository expresses the finite-spline evaluator, signed-natural arithmetic, precision selection, and continuity modulus in primitive-recursive terms. This supplies a constructive proof of Fabius computability stronger than an informal numerical algorithm.

Primitive-root block.

LEAN-PROVED

A primitive-root block groups factors whose exponents form a complete residue system modulo the order d of a root of unity. At a primitive d -th root, products over a complete nonzero block collapse to explicit constants, while the zero residue records vanishing order. Blocking isolates the singular factor before taking limits and avoids invalid termwise substitution into $0/0$ quotients.

Principal branch.

CONVENTION

A principal branch is a conventional single-valued choice from a multivalued complex function, such as the principal logarithm or W_0 branch of Lambert W . Branch choices determine cuts, arguments, and continuity domains. The Fabius saddle requires the nonprincipal real branch W_{-1} ; using the principal branch would solve a different root of the same equation. Complex shifts and logarithms in transform formulas likewise require consistent branch declarations.

Principal-Lambert phase.

PROJECT / MIXED

A principal-Lambert phase uses the branch W_0 in an inverse or saddle coordinate. It is appropriate when the implicit solution remains near zero or grows on the principal sheet. It must not replace the W_{-1} phase at a small positive Fabius endpoint, where the relevant solution tends to $-\infty$.

Probabilistic representation.

CLASSICAL

A probabilistic representation identifies a deterministic function or sequence with a probability, expectation, density, or law. It can make positivity, monotonicity, normalization, and convolution structure transparent. The Fabius function is the CDF of an infinite dyadic weighted sum of independent uniforms; the up-function is the density of a centered affine version. Transform products and finite spline approximants follow directly from this representation.

Probabilists' Hermite polynomial.

PROJECT / MIXED

The probabilists' Hermite polynomial is

$$\text{He}_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}.$$

It satisfies $(\phi(x) \text{He}_n(x))' = -\phi(x) \text{He}_{n+1}(x)$ for the standard normal density ϕ . This is the normalization used in Edgeworth and Cornish–Fisher formulas.

Probability density.

CLASSICAL

A probability density f is a nonnegative integrable function with total integral one; probabilities are $\mathbb{P}(X \in A) = \int_A f(x) dx$. Its cumulative distribution is $F(x) = \int_{-\infty}^x f(t) dt$. Under the chosen affine normalization, the up-function is the smooth compactly supported density associated with the Fabius distribution. Strict positivity in the interior gives strict monotonicity of the CDF.

Probability measure.

CLASSICAL

A probability measure is a countably additive measure of total mass one. It provides the abstract framework for random variables and distributions. Pushforwards, products, and convolutions construct new laws. The infinite product probability space carrying independent uniform variables and the pushforward under their weighted sum gives a rigorous construction of the Fabius measure.

Product measure.

CLASSICAL

Given probability spaces (Ω_n, μ_n) , their product measure is the canonical measure on the Cartesian product under which coordinate random variables are independent and have laws μ_n . Infinite product measures are constructed from consistent finite-dimensional distributions. The Fabius random series can be defined on a product of unit intervals, with each coordinate carrying a uniform law. Absolute summability of weights makes the sum measurable and convergent.

Profile generating function.

FRONTIER

For scale multiplicities c_h , the profile generating function is $C(z) = \sum_{h \geq 0} c_h z^h$. Evaluation at $z = 4^{-n}$ encodes normalized even cumulants, while substitution at 2^{-s} appears in the spectral Dirichlet transform. Its coefficient nonnegativity reflects geometric multiplicity.

Proof-backed exposition.

LEAN-PROVED

A proof-backed exposition is project prose whose mathematical claims are required to have counterparts in the compiled Lean development. Such a document may reorganize proofs and suppress implementation detail, but it must not introduce unproved analytic strengthening. In this repository the primary Fabius–Rvachev article has this role, unlike the frontier volumes, which explicitly mix proved, derived, conjectural, and exploratory material.

Proof-status boundary.

CONVENTION

A proof-status boundary separates Lean-proved results, classical background, frontier derivations, conjectures, refuted claims, and historical quotations. Crossing that boundary requires a proof or an explicit change of label. Numerical evidence, a formal power-series identity, or a manuscript derivation does not by itself become a theorem in the compiled library. The present glossary uses status badges to preserve this distinction.

Prouhet cancellation.

CLASSICAL

Prouhet cancellation is the vanishing of signed power sums generated by the Thue–Morse sequence. For a complete block of length 2^m , $\sum_{n=0}^{2^m-1} \varepsilon_n n^r = 0$ for $0 \leq r < m$. It follows from the order- m zero at $z = 1$ of $\prod_{j < m} (1 - z^{2^j})$. This annihilation removes lower-degree polynomial contributions in dyadic Fabius sums and underlies the exact leading-coefficient extraction.

Prouhet partition.

CLASSICAL

A Prouhet partition divides a finite set of integers into classes whose low power sums agree. The binary Thue–Morse partition of $\{0, \dots, 2^m - 1\}$ into even and odd digit-sum classes has equal moments through degree $m - 1$. Root-of-unity digit characters give higher-base generalizations.

Prouhet–Tarry–Escott problem.

CLASSICAL

The Prouhet–Tarry–Escott problem asks for disjoint multisets with equal power sums through a prescribed degree. Splitting a complete binary block by the Thue–Morse parity gives Prouhet’s classical solution: signed power sums vanish below the block level. These cancellations drive finite-difference, prefix-sum, and exact dyadic identities in the project.

Provenance ledger.

CONVENTION

A provenance ledger records where a formula entered the corpus, how its notation was normalized, whether it was corrected, and which formal declarations support it. It is especially valuable when multiple frontier drafts are merged or when a historical paper uses incompatible symbols. Provenance is not proof, but it makes proof obligations and source corrections auditable.

Public declaration.

CONVENTION

A public declaration is a definition, theorem, structure, instance, or other named Lean object exported without private visibility through the relevant facade. The repository audit counts declarations after taking a union, so a theorem imported through several modules is counted once. A declaration count is a coverage diagnostic, not a claim that every declaration deserves a separate headword.

Public facade.

CONVENTION

A public facade is an import module that re-exports the declarations intended for downstream use. The root import `FabiusFunction` is the complete facade for this development, while smaller thematic imports reduce dependency footprint. Corpus census numbers refer to the union visible through the facade, not to every private helper definition or local theorem in every source file.

Puiseux–logarithmic series.

PROJECT / MIXED

A Puiseux–logarithmic series permits $t^{r/s} p_r(\log t)$. It combines algebraic ramification with logarithmic corrections and is the natural local class when an inverse has a multiple critical point and the forward function also contains logarithms. A fixed denominator s is part of the formal ambient ring.

Puiseux series.

PROJECT / MIXED

A Puiseux series is a Laurent series in a fractional power $t^{1/s}$, so its exponents lie in $s^{-1}\mathbb{Z}$. It represents local algebraic inverse branches near a ramification point after a branch of the fractional power is chosen. The Lambert W branch point and cyclotomic critical points naturally produce square-root or higher Puiseux expansions.

Pushforward measure.

CLASSICAL

For a measurable map $T : X \rightarrow Y$ and a measure μ on X , the pushforward $T_{\#}\mu$ is defined by $T_{\#}\mu(A) = \mu(T^{-1}(A))$. It is the law of $T(X)$ when X has law μ . The Fabius measure is the pushforward of an infinite product of uniform measures under the weighted-sum map. Affine pushforwards relate the bounded distribution and centered up-density.

Pythagorean identity.

CLASSICAL

In an inner-product space, if $u \perp v$ then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$. For an orthogonal projection p of f , one has $\|f - q\|^2 = \|f - p\|^2 + \|p - q\|^2$ for every q in the subspace. This proves least-squares optimality and quantifies the error reduction of Fourier-Legendre truncation for the up-function.

17 Q **q -Bernstein polynomial.**

FRONTIER

A q -Bernstein basis deforms the classical Bernstein basis by replacing binomial coefficients and powers with Gaussian and q -shifted factors. Several inequivalent definitions occur. In geometric-comb approximation it preserves a chosen positive partition of unity and converges to the classical basis as $q \rightarrow 1$, subject to the declared normalization.

 q -beta function.

PROJECT / MIXED

The q -beta function is

$$B_q(x, y) = \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)}.$$

For positive real arguments and $0 < q < 1$, it has a Jackson-integral representation after choosing the matching normalization. It is symmetric in x, y , and its logarithmic derivatives are differences of q -digamma functions. Pole and branch conventions are inherited from Γ_q .

q -beta integral.

LEAN-PROVED

A q -beta integral evaluates a Jackson integral of a product of powers and q -shifted factors in terms of q -gamma functions. It is the geometric comb analogue of Euler's beta integral. The formal theorem states positivity, convergence, and endpoint conditions explicitly; the symbol $d_q t$ cannot be replaced by ordinary Lebesgue measure.

 q -binomial coefficient.

CLASSICAL

The q -binomial coefficient is the Gaussian coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$. It satisfies a q -Pascal recurrence and counts subspaces of a finite vector space when q is a prime power, while algebraically it is a polynomial in q . The Fabius directory uses the analytic value $q = 1/2$, where a normalized row gives interpolation and Toeplitz weights. The parameter is a scalar base, not a probability.

 q -binomial scalar extension.

CONVENTION

Scalar extension transports a polynomial or finite-sum identity proved over $\mathbb{Q}$ or $\mathbb{R}$ into a larger field such as $\mathbb{C}$. Since Gaussian binomial coefficients at $q = 1/2$ are rational, their identities can be mapped through the canonical embedding. The repository separates this algebraic step from analytic convergence, allowing the complex-shift q -binomial Taylor identity to reuse an exact real polynomial theorem.

 q -binomial theorem.

CLASSICAL

The q -binomial theorem comprises a finite polynomial identity and an infinite analytic identity. One finite form is

$$\prod_{j=0}^{n-1} (1 + zq^j) = \sum_{k=0}^n q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k.$$

For $|q| < 1$ and $|z| < 1$, the infinite form expresses a quotient of q -Pochhammer products as a basic hypergeometric series. The repository formalizes terminating identities, scalar and polynomial forms, reciprocal Euler specializations, and the infinite theorem with explicit summability hypotheses. Finite algebraic statements must not be silently upgraded to analytic convergence statements.

 q -calculus.

CLASSICAL

q -calculus is a deformation of ordinary calculus in which additive increments are replaced by multiplicative scaling by q . Its basic objects include q -integers, q -factorials, q -differences, Gaussian binomial coefficients, and q -Pochhammer symbols. The Fabius problem has a natural half-base $q = 1/2$ because every differential scaling step doubles the argument. Here q -calculus is finite and algebraic rather than a replacement for the ordinary derivative defining F .

 q -Catalan number.

LEAN-PROVED

One common normalization is

$$C_n(q) = \frac{1}{[n+1]_q} \begin{bmatrix} 2n \\ n \end{bmatrix}_q.$$

The displayed quotient is in fact a polynomial, but this requires a divisibility theorem. Different q -Catalan families exist, so the normalization must be stated. The project uses the Gaussian quotient

form and keeps its cyclotomic divisibility proof separate from combinatorial interpretations.

q -digamma function.

PROJECT / MIXED

The q -digamma function is $\psi_q(z) = \frac{d}{dz} \log \Gamma_q(z)$. For $0 < q < 1$,

$$\psi_q(z) = -\log(1-q) + (\log q) \sum_{n \geq 0} \frac{q^{n+z}}{1-q^{n+z}},$$

away from poles. It controls monotonicity, convexity, and parameter inversion for Γ_q , and tends to the ordinary digamma function as $q \uparrow 1$ under standard fixed- z hypotheses.

q -discrete limit.

PROJECT / MIXED

The complex q -discrete limit is a sequence of finite Gaussian-binomial combinations of complex-shift spline approximants that converges to the signed global Fabius extension. After reindexing, each approximant is a Toeplitz row with total mass one and uniformly bounded variation, sampling only sufficiently advanced splines. A finite row can depend on the complex shift Q ; the limit is independent of fixed Q . This limit is not termwise identical to the global q -series even though both converge to the same function.

q -factorial.

CLASSICAL

The q -integer is $[n]_q = (1 - q^n)/(1 - q)$, and the q -factorial is $[n]_q! = \prod_{j=1}^n [j]_q$. It is related to the q -Pochhammer product by $[n]_q! = (q; q)_n / (1 - q)^n$. Gaussian binomial coefficients are ratios of q -factorials. At $q = 1/2$, clearing powers of two converts these products into odd Mersenne products.

q -finite difference.

CLASSICAL

A q -finite difference compares values at multiplicatively related points, often through $D_q f(x) = (f(x) - f(qx))/((1 - q)x)$, or through a finite Gaussian-weighted node sum. The repository mainly uses the latter leading-coefficient functional on geometric nodes. Like an ordinary n -th difference, it annihilates polynomials of degree below n , but the weights and nodes encode powers of two.

q -gamma function.

PROJECT / MIXED

For $0 < q < 1$,

$$\Gamma_q(z) = (1 - q)^{1-z} \frac{(q; q)_\infty}{(q^z; q)_\infty}, \quad q^z = e^{z \log q}.$$

It satisfies $\Gamma_q(1) = 1$ and $\Gamma_q(z + 1) = [z]_q \Gamma_q(z)$. Its meromorphic pole lattice comes from the denominator. A separate reciprocal-base normalization is required for $q > 1$; the two conventions must not be mixed inside an inverse or asymptotic formula.

q -Gauss summation.

LEAN-PROVED

Under $|c/(ab)| < 1$ and denominator nonvanishing,

$${}_2\phi_1 \left[\begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab} \right] = \frac{(c/a, c/b; q)_\infty}{(c, c/(ab); q)_\infty}.$$

It is the basic-hypergeometric analogue of Gauss's ordinary hypergeometric summation. The infinite products encode the complete value; truncating them without an error estimate changes the theorem into an approximation.

q-integer.

CLASSICAL

The q -integer $[n]_q = 1 + q + \cdots + q^{n-1} = (1 - q^n)/(1 - q)$ deforms the ordinary integer n , recovered as $q \rightarrow 1$. Products of q -integers form q -factorials. Although the Fabius formulas are often written directly with $(q; q)_n$ and Gaussian coefficients, q -integers explain the parallel with ordinary binomial algebra and the appearance of factors $2^j - 1$ at half base.

q-kernel of a sequence.

CLASSICAL

The q -kernel replaces powers of two by powers of an integer base $q \geq 2$. Finiteness characterizes q -automatic sequences, and finite module generation characterizes q -regular sequences. Kernel recursion is the automata-theoretic counterpart of digitwise decomposition and Mahler functional equations.

q-Lucas theorem.

LEAN-PROVED

At a primitive d -th root of unity ζ , writing $n = n_1d + n_0$ and $k = k_1d + k_0$ with $0 \leq n_0, k_0 < d$, a q -Lucas formula reduces

$$\begin{bmatrix} n \\ k \end{bmatrix}_{\zeta}$$

to an ordinary binomial factor involving n_1, k_1 times a small Gaussian coefficient $\begin{bmatrix} n_0 \\ k_0 \end{bmatrix}_{\zeta}$. The theorem is a root-of-unity digit law and is distinct from the ordinary Lucas theorem modulo a prime.

q-multinomial coefficient.

LEAN-PROVED

For nonnegative $k_1, \dots, k_r$ with sum n ,

$$\begin{bmatrix} n \\ k_1, \dots, k_r \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_{k_1} \cdots (q; q)_{k_r}}.$$

It is a polynomial with nonnegative integer coefficients and counts flags or weighted words in standard interpretations. Iterating Gaussian binomial factorizations proves its polynomiality and yields quantum multinomial expansions for suitably q -commuting variables.

q-Pascal summation.

LEAN-PROVED

A q -Pascal summation is a finite identity obtained by summing a Gaussian Pascal recurrence, often a q -weighted analogue of the hockey-stick identity. Typical forms collapse a triangular Gaussian sum to a single Gaussian coefficient. Since several exponent placements are equivalent after reindexing, the exact power of q is included in the formula rather than hidden in the name.

q -Pfaff–Saalschütz summation.

LEAN-PROVED

A terminating balanced ${}_3\phi_2$ sum has the form

$${}_3\phi_2 \left[\begin{matrix} a, b, q^{-N} \\ c, abq^{1-N}/c \end{matrix}; q, q \right] = \frac{(c/a, c/b; q)_N}{(c, c/(ab); q)_N}.$$

The balance relation among the lower parameters is essential. The project proves the finite identity by a telescoping or inductive certificate suitable for formal verification.

 q -Pochhammer dissection.

LEAN-PROVED

For an integer $m \geq 1$, splitting indices by residue class gives

$$(a; q)_\infty = \prod_{r=0}^{m-1} (aq^r; q^m)_\infty$$

when $|q| < 1$, with the analogous finite formula after the terminal lengths are distributed among residue classes. Dissection is the product-side counterpart of digit decomposition. It supports root-of-unity analysis, theta identities, and multiscale factorizations of Fabius-related products.

 q -Pochhammer logarithmic derivative.

LEAN-PROVED

Away from the zeros of $(a; q)_\infty$,

$$\frac{\partial}{\partial a} \log(a; q)_\infty = - \sum_{j \geq 0} \frac{q^j}{1 - aq^j}.$$

The right side is a Lambert series and converges locally normally on compact sets avoiding the zero lattice. Higher derivatives are obtained by differentiating the summands. The formula is meromorphic, not an everywhere holomorphic identity, because each product zero becomes a pole.

 q -Pochhammer order derivative.

PROJECT / MIXED

A continuous-order extension replaces the integer length n by a complex parameter, commonly through

$$(a; q)_z = \frac{(a; q)_\infty}{(aq^z; q)_\infty}, \quad |q| < 1,$$

after fixing a branch of q^z . Differentiation with respect to z then produces a logarithmic Lambert series. This object agrees with the finite product at nonnegative integers but is branch-dependent for complex q ; the repository records the branch and pole lattice explicitly.

 q -Pochhammer reversal.

LEAN-PROVED

For nonzero a, q ,

$$(a; q)_n = (-a)^n q^{n(n-1)/2} (a^{-1} q^{1-n}; q)_n.$$

This reverses the order of the factors and exposes reciprocal-base symmetry. Specializing $a = q^{-N}$ makes termination explicit in basic hypergeometric series. The powers of $-a$ and q are essential; omitting either changes both the degree and the zero divisor.

q -Pochhammer shift law.

LEAN-PROVED

The elementary shift identity is

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

For the infinite product this becomes $(a; q)_\infty = (a; q)_m (aq^m; q)_\infty$ when $|q| < 1$. It is the multiplicative analogue of cutting a series into a finite head and a tail. Many quotient identities, contiguous relations, and numerical evaluators are obtained by applying the law before cancellation.

 q -Pochhammer symbol.

CLASSICAL

For a base q , parameter a , and natural length n , the finite q -Pochhammer symbol is

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_0 = 1.$$

When $|q| < 1$, the infinite symbol $(a; q)_\infty$ is the normally convergent product over $j \geq 0$. The current tree treats finite and infinite forms, dissection, base reversal, analytic dependence on a , logarithmic derivatives, and branch-dependent complex order. Every generalized-order formula states its logarithm branch and denominator nonvanishing conditions; the notation is never interpreted by informal substitution alone.

 q -product.

CLASSICAL

A q -product is a finite or infinite product with geometrically progressing factors, typically a q -Pochhammer symbol. Its zeros and ratios encode q -recurrences and basic hypergeometric coefficients. The Fabius half-base products link geometric interpolation denominators to Mersenne factors. They should be distinguished from the independent sinc product of the Fourier transform, although both reflect the same dyadic scale hierarchy.

 q -Richardson extrapolation.

FRONTIER

A q -Richardson scheme cancels leading error powers in data sampled at geometrically shrinking scales $h, qh, q^2h, \dots$. Its weights are Lagrange weights at the target scale 0 and therefore have explicit q -product forms. Stability depends on the base, order, and error model; exact cancellation does not by itself imply numerical robustness.

 q -series.

CLASSICAL

A q -series is a series whose coefficients or exponents are organized by powers of a base q , often using q -Pochhammer symbols. It may be formal, terminating, or analytically convergent. The global Fabius q -binomial series is an absolutely convergent sum of finite Taylor blocks expressed through Gaussian rows and a fixed complex shift. The convergence comes from binary reduction estimates, not merely from formal q -series algebra.

q -shifted factorial.

CLASSICAL

q -shifted factorial is a standard synonym for the q -Pochhammer symbol. The word “factorial” emphasizes the recurrence $(a; q)_{n+1} = (1 - aq^n)(a; q)_n$; it does not mean the special product $[n]_q!$. The glossary keeps a separate entry because both phrases occur in the literature and their collision is a common source of normalization errors.

 q -Taylor identity.

PROJECT / MIXED

The repository’s q -Taylor identity states that a finite Gaussian-binomial expression with arbitrary common complex shift Q equals a real Taylor block embedded in $\mathbb{C}$. Translation invariance follows because the order- n geometric finite difference annihilates every lower-degree term generated by expanding the shift. This identity converts local derivative data into the finite summands of the global complex q -series.

 $q \rightarrow 1$ limit.

PROJECT / MIXED

A $q \rightarrow 1$ limit recovers a classical object only after its scaling is fixed: $[n]_q \rightarrow n$, $\begin{bmatrix} n \\ k \end{bmatrix}_q \rightarrow \binom{n}{k}$, and a suitably scaled q -exponential tends to e^z . Infinite products often require renormalization or logarithmic asymptotics. The direction of approach and uniformity in other parameters are part of the theorem.

 q -Vandermonde convolution.

PROJECT / MIXED

The q -Vandermonde identity convolves two Gaussian rows with a quadratic or bilinear power of q to obtain a larger Gaussian coefficient. One form is

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_k q^{(m-k)(r-k)} \begin{bmatrix} m \\ k \end{bmatrix}_q \begin{bmatrix} n \\ r-k \end{bmatrix}_q.$$

Different but equivalent exponents correspond to reversing one row. It is the geometric-node analogue of the ordinary Vandermonde convolution.

Quadratic logarithmic law.

CLASSICAL

A quadratic logarithmic law states that the leading behavior of $\log F(x)$ is a negative constant times $(\log(1/x))^2$. It is a coarse consequence of the sharp endpoint expansion and ignores nested logs and periodic fluctuations. Such a law proves superpolynomial flatness and gives the correct broad decay class. It is insufficient for asymptotic equivalence or accurate numerical evaluation because lower terms in the exponent remain large or bounded but nonconvergent.

Quadrature weight.

CLASSICAL

In a rule $\int f d\mu \approx \sum_j w_j f(x_j)$, the numbers w_j are quadrature weights. Exactness through a polynomial degree gives moment equations $\sum_j w_j x_j^k = m_k$. Positive weights yield stability and a probabilistic interpretation; signed extrapolation weights can be exact but ill-conditioned.

Quantile coupling.

CLASSICAL

For real laws with generalized inverses Q_μ, Q_ν , taking one uniform U and setting $X = Q_\mu(U)$, $Y = Q_\nu(U)$ gives the monotone quantile coupling. It is optimal for one-dimensional Wasserstein costs. In the Fabius setting the ordinary inverse on $[0, 1]$ supplies the quantile exactly.

Quantile transport.

LEAN-PROVED

Quantile transport maps a reference probability $u \in (0, 1)$ through an inverse CDF to produce a target law. Identities for the geometric uniform CDF therefore induce identities for its inverse after monotonicity and range have been proved. Error transfer requires lower density or gap information, because small CDF error can be magnified near flat endpoints.

Quantum binomial theorem.

LEAN-PROVED

The quantum binomial theorem is the expansion of a power of a sum of q -commuting variables by Gaussian coefficients. Unlike the finite commutative q -binomial product identity, it is an identity in a noncommutative algebra. The project formalizes the commutation relation and then proves the expansion by induction using a q -Pascal recurrence.

Quantum multinomial theorem.

LEAN-PROVED

For a consistently ordered family of pairwise q -commuting variables, a power of their sum expands through q -multinomial coefficients. The precise power of q depends on the chosen ordering and commutation convention. Formal statements fix both, preventing a classical multinomial formula from being imported into a noncommutative algebra without the required weights.

Quantum plane.

LEAN-PROVED

The quantum plane is the algebra generated by x, y with $yx = qxy$. In this noncommutative setting,

$$(x + y)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q x^{n-k} y^k$$

for the displayed commutation convention. Reversing the relation or monomial order replaces q by q^{-1} and introduces powers of q . The relation therefore belongs to the statement, not merely to the proof.

Quotient-of-exponentials approximation.

CLASSICAL

A quotient-of-exponentials approximation replaces a complicated product, recurrence, or sum by a ratio of exponential expressions suggested by leading logarithms. Such an ansatz can capture a coarse scale while missing a linear saddle residue, a periodic factor, or a normalization constant. Repository modules named for quotient-exponential mismatch formalize why certain attractive informal proxies cannot achieve the claimed error. Substitution into the exact stationary equation is used to expose the residual term.

r -Stirling number.

FRONTIER

An r -Stirling number of the second kind counts set partitions in which r distinguished elements lie in pairwise distinct blocks. Boundary conditions depend on whether the total set size includes those distinguished elements. The notation catalogue requires the convention to be stated because several shifted definitions occur in the literature.

Rademacher sign.

CLASSICAL

The j -th Rademacher function alternates between $+1$ and -1 on dyadic intervals of length 2^{-j} , equivalently reading one binary digit of the argument. Products of Rademacher functions form the Walsh system. They are continuous analogues of digit characters and connect Thue–Morse parity with dyadic harmonic analysis.

Rademacher sine-product identity.

FRONTIER

A Rademacher sine-product identity rewrites signs of dyadic sine or sinc factors in terms of binary digit functions. Magnitudes and signs are separated so that the real Fourier lobes become a Walsh-like product. Such identities support exact sign classification and phase-aware extrapolation.

Radius asymptotic.

CLASSICAL

A radius asymptotic describes the size of the neighborhood around a saddle on which a local expansion is used, often as a function of the large parameter. The radius must grow enough to capture Gaussian mass but remain small enough that higher phase terms are controlled. Optimizing this tradeoff is part of a quantitative saddle proof. The Fabius modules derive admissible central radii and show that tail and Taylor remainders meet the target inverse- λ order.

Radon–Nikodym derivative.

CLASSICAL

If a σ -finite measure ν is absolutely continuous with respect to μ , the Radon–Nikodym theorem gives a measurable density $d\nu/d\mu$, unique almost everywhere, such that $\nu(B) = \int_B (d\nu/d\mu) d\mu$. Canonical continuous densities in the Fabius development are selected representatives of this a.e. class.

Radon transform.

FRONTIER

The Radon transform integrates a multivariate function over affine hyperplanes. For a probability density it gives the densities of linear projections. Projecting a product or infinite convolution of segment laws reduces multivariate Rvachev constructions to one-dimensional direction-profile sinc products.

Ramification index.

PROJECT / MIXED

If $f(x) - f(x_0) = c(x - x_0)^m + \dots$ with $c \neq 0$, then m is the ramification index of the local map. The inverse has m Puiseux branches with leading exponent $1/m$. The case $m = 1$ is an ordinary analytic inverse; $m = 2$ gives a square-root branch point.

Random series.

CLASSICAL

A random series is a series of random variables. Its convergence can be studied almost surely, in probability, or in L^p . The Fabius random series has bounded independent terms with geometrically summable weights, so it converges absolutely for every outcome. Its distributional self-similarity under removal of the first summand leads to the fixed-point and differential equations.

Rapid decay.

CLASSICAL

A function $g(t)$ has rapid, or Schwartz, decay if $|t|^N g(t) \rightarrow 0$ for every N , usually together with analogous bounds for derivatives. The Fourier transform of a compactly supported C^∞ function decays rapidly along the real axis. This property ensures absolute convergence of inversion integrals and Fourier coefficient series. The explicit sinc product supplies a direct route to rapid-decay estimates for the up-function transform.

Rarefaction of a sequence.

FRONTIER

Rarefaction extracts terms in one or more residue classes, such as $a(qn + r)$, or sums a sequence over those classes. For automatic sequences the rarefied subsequences belong to the kernel and satisfy a finite linear system. Thue–Morse rarefaction exposes spectral and arithmetic structure hidden in the full sequence.

Rate–distortion function.

FRONTIER

For a source X and distortion d , the rate–distortion function is

$$R(D) = \inf I(X; \hat{X})$$

over couplings with expected distortion at most D . It describes the optimal information rate, not the error of one prescribed spline or prefix quantizer. Compact-support geometric sources yield structured candidate couplings but still require an optimization theorem.

Rate of convergence.

CLASSICAL

A rate of convergence quantifies how fast an error tends to zero, for example geometrically, polynomially, or as an inverse logarithm. A certified rate provides a function mapping tolerance to a sufficient index. Fabius fixed-point iterates converge geometrically, centered splines have a dyadic uniform rate, Fourier-Legendre coefficients decay rapidly, and the endpoint saddle expansion has inverse-Lambert orders. These rates serve different algorithms and should not be compared without matching cost models.

Ratio set.

FRONTIER

A ratio set is a finite family of small monomials used to witness that one transseries monomial is asymptotically smaller than another. Products of the ratios generate a grid containing all relevant quotients. It gives an effective certificate for dominance and remainder control in grid-based transseries calculations.

Rational Legendre coefficient.

LEAN-PROVED

A rational Legendre coefficient is obtained by applying the exact rational Rvachev moment functional to the rational polynomial P_n :

$$c_n^{\mathbb{Q}} = \frac{2n+1}{2} \mathcal{M}_{\text{Rv}}(P_n).$$

After casting to $\mathbb{R}$, it agrees with the analytic integral coefficient under the established moment bridge. This two-stage construction keeps exact arithmetic independent of integration theory.

Rational recurrence.

CLASSICAL

A rational recurrence determines each term using rational operations on earlier terms. Triangular moment recurrences are typical. They prove rationality inductively but may obscure integrality and denominator growth. The Fabius arithmetic converts selected rational recurrences into division-free integer recurrences by multiplying through canonical factors, allowing parity and valuation proofs unavailable from the raw recurrence alone.

Rationality.

CLASSICAL

A quantity is rational when it belongs to $\mathbb{Q}$. Rationality at all dyadic arguments is a remarkable property of the Fabius function, despite its definition through an infinite random series or transform product. It follows from terminating Taylor data and explicit finite Thue–Morse/moment formulas. Rationality alone does not identify the reduced denominator; the valuation and common-denominator theorems refine it.

Real-analytic function.

CLASSICAL

A real function is analytic at x_0 if it agrees with a convergent power series on some neighborhood of x_0 . Real analyticity is stronger than C^∞ smoothness and entails local determination by derivatives. The Fabius function is analytic on the interiors of its constant tails but nowhere analytic on its transition interval; the up-function is analytic where it is identically zero outside its support and nowhere analytic inside. Endpoint flatness does not extend analyticity across the boundary.

Reciprocal-base q -duality.

PROJECT / MIXED

Reciprocal-base duality relates a formula at q to one at q^{-1} after extracting a monomial power and sometimes a sign. Gaussian palindromy and q -Pochhammer reversal are canonical examples. For probability laws, an additional affine normalization may be needed. The duality is algebraic for finite products and analytic only on domains where the infinite objects are defined.

Reciprocal gamma function.

PROJECT / MIXED

The reciprocal gamma function $1/\Gamma(z)$ is entire and has simple zeros at $0, -1, -2, \dots$. Its Weierstrass product and Taylor coefficients encode Euler's constant and zeta values. Entire-function treatment avoids carrying a meromorphic denominator in products and derivative towers.

Reciprocal-power table.

PROJECT / MIXED

A reciprocal-power table is a list of exact values, normalized numerators, or asymptotic data for $F(2^{-n})$. These samples sit at the natural dyadic endpoint scales and are linked directly to half moments. The table is useful for discovering parity patterns, checking denominator formulas, and validating the periodic saddle asymptotic. Exact rational entries avoid the catastrophic underflow that affects direct floating-point evaluation for large n .

Recurrence relation.

CLASSICAL

A recurrence relation defines terms of a sequence or family from earlier terms or smaller parameters. It is triangular when the new term appears with an invertible coefficient and only lower indices occur elsewhere. Recurrences can be linear or nonlinear, homogeneous or inhomogeneous, scalar or polynomial-valued. The Fabius directory uses them for moments, numerators, Bernoulli coefficients, saddle jets, Lambert coefficients, and exact dyadic evaluation.

Recursive fast name.

PROJECT / MIXED

A recursive fast name is a computable sequence of rational approximations whose error decreases at a prescribed exponential rate. Recursive means there is an effective algorithm for every precision, while fast refers to the convergence guarantee. The signed-natural dyadic representation in the repository is a concrete fast name. Composition with effective arithmetic and the finite-spline evaluator yields names for Fabius values.

Recursive modulus.

CLASSICAL

A recursive modulus is a computable rate function for continuity, convergence, or approximation. Given a desired output accuracy, it returns a sufficient input accuracy or truncation level. The Fabius computability theorem provides both a uniform-continuity modulus from the Lipschitz constant and an approximation modulus from the centered spline error. Together they make the qualitative algorithm uniform over real inputs.

Reduced dyadic level.

LEAN-PROVED

An interior dyadic rational is reduced at positive level n when it is written $m/2^n$ with $0 < m < 2^n$ and m odd. Oddness prevents the same point from belonging to a coarser positive denominator level. Exact derivative-order, sign, denominator, and recursion statements use the reduced level rather than an arbitrary presentation.

Reference tail.

FRONTIER

A reference tail is a simpler majorant or comparison integral used to bound the remote part of a contour integral. Its decay is chosen to dominate the true integrand after the central saddle region, while remaining explicitly integrable. Fabius saddle modules separate reference-tail estimates from local phase algebra, allowing the proof to show that omitted contour portions are below every retained inverse-saddle order.

Reference weight.

FRONTIER

A reference weight is a normalized model density, usually Gaussian, against which the local saddle integrand is compared. Moments of the reference weight generate universal double-factorial coefficients. The true normalized integrand is expanded as the reference weight times phase and amplitude corrections. Controlling the ratio on the central region and the tails produces both the leading mass and all higher saddle coefficients.

Refinement equation.

CLASSICAL

A refinement equation expresses a function in terms of scaled and translated copies of itself, typically $f(x) = \sum_k c_k f(2x - k)$. It is fundamental in wavelets, splines, and subdivision schemes. The up-function satisfies a differential/refinement identity derived from the Fabius equation; its Fourier transform then satisfies an infinite product relation. Support and normalization select the canonical compactly supported solution.

Refinement operator.

CLASSICAL

A refinement operator maps a function to a combination or integral of its dilates and translates. Fixed points solve the associated refinement equation. Depending on the norm and normalization, the operator may be a contraction or preserve mass and positivity. Iterating the Fabius refinement operator produces increasingly fine probability or spline approximants and gives a constructive existence proof.

Reflection symmetry.

CLASSICAL

Reflection symmetry about c means $f(c + x) = f(c - x)$ for an even-type function, or $F(c + x) = 1 - F(c - x)$ for a symmetric CDF. The bounded Fabius function satisfies $F(1 - x) = 1 - F(x)$, while the up-function is even about zero. Symmetry halves many integrals, kills odd moments or Legendre coefficients, and transfers endpoint and convexity results from one side to the other.

Regular sequence.

CLASSICAL

A q -regular sequence has a q -kernel whose elements generate a finitely generated module over the coefficient ring. It need not take finitely many values. Repeated sums, polynomially weighted Thue–Morse sequences, and many digit-counting functions naturally fall into this class, which is stable under several algebraic operations.

Relative entropy.

FRONTIER

For probability densities f, g ,

$$D(f||g) = \int f \log(f/g)$$

with the usual absolute-continuity convention. It is nonnegative and vanishes only for equality almost everywhere. Comparisons between finite sinc-prefix splines and the limiting up-density must control zeros before this integral is asserted finite.

Relative error.

CLASSICAL

The relative error of an approximation g to a nonzero quantity f is $|g - f|/|f|$. For extremely small Fabius endpoint values, a tiny absolute error can still be a huge relative error. Logarithmic asymptotics are often analyzed first because direct values underflow, but exponentiating a bounded missing term causes an order-one relative oscillation. Exact rational benchmarks make relative-error assessment reliable.

Relative saddle mass.

PROJECT / MIXED

Relative saddle mass is the ratio of the exact localized saddle integral to its leading Gaussian approximation. It tends to one and admits an inverse-saddle expansion. Expressing corrections relatively avoids repeatedly carrying the dominant exponential and square-root prefactors. The repository's mass-all-orders modules compute this ratio first and then take its formal logarithm to obtain the endpoint expansion for $\log F$.

Relative tail logarithm.

FRONTIER

For an infinite product $\Phi = \Phi_N T_N$, the relative-tail logarithm is $\log T_N$ on a zero-free region with a specified logarithm branch. Expanding it in powers of the small tail scales yields multiplicative extrapolation corrections. Factoring out Φ_N preserves the zeros already present in the finite product.

Remainder estimate.

CLASSICAL

A remainder estimate bounds the error after truncating a series, Taylor expansion, contour localization, or asymptotic formula. A formal coefficient identity becomes an analytic theorem only when its remainder is controlled uniformly in the claimed regime. The Fabius development has separate modules for Taylor remainders, saddle central errors, minor arcs, tail flatness, and inverse-Lambert transfer. Their combination certifies each stated $\mathcal{O}$ -term.

Removable singularity.

CLASSICAL

A removable singularity is an isolated point where a function is initially undefined or apparently singular but has a finite limit and extends holomorphically or continuously. The sinc function $\sin z/z$ has the removable value 1 at $z = 0$. Regularized Bose kernels likewise subtract a principal singular term to produce a smooth remainder, though the original integral may still require a finite part. Explicitly assigning removable values avoids domain defects in formal statements.

Removable-value convention.

CONVENTION

A removable-value convention fills a point where a displayed quotient has both numerator and denominator zero by its limiting analytic value. Examples include $\text{sinc}(0) = 1$, Jackson quotients at special parameters, and polynomial quotients constructed after cyclotomic cancellation. The value must be declared or proved removable before substitution; writing $0/0$ and informally cancelling at the evaluation point is not a definition.

Report finite Fabius approximant.

LEAN-PROVED

The report finite Fabius approximant is the explicitly named finite spline used to mirror formulas in the exposition. It is constructed from finitely many uniform or dyadic convolution factors and has exact piecewise-polynomial behavior. Quarter-cell and related finite theorems concern this object; they do not construct a finite inverse approximant unless a separate definition does so.

Representation independence.

CLASSICAL

Representation independence means that a mathematical output depends only on the represented object, not on a nonunique encoding. Dyadic rationals can be written at multiple levels, real numbers can have two binary expansions, and signed dyadic numerators can have redundant positive-negative pairs. Exact Fabius evaluators prove compatibility under refinement and coding equivalence. This is necessary before an implementation can be regarded as a function on rationals or reals rather than on syntax.

Representation operator.

PROJECT / MIXED

A representation operator maps coefficient data to a function, for example $(Tc)(x) = \sum_k c_k \phi_k(x)$. Boundedness, injectivity, surjectivity, and stability depend on the coefficient and function spaces. In the project, Legendre synthesis, shifted-up synthesis, and sampling reconstruction are different operators even when they represent the same polynomial.

Reproducing-kernel Hilbert space.

FRONTIER

An RKHS is a Hilbert space of functions in which each point-evaluation map is continuous. There is a unique kernel K with $f(x) = \langle f, K(\cdot, x) \rangle$. Sampling stability, interpolation matrices, and minimum-norm reconstruction can be phrased in this framework, but a kernel's smoothness alone does not establish completeness or universality.

Reshetnikov integers.

PROJECT / MIXED

Reshetnikov's integers are normalized quantities R_n formed from Fabius endpoint values or half moments by multiplying by factorial, power-of-two, double-factorial, and odd Mersenne factors according to the repository's displayed definition. The main arithmetic theorem proves that every R_n is a positive odd integer. Their integrality refines earlier conjectural observations and packages exact 2-adic cancellation into a natural integer sequence.

Residue.

CLASSICAL

The residue of a meromorphic function at z_0 is the coefficient of $(z - z_0)^{-1}$ in its Laurent expansion. The residue theorem converts a contour shift into the sum of residues of crossed poles. In Mellin analysis, residues produce asymptotic terms. For the Fabius dyadic harmonic sum, residues at the imaginary lattice yield the Fourier coefficients of the periodic correction.

Resolvent-order hierarchy.

LEAN-PROVED

A resolvent-order hierarchy studies $\int (z - x)^{-\alpha} d\mu(x)$ as the positive order α varies. Differentiation in z raises integer order, while integral identities lower general real order under explicit hypotheses. The current formal results are real-variable statements for $z > 1$, not a complex-order branch theory.

Retired alias.

HISTORICAL

A retired alias is a former command or spelling that may still occur in archives or source-faithful quotations but is no longer emitted in normative project prose. The notation catalogue records a lossless mapping from retired forms to canonical symbols when one exists. An alias that merged genuinely different normalizations is not mechanically replaceable and requires a local semantic check.

Riemann–Liouville fractional integral.

CLASSICAL

For $\alpha > 0$, the left-sided Riemann–Liouville integral is

$$(I_{a+}^{\alpha} f)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x (x - t)^{\alpha-1} f(t) dt.$$

A right-sided version reverses the interval. Integer orders recover repeated integration, and suitable function classes satisfy $I^{\alpha} I^{\beta} = I^{\alpha+\beta}$.

Riemann zeta function.

CLASSICAL

The Riemann zeta function is $\zeta(s) = \sum_{n \geq 1} n^{-s}$ for $\Re s > 1$, continued meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$. It appears in Mellin transforms of Bose-type kernels and in Euler–Maclaurin constants. The Fabius periodic Fourier coefficients use $\zeta(1 - \chi_k)$ at nonreal points. Nonvanishing on the line $\Re s = 1$ at these frequencies helps establish nonconstancy.

Riesz basis.

FRONTIER

A Riesz basis is the image of an orthonormal basis under a bounded invertible operator. Equivalently, finite coefficient combinations satisfy two-sided ℓ^2 norm bounds and the span is complete. Exact reproduction by translated up-functions does not by itself establish a Riesz basis; global stability and completeness require separate proofs.

Rising factorial.

CLASSICAL

The rising factorial is $x^{\overline{n}} = x(x+1) \cdots (x+n-1)$, with $x^{\overline{0}} = 1$. It is also the ordinary Pochhammer symbol $(x)_n$ in hypergeometric notation. The project keeps it distinct from the q -Pochhammer symbol $(a; q)_n$ and from falling factorials.

Rodrigues formula.

CLASSICAL

Rodrigues' formula constructs Legendre polynomials by $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$. It proves degree, parity, endpoint normalization, and orthogonality after integration by parts. In the up-function Legendre analysis, it converts coefficients into integrals involving derivatives or moments, which can then be evaluated from the Fabius recurrence data.

Rogers–Szegő polynomial.

LEAN-PROVED

The Rogers–Szegő polynomial is

$$H_n(z; q) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q z^k.$$

It is a generating polynomial for one Gaussian row. Its recurrence, symmetry, and special values package several q -binomial identities. It should not be confused with the Hermite polynomial despite the common letter H .

Root-of-unity collision.

PROJECT / MIXED

A root-of-unity collision occurs when several factors $1 - q^m$ vanish at the same base while numerator and denominator zeros interact. The correct local object is obtained by cyclotomic valuation or a Laurent expansion, not by substituting into individual quotient factors. Collisions govern zeros, finite limits, derivative values, and ramification of inverse q -functions.

Rounding to nearest dyadic.

PROJECT / MIXED

Rounding to the nearest dyadic at precision p selects an integer a so that $|x - a/2^p| \leq 2^{-p-1}$, with a specified tie convention. Combined with a Lipschitz bound, it converts input approximation error into output error. The computable Fabius algorithm can round a real name to a dyadic grid, evaluate a finite spline exactly there, and add the certified spline and input-propagation errors.

Rvachev Appell polynomial.

LEAN-PROVED

The Rvachev Appell polynomial is the centered Appell sequence associated with the up-function probability law. Its generating function is the exponential kernel divided by the Rvachev moment-generating function. Exact rational moments determine all coefficients, and averaging a translated polynomial against the Rvachev density recovers the corresponding monomial.

Rvachev atom.

LEAN-PROVED

A Rvachev atom is a translate and scale of up, multiplied by a coefficient supplied by polynomial deconvolution or interpolation. Compact support makes only finitely many atoms active on a bounded interval. Atom families provide exact polynomial, Lagrange, and Legendre synthesis rather than merely approximate basis expansions.

Rvachev decoder.

LEAN-PROVED

A Rvachev decoder is the coefficient functional or finite sum that combines up-atoms to recover a target value or interpolating polynomial. Its normalization is chosen so that the sum of cardinal basis functions is one. Decoder theorems state degree bounds, mesh divisibility, and exact finite index ranges.

Rvachev differential–difference law.

LEAN-PROVED

Rvachev’s up-function satisfies

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1))$$

for all real x . The equation combines differentiation, dilation, and two translations. Together with compact support, smoothness, evenness, and normalization it forms the original characterization and generates higher derivative and refinement identities.

Rvachev polynomial deconvolution.

LEAN-PROVED

Rvachev polynomial deconvolution expresses a target polynomial as an average or finite combination of translated Rvachev Appell polynomials. It is exact on each finite polynomial degree and should not be confused with unstable deconvolution of arbitrary noisy functions. The compact-support law supplies the convolution operator; the Appell sequence supplies its polynomial inverse.

Rvachev up-function.

PROJECT / MIXED

Rvachev’s up-function is the even compactly supported bump $\text{up}(x) = F(1 - |x|)$ obtained from the bounded Fabius function. It is nonnegative, supported exactly on $[-1, 1]$, normalized by $\text{up}(0) = 1$, strictly increasing on $[-1, 0]$, and strictly decreasing on $[0, 1]$. It is the centered density form of the same probability law, has an entire sinc-product Fourier transform, satisfies refinement identities, and is a canonical atomic function.

19 S**Saddle central arc.**

LEAN-PROVED

The central arc is the portion of a contour within a shrinking neighborhood of the saddle where a finite Taylor expansion of the phase and amplitude is uniformly accurate. Its width balances Taylor error against Gaussian tail loss. The central contribution yields the coefficient expansion; it is only one part of the full inversion proof.

Saddle coefficient recurrence.

PROJECT / MIXED

A saddle coefficient recurrence computes the coefficients of a local integral expansion from lower-order phase and amplitude data. Gaussian parity eliminates odd total powers, while each surviving coefficient is a finite combination of jets and double-factorial moments. The repository supplies recurrences suitable for Lean induction as well as explicit closed forms indexed by ordered compositions and subsets. The logarithmic coefficients are then obtained from the mass coefficients by a formal-log recurrence.

Saddle equation.

LEAN-PROVED

For an integral with exponent $\Phi(z; \lambda)$, the saddle equation is $\partial_z \Phi = 0$. Solving it exactly or asymptotically selects the local center of mass for steepest descent. In the Fabius Bromwich problem it reduces to a Lambert equation after isolating the dominant negative-Laplace terms.

Saddle Hessian.

CLASSICAL

In several variables the Hessian is the matrix of second derivatives of the phase at the saddle. A nonsingular Hessian gives a multidimensional Gaussian main term with determinant factor $(\det(-\nabla^2\Phi))^{-1/2}$, after a branch and orientation are chosen. In one dimension it reduces to the second derivative.

Saddle jet.

PROJECT / MIXED

A saddle jet is the finite list of derivatives of the phase or amplitude at the saddle, usually normalized by powers of the large parameter so that each component has a finite limit or periodic dependence. These jets determine every fixed order of the steepest-descent expansion. Repository modules derive closed forms for the Fabius saddle jets using Stirling-number algebra and the periodic Mellin correction, then feed them into generic coefficient machinery.

Saddle mass.

PROJECT / MIXED

The saddle mass is the integral contribution from the central neighborhood after the dominant exponential has been extracted. Its leading term is Gaussian, proportional to the inverse square root of the phase curvature. Higher corrections account for nonquadratic phase terms and amplitude variation. In the Fabius problem the normalized saddle mass has an all-orders inverse- λ expansion with periodic coefficients inherited from the negative-Laplace decomposition.

Saddle point.

PROJECT / MIXED

A saddle point of a complex phase Φ is a critical point where $\Phi'(z_0) = 0$. Along suitable directions the real part decreases away from the point, concentrating a contour integral. The Fabius saddle is a real negative point in the Bromwich variable, or equivalently a positive large parameter λ satisfying $\lambda 2^{-\lambda} = x$. Its exact location is expressed through W_{-1} .

Saddle-point method.

PROJECT / MIXED

The saddle-point method estimates integrals or coefficients with a large parameter by expanding the phase around a critical point. A complete proof identifies the saddle, controls contour deformation, evaluates the Gaussian central contribution, expands corrections, and bounds tails uniformly. The repository implements each of these steps for the Fabius endpoint and extends the calculation to all orders. The method is closely related to steepest descent and Laplace's method.

Saddle radius.

PROJECT / MIXED

The saddle radius is the cutoff separating the central region, where a Taylor expansion of the phase is accurate, from the tail or minor arc. It is typically a slowly growing function of the saddle parameter. Choosing it too small discards Gaussian mass; choosing it too large invalidates local remainder bounds. The Fabius radius modules prove explicit inequalities that balance these effects at each desired expansion order.

Saddle tail bound.

LEAN-PROVED

A saddle tail bound controls the contour outside the central arc by proving a strict drop in the real part of the phase or a transform decay estimate. Without it, a local Gaussian calculation does not establish an asymptotic for the original integral. The current Fabius proof separates central, intermediate, and remote arcs.

Saddlepoint density approximation.

FRONTIER

A saddlepoint approximation evaluates a density through the solution $\Lambda'(t_x) = x$ of the tilted log-MGF equation:

$$f(x) \approx \frac{\exp(\Lambda(t_x) - t_x x)}{\sqrt{2\pi\Lambda''(t_x)}}.$$

Higher corrections use tilted cumulants. Endpoint degeneracy and lattice effects require modifications.

Sampling comb.

PROJECT / MIXED

A sampling comb is a discrete family of nodes, often an affine lattice $(k + \theta)/M$, together with weights. In the Rvachev reconstruction theorems, up-function atoms centered at the comb nodes reproduce polynomials after deconvolution. Mesh, phase, endpoint truncation, and normalization are all part of the object.

Sampling identity.

CLASSICAL

A sampling identity relates sums of function values on a lattice to sums of transform values on the reciprocal lattice. Poisson summation gives $\sum_{m \in \mathbb{Z}} \text{up}(am) = a^{-1} \sum_{m \in \mathbb{Z}} \widehat{\text{up}}(m/a)$ under the repository's convention. Compact support can reduce the spatial sum to only a few terms for large spacing, while the transform side remains an infinite rapidly convergent series. This yields exact identities for selected up-values.

Scalar extension.

CONVENTION

Scalar extension maps an algebraic object over a field or ring into one over a larger scalar system, such as embedding $\mathbb{Q}[x]$ into $\mathbb{C}[x]$. Polynomial identities, finite sums, and recurrences are preserved by ring homomorphisms. The Fabius complex-shift modules prove rational identities once and extend them to complex parameters, keeping analytic convergence arguments separate from algebraic coercions.

Scaling law.

CLASSICAL

A scaling law relates values of a function or distribution under dilation. The global Fabius law $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ and its derivative iterates are exact scaling laws. In transforms, dilation changes frequency inversely and contributes a Jacobian factor. Discrete-scale invariance at ratio two is the common source of dyadic arithmetic, sinc products, and log-periodic endpoint corrections.

Schröder equation.

FRONTIER

Schröder's equation

$$\phi(f(x)) = \lambda\phi(x)$$

conjugates a nonlinear map near a fixed point to multiplication by $\lambda = f'(x_0)$. When ϕ is locally invertible, $f^{\circ t}(x) = \phi^{-1}(\lambda^t \phi(x))$ suggests fractional iterates. Parabolic or flat fixed points require different coordinates.

Schur complement.

PROJECT / MIXED

For a block matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with A invertible, the Schur complement is $D - CA^{-1}B$. It controls determinants, positive definiteness, conditional covariance, and least-squares residuals. Exact product formulas for geometric-family prediction errors arise from successive Schur complements of covariance matrices.

Schwartz function.

CLASSICAL

A Schwartz function is C^∞ and, together with every derivative, decays faster than every polynomial: $\sup_x |x^m f^{(n)}(x)| < \infty$ for all m, n . Every compactly supported smooth function is Schwartz. Thus the up-function belongs to the Schwartz space, and its Fourier transform is also Schwartz on the real axis. This guarantees strong inversion and Poisson-summation properties.

Second-order Eulerian number.

FRONTIER

Second-order Eulerian numbers count Stirling permutations by descents and occur in expansions related to increasing trees and repeated derivative operators. They are not squares or iterates of ordinary Eulerian numbers. The indexing and boundary convention are declared with the family.

Second saddle correction.

PROJECT / MIXED

The second saddle correction is the coefficient of $1/\lambda^2$ in the normalized or logarithmic saddle expansion. It combines phase and amplitude derivatives through higher order and includes cross terms involving the first correction. The repository isolates this coefficient in dedicated modules both as a check on the generic all-orders machinery and as a sharper practical endpoint approximation.

Sectorial asymptotic expansion.

CLASSICAL

A sectorial expansion is valid as a complex variable tends to a limit inside a specified sector. Powers, logarithms, and exponentials use branches compatible with that sector. Crossing a boundary ray can change exponentially small terms or activate a Stokes phenomenon, even when the Poincaré power series remains unchanged.

Self-convolution.

CLASSICAL

Self-convolution is $f * f$, representing the sum of two independent variables with density f . A self-similar density may satisfy an exact relation between its self-convolution and a scaled or shifted version of itself. The up-function has such identities inherited from splitting its infinite random series into interlaced dyadic scales. Fourier transforms turn them into elementary product identities.

Self-sampling identity.

FRONTIER

A self-sampling identity reconstructs a function, transform, or polynomial using samples of the same distinguished kernel rather than an unrelated basis. For the up-function, partition-of-unity and translate-synthesis formulas supply exact self-referential sampling patterns. Convergence and finite support depend on the precise identity.

Self-similarity.

CLASSICAL

Self-similarity means that an object is reproduced, exactly or statistically, after rescaling and possibly translation or averaging. The Fabius random series splits into a first uniform component plus a scaled independent copy of itself. This distributional identity yields the contraction equation, differential scaling, and refinement structure. Because the scale ratio is discrete, the endpoint asymptotic retains periodic information in a logarithmic phase.

Semantic API.

CONVENTION

A semantic API is a collection of notation commands named by mathematical meaning rather than visual appearance. Commands such as a specified Fourier transform, binary digit sum, or generalized q -Pochhammer symbol encode arity and normalization in their names. This makes large-scale migration and collision auditing possible and reduces the risk that identical glyphs conceal different objects.

Sequential computability.

CLASSICAL

A real function is sequentially computable if it maps every computable input sequence to a computable output sequence by one uniform algorithm. This formulation avoids choosing a particular representation for a single arbitrary real while still enforcing effective dependence on input data. In Grzegorzczuk's framework, sequential computability plus effective uniform continuity defines computability of a continuous real function. The finite-spline Fabius algorithm supplies the sequential component.

Series-reversion coefficient.

PROJECT / MIXED

For $f(w) = \sum_{j \geq 1} f_j w^j / j!$ with $f_1 \neq 0$ and $f^{(-1)}(z) = \sum g_n z^n / n!$, each g_n is a rational polynomial in $f_1^{-1}, f_2, \dots, f_n$. Bell-polynomial and partition formulas encode these coefficients. Factorial normalization must be fixed before comparing formulas.

Sharp asymptotic.

CLASSICAL

A sharp asymptotic includes every contribution needed to attain a specified small remainder, not merely the leading growth order. For $F(x)$ near zero, sharpness requires the exact lower-Lambert phase, the nonconstant periodic correction, the Gaussian prefactor, and stated inverse-saddle errors. A coarser quadratic logarithmic law remains true but is not sharp enough for asymptotic equivalence or first correction terms.

Sharp constant.

CLASSICAL

A constant in an inequality is sharp if it cannot be decreased while preserving the statement. Sharpness is usually proved by exhibiting points or sequences where equality holds or the ratio approaches the constant. The derivative bounds $2^{\binom{m+1}{2}}$ for the Fabius-related functions are sharp because exact dyadic points attain them. The global Lipschitz constant 2 is similarly attained at the midpoint.

Sharp derivative bound.

PROJECT / MIXED

The repository proves $\|F^{(m)}\|_\infty, \|\text{up}^{(m)}\|_\infty, \|\mathcal{F}^{(m)}\|_\infty \leq 2^{\binom{m+1}{2}}$, with equality in each case. The bound follows from derivative scaling and the unit bound on the signed extension; equality follows at carefully chosen dyadic arguments. These estimates imply effective flatness, coefficient decay, and stable uniform approximation. They are exact rather than merely exponential-order bounds.

Sharp transfer.

CLASSICAL

A sharp transfer theorem converts asymptotic information for one analytic object into equally precise information for another while preserving constants, periodic phases, and error orders. In the Fabius directory, transfer modules pass from negative-Laplace and saddle estimates to endpoint values, from the continuous endpoint formula to inverse-power sequences, and from the Lambert coordinate to elementary logarithms. The adjective sharp means no term at the target error scale is lost.

Sheffer sequence.

LEAN-PROVED

A Sheffer sequence has exponential generating function $A(z)e^{xB(z)}$, where $A(0) \neq 0$, $B(0) = 0$, and $B'(0) \neq 0$. Appell sequences are the case $B(z) = z$. Sheffer addition, lowering operators, and exponential Riordan arrays organize changes among polynomial bases used in coefficient and reversion calculus.

Shifted spline.

CLASSICAL

A shifted spline is a spline evaluated after translation, possibly by a complex parameter in a polynomial extension. Real shifts move support and knots; complex shifts are algebraic evaluations of the polynomial pieces rather than probability densities. The Fabius complex-shift spline modules prove convergence for fixed shifts and then combine the approximants with Gaussian q -binomial weights. Common-shift invariance belongs to the finite Taylor functional, not necessarily to each finite spline limit row.

Signed global series.

PROJECT / MIXED

The signed global Fabius representation writes the global extension as a Thue–Morse-signed sum of integer translates of the bounded function, with the normalization fixed in the defining module. Because the bounded Fabius function has constant tails and the translate coefficients cancel in complete binary blocks, the formal expression must be read through the proved locally-finite or finite-reduction identity appropriate to the theorem. It agrees with the bounded function on $[0, 1]$, satisfies the pure global differential-dilation law, and is not itself a cumulative distribution function.

Signed measure.

CLASSICAL

A signed measure is a countably additive set function that may take positive and negative values but has finite total variation on the relevant space. It decomposes into positive and negative parts by the Jordan decomposition. Thue–Morse translate sums may be viewed as signed combinations of compactly supported pieces. The signed global Fabius function is not itself a probability CDF, even though it is built from the same local density.

Signed-natural numerator.

CLASSICAL

A signed-natural numerator is a pair $(c^+, c^-) \in \mathbb{N}^2$ representing the integer $c^+ - c^-$. It is redundant because many pairs represent the same integer, but addition, subtraction, and primitive recursion can be implemented using only natural numbers. At dyadic precision p , the represented rational is $(c^+ - c^-)/2^p$. This coding is used in the repository's fast names for computable real sequences.

Signed Stirling numbers of the first kind.

CLASSICAL

The signed Stirling number $s(n, k)$ is the coefficient of x^k in x^n . It obeys $s(n+1, k) = s(n, k-1) - n s(n, k)$ and $s(0, 0) = 1$. The sign convention is part of the name; unsigned first-kind numbers are absolute values and count permutations by cycles.

Signed Thue–Morse prefix.

LEAN-PROVED

A signed Thue–Morse prefix is the finite list $(\varepsilon_0, \dots, \varepsilon_{N-1})$ or its partial sum. Complete powers-of-two prefixes sum to zero after the first level, while incomplete prefixes obey a reflection recursion. Repeated summation of these prefixes produces polynomial arrays connected to Fabius dyadic values.

Signed-weight random series.

LEAN-PROVED

A signed-weight random series permits positive and negative deterministic coefficients in $\sum a_j U_j$. Absolute summability of the weights gives almost-sure and L^p convergence for bounded summands and determines a compact support interval by extremal choices. Sign changes affect the affine center and support but not the product structure of centered characteristic functions.

Sinc function.

CLASSICAL

The sinc function is $\text{sinc } z = \sin z / z$ with removable value 1 at $z = 0$. It is entire and has simple zeros at nonzero integer multiples of π before normalization. The Fourier transform of a centered uniform density is a sinc factor. Multiplying the factors from all dyadic-scale uniforms gives the up-function's entire Fourier product.

Sinc-lobe sign.

PROJECT / MIXED

Between consecutive real zeros, a sinc factor has constant sign. The sign of a product is therefore determined by how many factors lie in negative lobes. For the dyadic up product this parity can be expressed through binary or Thue–Morse data on integer-separated intervals. Lobe sign is pointwise real information, not a complex phase convention.

Sinc product.

CLASSICAL

The up-function Fourier transform is $\widehat{\text{up}}(z) = \prod_{n=0}^{\infty} \text{sinc}(\pi z/2^n)$. The product converges locally uniformly and defines an entire function. Its zeros at nonzero integers imply exact periodization and partition-of-unity identities; repeated factors and scale splitting encode convolution laws. Along the real axis, the accumulation of factors gives rapid decay.

Small-argument asymptotic.

CLASSICAL

A small-argument asymptotic describes a function as $x \rightarrow 0$. For the Fabius function this regime is singular because all endpoint derivatives vanish, so no nontrivial Taylor series exists. The correct expansion is obtained indirectly through a negative-Laplace transform and a moving saddle. It includes a quadratic logarithmic main scale, nested-log corrections, a lower-Lambert phase, and smooth periodic coefficients.

Smooth compact support.

CLASSICAL

A function has smooth compact support when it is C^∞ and vanishes outside a compact set. Such functions form the test-function space $\mathcal{D}(\mathbb{R})$ and are automatically Schwartz. Their Fourier transforms are entire of exponential type and rapidly decaying on the real axis. The up-function is a highly structured example, distinguished by its refinement equation and exact arithmetic.

Source checkpoint.

CONVENTION

A source checkpoint is a recorded commit identifier at which a module inventory or theorem list was compiled. It makes a statement reproducible even when the main branch continues to evolve. A short hash is adequate as a locator only when its repository is clear. Checkpoints attached to individual tranches do not imply that every current file was frozen at the same commit.

Source module.

CONVENTION

A source module is a Lean file participating in the imported development. Its mathematical subject may be narrow—for example one q -identity, one moment recurrence, or one approximation bridge—even when it exports several public declarations. Module counts therefore measure formal organization, not the number of independent mathematical theories or glossary concepts.

Source–PDF parity.

CONVENTION

Source–PDF parity means that a rendered PDF incorporates the mathematical content currently present in its LaTeX source. A live research repository can contain post-render source additions, so a PDF may be valid yet incomplete relative to the head source. Audits record this condition explicitly; a glossary based on current state must inspect source as well as retained PDFs.

Spectral measure.

CLASSICAL

A spectral measure represents a self-adjoint operator or a moment functional by integration against its spectrum. For a cyclic vector, moments are matrix elements of powers of the operator. Orthogonal polynomials, Jacobi matrices, Stieltjes transforms, and quadrature nodes are different views of the same measure.

Spectral profile inversion.

FRONTIER

If integer-zero multiplicities satisfy $m_v = \sum_{h \leq v} c_h(v - h + 1)$, then the profile is recovered by the second difference $c_v = m_v - 2m_{v-1} + m_{v-2}$. Thus admissible zero ledgers are exactly the nonnegative discrete-convex multiplicity sequences with the stated boundary conventions.

Spectral zeta function.

FRONTIER

A spectral zeta function is $\zeta_L(s) = \sum_{\lambda_j > 0} \lambda_j^{-s}$ in its convergence half-plane, continued when possible. Poles encode eigenvalue asymptotics, and $-\zeta'_L(0)$ defines a regularized determinant when continuation is regular at zero. Any Fabius-transform version must state the chosen multiplicity and zero enumeration.

Spline.

CLASSICAL

A spline is a piecewise-polynomial function with prescribed continuity of derivatives across knots. Repeated convolution of box functions produces B-splines, whose degree and smoothness increase together. The finite Fabius approximants are explicit centered splines on dyadic knots. They converge uniformly to a C^∞ limit even though each finite approximant has only finite smoothness.

Square-summable activation profile.

LEAN-PROVED

A square-summable activation statement controls a sequence of scale differences, derivatives, or deficits in ℓ^2 . Such estimates are stronger than pointwise convergence and support energy identities, Hilbert space arguments, and stable reconstruction. In the Fabius setting the scale recursion and compact transition make the relevant activation increments explicitly summable.

Stable endpoint evaluation.

PROJECT / MIXED

Stable endpoint evaluation computes extremely small Fabius values without losing all relative accuracy. Direct subtraction, naive transform inversion, or standard floating-point arithmetic may underflow or suffer cancellation. Reliable methods use exact dyadic formulas, high-precision logarithmic asymptotics with the periodic phase, or monotone recurrences. The repository discusses operation choice, residual checks, and Newton refinement in a regime where the value itself may be far below machine range.

Standardized cumulant.

PROJECT / MIXED

For variance $\sigma^2 > 0$, the standardized cumulant is $\lambda_r = \kappa_r / \sigma^r$, $r \geq 3$. It is invariant under positive rescaling and measures non-Gaussian shape. In geometric-uniform limits these cumulants decay with explicit powers of the small mesh parameter and generate Edgeworth corrections.

Standardized geometric uniform law.

FRONTIER

A standardized geometric uniform variable subtracts its mean and divides by its standard deviation. As the ratio approaches the Gaussian boundary $q \uparrow 1$, standardized cumulants organize Edgeworth and Cornish–Fisher expansions. The standardization constant must be stated, because source drafts also use centered variables on fixed support without variance normalization.

Stars-and-bars principle.

CLASSICAL

Stars and bars is the bijection proving that weak compositions of n into r parts are counted by $\binom{n+r-1}{r-1}$. Positive compositions are obtained by subtracting one from every part. The principle explains binomial kernels in repeated summation and simplex-volume factors in iterated integration.

Stationary point.

CLASSICAL

A stationary point of a differentiable real or complex phase is a point where the first derivative vanishes. It becomes a saddle suitable for asymptotics only when curvature and contour geometry make the contribution dominant. The Fabius audit examines an informal stationary substitution into a Stirling proxy and shows that the exact stationarity equation leaves a linear residual. Thus locating a stationary point does not by itself validate a claimed asymptotic simplification.

Steepest descent.

CLASSICAL

Steepest descent is a contour method in which the path is chosen through a saddle along directions where the imaginary part of the phase is constant and the real part decreases most rapidly. It turns the local integral into a Gaussian model with controlled corrections. The Bromwich saddle method used for Fabius is a specialized steepest-descent analysis adapted to inverse Laplace integration and a dyadic transform product.

Steepest-descent contour.

PROJECT / MIXED

A steepest-descent contour passes through a saddle along directions where the imaginary part of the phase is constant and the real part decreases most rapidly. Deforming to it requires holomorphy and control of connecting arcs. In a vertical Bromwich problem one may instead prove equivalent central and tail estimates without naming a global contour.

Stein equation.

FRONTIER

For a test function h , the Stein equation is

$$\mathcal{A}f_h = h - \int h \, d\mu.$$

Bounds on the solution f_h convert expectation identities into probability metrics. The norm, derivative order, and test class determine which distance is controlled.

Stein operator.

FRONTIER

A Stein operator $\mathcal{A}$ characterizes a probability law μ on a test class $\mathcal{T}$ when $\int \mathcal{A}f \, d\mu = 0$ for every $f \in \mathcal{T}$, together with a uniqueness theorem. An integration-by-parts identity alone is not a complete characterization. Proposed Fabius/Rvachev Stein operators retain frontier status until the class and uniqueness are established.

Stieltjes constant.

CLASSICAL

The Stieltjes constants γ_n are defined by the Laurent expansion $\zeta(s) = 1/(s-1) + \sum_{n \geq 0} (-1)^n \gamma_n (s-1)^{-n-1}/n!$, with $\gamma_0 = \gamma$. They appear when Mellin-zeta expressions are expanded near a pole to higher order. Fabius finite-part and all-orders calculations may involve them in constant or derivative coefficients of Gamma-zeta products.

Stieltjes inversion formula.

LEAN-PROVED

Stieltjes inversion reconstructs a measure from boundary values of its Cauchy transform. In the absolutely continuous case, the density is a constant-sign multiple of $\lim_{\eta \downarrow 0} \operatorname{Im} G(x + i\eta)$, determined by the denominator convention. Interval versions recover mass even when a density is absent.

Stieltjes moment sequence.

PROJECT / MIXED

A sequence (m_n) is a Stieltjes moment sequence when $m_n = \int_0^\infty x^n d\mu(x)$ for a positive measure μ . Equivalently, both the Hankel matrices (m_{i+j}) and shifted Hankel matrices (m_{i+j+1}) are positive semidefinite. Compactly supported Fabius/Rvachev moments are Hausdorff moments after an affine change, hence automatically determinate.

Stieltjes–Padé approximation.

FRONTIER

For a Stieltjes function, continued-fraction convergents and diagonal Padé approximants give alternating or monotone bounds on positive real rays under standard hypotheses. Moment positivity prevents spurious nonreal poles. Applying this framework to transforms built from Rvachev moments yields certified approximants distinct from direct spline truncations.

Stieltjes–Perron inversion.

CLASSICAL

The Stieltjes–Perron formula recovers interval masses from the jump or imaginary boundary values of a Cauchy transform. Endpoint atoms receive convention-dependent half weights in symmetric limits. It connects analytic resolvents to CDFs and spectral distributions.

Stieltjes transform.

PROJECT / MIXED

A Stieltjes transform commonly means the Cauchy transform of a measure on a half-line, or the variant $\int (x+z)^{-1} d\mu(x)$. Sign and domain conventions vary. The repository prints the denominator and develops fixed-point, logarithmic, and generalized-order resolvent hierarchies for Fabius-related measures.

Stirling formula.

CLASSICAL

Stirling’s formula approximates the factorial: $n! = \sqrt{2\pi n}(n/e)^n(1 + O(1/n))$, or logarithmically $\log n! = n \log n - n + \frac{1}{2} \log(2\pi n) + O(1/n)$. Explicit Robbins bounds provide a rigorous two-sided error. The Fabius audit uses Stirling’s formula both in endpoint sequence asymptotics and to expose a residual linear term missed by an informal stationary substitution.

Stirling numbers.

CLASSICAL

Stirling numbers of the first kind convert falling factorials to ordinary powers, while those of the second kind convert ordinary powers to falling factorials. They also encode derivatives of logarithmic and exponential composites. The Fabius saddle-jet modules use Stirling-number identities to obtain closed forms for repeated derivatives of dyadic kernel expressions and Lambert-dependent phases. Sign conventions for first-kind numbers must be specified.

Stirling numbers of the second kind.

CLASSICAL

The Stirling number $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ counts partitions of an n -element set into k nonempty blocks. It is the coefficient in $x^n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^{\underline{k}}$. Its triangular recurrence and zero extensions are fixed at the boundaries.

Stokes line.

FRONTIER

A Stokes line is a ray or curve across which an exponentially small contribution changes its asymptotic multiplier while dominant exponential orders exchange activity. The underlying analytic function remains continuous after correct continuation. A Stokes statement requires a chosen phase, branch, and sector; it cannot be inferred from a real-axis expansion alone.

Stopped primitive.

LEAN-PROVED

A stopped primitive integrates only after a threshold, for example

$$(Pf)(x) = \int_a^x f(t) \, dt$$

on $x > a$ and assigns a fixed exterior value on the other side. Stopping preserves support or tail conventions while solving a derivative equation. Iterated stopped primitives appear in CDF construction and compact-support integration ladders.

Strang–Fix condition.

CLASSICAL

A compactly supported generator reproduces polynomials through degree d on a lattice when its Fourier transform has zeros of order at least $d + 1$ at all nonzero dual-lattice points, together with a normalization at zero. These Strang–Fix conditions translate transform zero multiplicities into approximation order and explain the role of $v_2(M)$ in Rvachev comb exactness.

Strict interior positivity.

LEAN-PROVED

Strict interior positivity means $\text{up}(x) > 0$ exactly for $-1 < x < 1$, with zero values outside and at the endpoints. It is stronger than nonnegativity almost everywhere. The result identifies the exact pointwise support interior, proves strict increase of F , and ensures that the density relation has no hidden zero plateaus.

Strict interior range.

LEAN-PROVED

The strict interior range statement says $0 < F(x) < 1$ for $0 < x < 1$. It follows from the strict positivity of the up-density and monotonicity of the CDF. This excludes flat subintervals, supplies

injectivity on the open unit interval, and makes the canonical inverse a genuine order isomorphism there.

Strict unimodality.

CLASSICAL

A function is strictly unimodal if it increases strictly up to a unique mode and decreases strictly afterward. Rvachev's up-function is strictly increasing on $[-1, 0]$, attains its unique maximum 1 at zero, and is strictly decreasing on $[0, 1]$. Its derivative signs follow from the positive bounded Fabius function under affine scaling. Strictness strengthens mere symmetry and nonincreasing shape.

Strip of convergence.

CLASSICAL

For a Mellin transform, the strip of convergence is the vertical region $a < \Re s < b$ where the defining integral converges absolutely. Endpoint behavior at zero controls one boundary and behavior at infinity controls the other. Mellin inversion begins on a vertical line inside this strip; asymptotic analysis then uses meromorphic continuation and contour shifting. The regularized Fabius kernels enlarge the workable region by subtracting singular endpoint terms.

Sturm-Liouville equation.

CLASSICAL

A Sturm-Liouville equation has the form $-(p(x)y')' + q(x)y = \lambda w(x)y$ with boundary conditions. Its eigenfunctions are orthogonal with respect to the weight w . Legendre polynomials solve the singular equation $-((1 - x^2)y')' = n(n + 1)y$ on $(-1, 1)$. This differential characterization yields recurrences and integration-by-parts identities used in the up-function's Legendre expansion.

Subgraph Fubini identity.

CLASSICAL

The measure of the subgraph $\{(x, t) : 0 < t < f(x)\}$ can be computed by integrating first in t or first in x . The two orders give $\int f$ and an integral of level-set measures. Applied to a monotone CDF or inverse, this produces geometric area identities and moment relations with endpoint terms made explicit.

Subspace lattice.

CLASSICAL

The subspaces of a finite-dimensional vector space over $\mathbb{F}_q$, ordered by inclusion, form a finite ranked lattice. The number of k -dimensional subspaces is $\begin{bmatrix} n \\ k \end{bmatrix}_q$. Intervals are themselves subspace lattices of quotient spaces, which yields a closed Möbius function.

Summability.

CLASSICAL

Summability concerns assigning or proving limits for series, sometimes by methods extending ordinary convergence. In this directory most series are shown to converge in the ordinary sense, often absolutely or uniformly; Toeplitz rows are regular summability transformations carrying convergent sequences to the same limit. The term should not be read as permission to manipulate divergent series without a named regularization such as a finite part.

Superasymptotic approximation.

FRONTIER

A superasymptotic approximation is an optimally or near-optimally truncated asymptotic expansion whose error reaches the scale of the least term. It improves fixed-order asymptotics without introducing the next exponential sector. Hyperasymptotic methods go further by expanding that exponentially small remainder.

Supergraph Fubini identity.

CLASSICAL

A supergraph identity integrates the set $\{(x, t) : f(x) < t\}$ inside a bounded rectangle. It is complementary to the subgraph formula and naturally expresses deficits such as $1 - F(x)$. Reflection of the Fabius CDF makes the subgraph and supergraph descriptions equivalent after an affine change of variables.

Support.

CLASSICAL

The support of a continuous function is the closure of the set where it is nonzero. For a measure it is the smallest closed set of full measure, equivalently the points every neighborhood of which has positive mass. The up-function has support exactly $[-1, 1]$, while the bounded Fabius probability law has support $[0, 1]$. Endpoints belong to the support despite zero function values there.

Support-as-range principle.

LEAN-PROVED

For a continuous nonnegative density, the topological support is the closure of the set where the density is positive. In the Fabius system the exact interior positivity and endpoint zeros identify the support of up as $[-1, 1]$, while strict monotonicity identifies the range of F on $[0, 1]$ as $[0, 1]$. These facts justify canonical inverse domains rather than merely almost-everywhere statements.

Support of an indexed family.

CLASSICAL

For an indexed family $(a_i)_{i \in I}$, its support is $\{i \in I : a_i \neq 0\}$. Finite support turns an apparently infinite sum into a finite one, while local finiteness says only finitely many terms affect each evaluation point. This discrete notion is distinct from topological support of a function or measure, though translate constructions use both simultaneously.

Supremum norm.

CLASSICAL

The supremum norm is $\|f\|_\infty = \sup_x |f(x)|$ on a specified domain. Spaces of bounded continuous functions are complete in this norm, enabling the Banach fixed-point construction. Uniform convergence is convergence in supremum norm. Sharp derivative bounds and finite-spline approximation errors in the Fabius development are stated in this norm.

Survival function.

CLASSICAL

The survival function of a real random variable is $\bar{F}(x) = \mathbb{P}(X > x) = 1 - F(x)$ under the stated endpoint convention. Reflection can identify a survival tail with a CDF tail at a mirrored point. Layer-cake formulas express positive moments as integrals of survival probabilities.

Symmetric mixed difference.

LEAN-PROVED

A symmetric mixed difference uses paired shifts $x \pm h_j/2$ or $x \pm h_j$, producing parity-adapted cancellation. Odd or even polynomial components then vanish automatically. Such operators connect centered moments, even up-function symmetry, and factorized sinc products.

Symmetry.

CLASSICAL

Symmetry is invariance or a simple covariance under a transformation. The bounded Fabius CDF has complement reflection symmetry about $1/2$, and the up-function has even symmetry about zero. Symmetry determines means, cancels odd moments and coefficients, transfers endpoint formulas, and locates the unique inflection point. It also reduces computations to half of the support.

20 T**Tail estimate.**

CLASSICAL

A tail estimate bounds the contribution from indices or integration regions beyond a cutoff. For infinite random sums it controls the omitted variables; for series it controls the remainder; for contour integrals it controls remote arcs. The dyadic random tail has an explicit deterministic bound, yielding the centered spline sandwich. Saddle tail estimates show that regions outside the central radius are negligible at the desired inverse- λ order.

Tail flatness.

CLASSICAL

Tail flatness refers to rapid vanishing or derivative control in the remote part of a scaled kernel or contour integrand. In the Fabius saddle proof, it supplies uniform bounds when the local Taylor approximation is no longer used. The phrase is repository-specific: it packages estimates showing that the reference tail and true tail differ by less than the target asymptotic remainder.

Tanh divided slope.

LEAN-PROVED

The tanh divided slope is the regularized quotient $\tanh x/x$, assigned the removable value 1 at $x = 0$, or an affinely scaled analogue used to compare activation derivatives. Positivity, evenness, and monotonicity properties turn pointwise slope estimates into summable scale bounds. The current primary exposition includes this calculus as a bridge between Fabius activation profiles and classical hyperbolic functions.

Taylor block.

CLASSICAL

A Taylor block is a finite Taylor polynomial contribution associated with one dyadic scale in the binary reduction. Because derivatives at a dyadic center vanish above a known order, the block is exact rather than an approximation. Its coefficients are expressed through scaled values of the signed global extension. Gaussian q -binomial identities provide an alternative complex-shift representation of the same block.

Taylor polynomial.

CLASSICAL

The degree- n Taylor polynomial of f at x_0 is $T_n(x) = \sum_{k=0}^n f^{(k)}(x_0)(x - x_0)^k/k!$. It approximates f locally when a remainder estimate is available. At reduced dyadic Fabius points, the formal Taylor series terminates exactly at the denominator exponent, yet the polynomial does not equal the function on any neighborhood. This is a striking separation between jet data and analyticity.

Taylor series.

CLASSICAL

The Taylor series of a smooth function at x_0 is the formal series of its derivatives. Analyticity requires that this series converge to the function locally; smoothness alone does not. Fabius endpoint Taylor series are constant because of flatness, and interior dyadic Taylor series are finite polynomials, but neither represents the nonconstant function nearby. Transform Taylor series, by contrast, are entire and valid globally.

Terminating basic hypergeometric series.

LEAN-PROVED

A basic hypergeometric series terminates when an upper parameter equals q^{-N} for some $N \in \mathbb{N}$, because $(q^{-N}; q)_n = 0$ for $n > N$. It is then a finite sum, so analytic convergence is irrelevant, though denominator nonvanishing remains necessary. Many Gaussian and Fabius–Thue–Morse identities are best proved in this finite setting.

Terminating Taylor series.

CLASSICAL

A Taylor series terminates when all derivatives above some order vanish at the center, so the formal series is a polynomial. At $x_0 = a/2^n$ with odd a , the Fabius derivatives vanish above order n because repeated dyadic scaling lands outside the active integer pattern. The order- n derivative is nonzero, so the degree is exactly n . Termination does not imply local polynomiality for a merely smooth function.

Termination index.

CONVENTION

The termination index is the largest n for which a terminating q -hypergeometric summand may be nonzero. An upper parameter q^{-N} usually gives index N , but denominator zeros can shorten the meaningful range or make a term undefined. Explicitly recording the index keeps finite proofs independent of convergence arguments.

Termwise differentiation.

CLASSICAL

Termwise differentiation replaces the derivative of a series by the series of derivatives. It is valid under conditions such as locally uniform convergence of the derivative series together with convergence at one point, or normal convergence for holomorphic series. The rapid Fourier and periodic-coefficient convergence in the repository permits repeated differentiation. Formal power-series differentiation is algebraic and requires no analytic hypothesis, so the two settings must be distinguished.

Termwise integration.

CLASSICAL

Termwise integration interchanges a sum and an integral. Tonelli's theorem covers nonnegative terms, Fubini covers absolute integrability, and dominated convergence covers limits of partial sums under a common majorant. Fabius moment, Fourier, and Laplace product derivations use these principles.

A merely conditionally convergent signed series requires additional care and is not automatically integrable term by term.

Test-function space.

CLASSICAL

The test-function space $\mathcal{D}(\mathbb{R})$ consists of infinitely differentiable compactly supported functions, equipped with a standard locally convex topology. Its elements are used to define distributions and weak derivatives. The up-function is an explicit member of $\mathcal{D}(\mathbb{R})$. In the original literature it is notable as a highly structured test function satisfying exact refinement and partition identities.

Theta quasi-periodicity.

LEAN-PROVED

For the multiplicative theta normalization,

$$\theta(qz; q) = -z^{-1}\theta(z; q).$$

Iteration gives a monomial factor times $\theta(z; q)$. This is quasi-periodicity rather than ordinary periodicity: the function changes by an explicit multiplier. The multiplier controls zero propagation and root-of-unity scaling.

Three-term recurrence for orthogonal polynomials.

PROJECT / MIXED

Monic orthogonal polynomials obey

$$\pi_{n+1}(x) = (x - \alpha_n)\pi_n(x) - \beta_n\pi_{n-1}(x), \quad \beta_n > 0,$$

for a positive measure with infinite support. The coefficients are obtained from inner products or Hankel determinants. Symmetry of the measure forces $\alpha_n = 0$. This recurrence is distinct from the functional-differential recurrences defining Fabius objects.

Thue–Morse bit.

LEAN-PROVED

The Thue–Morse bit is $t_n = s_2(n) \bmod 2$, where $s_2(n)$ is the binary digit sum. The sign convention is $\varepsilon_n = (-1)^{t_n}$. Bits satisfy $t_{2n} = t_n$ and $t_{2n+1} = 1 - t_n$, while signs satisfy the multiplicative versions. The project uses both forms and never identifies a bit in $\{0, 1\}$ with a sign in $\{\pm 1\}$.

Thue–Morse block product.

CLASSICAL

The generating polynomial of a complete Thue–Morse block factors as

$$\sum_{n=0}^{2^m-1} \varepsilon_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}).$$

The factorization follows by binary digit independence. Its zero of order m at $z = 1$ is equivalent to vanishing signed power sums through degree $m - 1$.

Thue–Morse gamma product.

LEAN-PROVED

A Thue–Morse gamma product is a finite or convergent product of shifted gamma factors whose exponents are Thue–Morse signs. Prouhet cancellation removes low-order terms in its logarithmic expansion. Reciprocal-gamma normalization turns pole bookkeeping into an entire zero divisor when possible.

Thue–Morse sequence.

PROJECT / MIXED

The Thue–Morse sequence is the binary automatic sequence $t_n = \text{wt}_2(n) \bmod 2$, satisfying $t_{2n} = t_n$ and $t_{2n+1} = 1 - t_n$. Its sign version is $\varepsilon_n = (-1)^{t_n}$. Finite generating polynomials factor digitwise and have high-order zeros at one, producing Prouhet cancellation. The sequence governs the translate signs of the global Fabius extension and exact dyadic formulas.

Thue–Morse sign.

PROJECT / MIXED

The Thue–Morse sign is $\varepsilon_n = (-1)^{\text{wt}_2(n)}$. It obeys $\varepsilon_{2n} = \varepsilon_n$, $\varepsilon_{2n+1} = -\varepsilon_n$, and multiplicative block identities when binary digits do not overlap. Integer values of the signed global Fabius extension are tied to these signs. They also determine the signs of exact highest dyadic derivatives.

Tightness.

CLASSICAL

A family of probability measures on $\mathbb{R}$ is tight if for every $\varepsilon > 0$ some compact set carries mass at least $1 - \varepsilon$ for every measure in the family. Tightness prevents mass from escaping to infinity and is central to weak compactness. Finite Fabius convolution laws have uniformly bounded compact supports, so tightness is immediate. This supports passage to the infinite convolution law.

Toeplitz lemma.

CLASSICAL

A Toeplitz lemma states that a triangular array of weights preserves the limit of a convergent sequence when row sums tend to one, individual fixed-column weights vanish or rows sample sufficiently late terms, and row total variations are uniformly bounded. The repository uses a finite-row variant: after reindexing, all sampled spline degrees are at least the row index, the mass is exactly one, and variation is uniformly bounded by a q -product ratio. Hence the weighted complex approximants converge to $\mathcal{F}$.

Toeplitz row.

CLASSICAL

A Toeplitz row is one row $(\omega_{n,j})_j$ of a triangular summability matrix applied to a sequence. Regular rows have mass tending to one, vanishing influence from fixed early indices, and uniformly bounded total variation. The Fabius complex discrete limit obtains exact row mass and variation from two evaluations of the finite q -binomial theorem. Reindexing also guarantees that every row samples only spline degrees at least n .

Tonelli theorem.

CLASSICAL

Tonelli's theorem allows the order of integration or summation to be exchanged for nonnegative measurable functions, even before finiteness is known. It proves that the iterated integrals and product integral have the same extended value. In probability constructions with nonnegative densities and

moments, it is often the cleanest justification. Alternating Thue–Morse sums require Fubini or a separate absolute-convergence argument instead.

Topological support.

CLASSICAL

Topological support is the closed set of points at which an object cannot be locally discarded. For a continuous function it is the closure of its nonzero set; for a measure it consists of points whose every neighborhood has positive measure. The distinction from the set of nonzero values matters at flat endpoints. Thus $\text{up}(\pm 1) = 0$ but $\pm 1 \in \text{supp up}$.

Total variation.

CLASSICAL

For a signed or complex measure, total variation is the mass of its variation measure. For a finite weight row it is $\sum_j |\omega_j|$, and for a function it is the supremum over partition increments. Uniform total-variation bounds make weighted averaging stable even with alternating coefficients. The q -binomial Toeplitz rows have an explicit uniform bound obtained by evaluating the finite q -binomial theorem at a negative argument.

Total-variation distance.

CLASSICAL

For probability measures μ, ν ,

$$d_{\text{TV}}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|.$$

If both have densities f, g with respect to a common measure, it equals $\frac{1}{2} \int |f - g|$. This is stronger than weak or Kolmogorov convergence in many settings; the factor $1/2$ is part of the project's convention.

Totalized definition.

CONVENTION

A totalized definition assigns a value at every input, including points where the motivating partial formula would be undefined. Lean often uses totalized division, inverse, infinite sums, or branch placeholders; useful theorems then state hypotheses under which the value matches the intended analytic object. The repository pairs total definitions with explicit zero, endpoint, and denominator lemmas so that totality is never mistaken for an unconditional identity.

Touchard polynomial.

CLASSICAL

The Touchard polynomial is $T_n(x) = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^k$. It is the n -th moment polynomial of a Poisson law with mean x and has exponential generating function $\exp(x(e^z - 1))$. At $x = 1$ it gives the Bell number.

Transfer theorem.

CLASSICAL

A transfer theorem carries a property or asymptotic from one representation to another. Examples include transferring transform decay to smoothness, saddle estimates to coefficient asymptotics, or endpoint formulas to moment sequences. The Fabius directory contains explicit transfer modules because constants, normalizations, and periodic phases can be lost in an informal passage. A sharp transfer preserves the stated remainder order.

Translate block.

LEAN-PROVED

A translate block is a finite linear combination of consecutive translates of one kernel, usually at a fixed scale. Polynomial reproduction, finite-difference annihilation, and local support are analyzed blockwise before blocks are summed across scales or modes. The term records geometry; it does not imply orthogonality.

Translate sum.

CLASSICAL

A translate sum has the form $\sum_k c_k f(x - k)$ or a scaled variant. With compact support it may be locally finite. Constant positive coefficients can form partitions of unity; Thue–Morse signs create a signed global extension. Fourier transformation converts translation into multiplication by phases, making periodicity and cancellation visible.

Translated Legendre series.

CLASSICAL

A translated Legendre series expands a function on an interval other than $[-1, 1]$ by composing Legendre polynomials with the affine map to the standard interval. The bounded Fabius function on $[0, 1]$ can be related to the up-function series on $[-1, 1]$ through translation and reflection. Coefficients and convergence must include the Jacobian and normalization of this change of variables.

Transmonomial.

CLASSICAL

A transmonomial is one multiplicative scale in a transseries, such as $x^\alpha (\log x)^m e^{-Ax}$. Transmonomials form an ordered multiplicative group. The ordering records asymptotic dominance, not numerical size at one finite point.

Transseries.

PROJECT / MIXED

A transseries extends ordinary series by allowing ordered monomials involving powers, logarithms, exponentials, and iterated combinations, for example $x^{-a} (\log x)^m e^{-bx}$. Its support must be well based so that every coefficient operation is locally finite. Inversion and composition require both formal support conditions and, for analytic use, branch and sector conditions.

Transseries composition.

PROJECT / MIXED

Composition substitutes one large or small transseries into another. Formally it requires support conditions making each output coefficient finite; analytically the inner map must remain in the outer expansion's sector and avoid branch cuts. For $t = 1/X$ and $L = \log X$, substituting $G = X + H$ changes both generators: $t \mapsto t/(1 + tH)$ and $L \mapsto L + \log(1 + tH)$.

Transseries reversion.

PROJECT / MIXED

Reversion constructs a compositional inverse in an ordered transseries class. The leading monomial must be invertible and the correction must be smaller in a way that makes coefficient determination triangular. Polynomial–logarithmic reversion solves successively for logarithmic coefficient blocks; multiple critical points require Puiseux rather than ordinary Laurent powers.

Triangle admissibility.

LEAN-PROVED

A triple (ℓ, m, n) is triangle-admissible when each index is at most the sum of the other two. For Legendre products one also requires even total degree. These finite arithmetic conditions decide whether a Gaunt coefficient can be nonzero before any factorial expression is evaluated.

Triangular cancellation.

CLASSICAL

Triangular cancellation is a layered vanishing mechanism in which lower-degree contributions disappear successively, leaving a top-degree term. In the Fabius dyadic formula, Gaussian interpolation weights annihilate node polynomials below degree n , while Thue–Morse signs independently annihilate lower power sums inside dyadic blocks. The surviving term has a triangular power-of-two normalization. Recognizing both layers explains why apparently complicated finite sums collapse exactly.

Triangular recurrence.

CLASSICAL

A recurrence is triangular when the equation for index n involves the new term with a nonzero coefficient and only terms of smaller index otherwise. It can be solved inductively and often determines uniqueness. Moment, Bernoulli, numerator, and saddle coefficient recurrences in the repository have this form. Clearing denominators may convert a rational triangular recurrence into an integer division-free one.

Triangular reduction.

CLASSICAL

Triangular reduction is the exact decomposition of a dyadic numerator into a large block, a reflected remainder, and lower-level data so that evaluation proceeds inductively. The word triangular reflects both decreasing level and the triangular exponents $\binom{n}{2}$ appearing in normalization. This reduction proves correctness of fast dyadic recursion and exposes binary reflection symmetries.

Truncated-power basis.

PROJECT / MIXED

The truncated-power function is $(x - a)_+^m = \max(x - a, 0)^m$. Finite linear combinations of such functions, plus a polynomial, represent univariate splines. Inverse Fourier transforms of finite sinc products admit exact alternating truncated-power formulas indexed by signed subset sums, making knots and local polynomials explicit.

Truncation error.

CLASSICAL

Truncation error is the difference between an infinite object and a finite partial approximation. It may arise from omitting random summands, series terms, spline levels, or asymptotic corrections. The Fabius finite random tail gives a deterministic support bound and hence a uniform CDF error; Fourier and saddle truncations use summable or exponential estimates. Explicit truncation errors turn formulas into certified algorithms.

2-adic unit.

CLASSICAL

A rational or 2-adic integer is a 2-adic unit when its 2-adic valuation is zero. For ordinary integers this means oddness. Separating powers of two from unit factors clarifies exact dyadic denominators, Reshetnikov odd numerators, and central-binomial valuations.

2-kernel of a sequence.

CLASSICAL

The 2-kernel of a sequence $a(n)$ is the family

$$\{ n \mapsto a(2^e n + r) : e \geq 0, 0 \leq r < 2^e \}.$$

A finite-valued sequence is 2-automatic exactly when this kernel is finite. The Thue–Morse sequence has a two-state kernel, while iterated prefix sums typically produce finite-dimensional 2-regular rather than finite-valued behavior.

 2π -frequency Fourier transform.

CONVENTION

The 2π -frequency convention is

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx.$$

Under it, translation contributes $e^{-2\pi i a \xi}$, differentiation contributes $2\pi i \xi$, and integer sampling is naturally paired with integer frequencies. The up-function product then uses $\text{sinc}(\pi \xi / 2^n)$ in the radian-normalized sinc.

Type-B Eulerian number.

FRONTIER

Type-B Eulerian numbers count signed permutations by a type-B descent statistic. Their polynomials have different degree, symmetry, and recurrence from the ordinary type-A Eulerian family. The semantic family label prevents a bare bracket notation from merging them.

Typed map.

CLASSICAL

A typed map declaration $f : A \rightarrow B$ includes both domain and codomain. Restrictions, inverses, compositions, and supports depend on these types. Many apparent formula conflicts in the Fabius corpus disappear once one distinguishes a bounded real function, a function on a compact interval, a formal power series, and a polynomial evaluated in a ring. The type is part of the mathematical statement, not implementation decoration.

21 U**Unconditional convergence.**

CLASSICAL

A series converges unconditionally when every permutation of its terms converges to the same value. In finite-dimensional real or complex spaces this is equivalent to absolute convergence, while in general Banach spaces the notions differ. Unconditional convergence licenses regrouping without choosing a privileged order. The repository invokes it for genuinely infinite analytic and Hilbert-space

expansions; locally finite Fabius translate identities are order-independent for the simpler reason that only finitely many summands are nonzero at each point.

Uniform approximation.

CLASSICAL

Uniform approximation means convergence in supremum norm: $\sup_x |f_n(x) - f(x)| \rightarrow 0$. It preserves continuity and provides one error bound for all inputs. The centered finite splines approximate the bounded Fabius function uniformly with an explicit dyadic rate, and the Fourier-Legendre series of the up-function converges uniformly as well. The proofs use different mechanisms: probabilistic coupling for splines and coefficient bounds for polynomials.

Uniform asymptotic expansion.

PROJECT / MIXED

An expansion is uniform in a parameter $u \in U$ when its remainder bound holds with constants independent of u . Uniformity is indispensable near moving saddles, for quantiles away from the endpoints, and for q -limits on compact parameter sets. A pointwise expansion for every fixed u does not imply a uniform expansion.

Uniform convergence.

CLASSICAL

A sequence f_n converges uniformly to f if for every $\varepsilon > 0$ one index works for all points in the domain. Uniform limits of continuous functions are continuous, and uniform convergence permits integration interchange on finite measure domains. It is stronger than pointwise convergence. The Fabius development repeatedly upgrades structural approximations to uniform ones so they can support computability and exact shape arguments.

Uniform distribution.

CLASSICAL

The uniform distribution on $[a, b]$ has constant density $1/(b - a)$ on that interval. Its characteristic function is a phase times a sinc factor, and its moments are elementary rational expressions. Independent dyadically scaled uniforms are the building blocks of the Fabius random series. Centering removes the phases in the product transform and yields the even up-density.

Uniform Legendre convergence.

LEAN-PROVED

Uniform Legendre convergence means

$$\sup_{x \in [-1, 1]} \left| f(x) - \sum_{n=0}^N c_n P_n(x) \right| \rightarrow 0.$$

It permits termwise evaluation and preserves continuity. It is stronger than L^2 convergence; in the repository it follows from an explicit summable majorant for the coefficients, not merely from completeness.

Uniform norm.

CLASSICAL

Uniform norm is another name for the supremum norm on a function space. Convergence in this norm is uniform convergence. It is the metric used for the contraction fixed-point theorem, finite-spline error, and sharp derivative bounds in the Fabius development. On unbounded domains, boundedness or constant tails ensure the norm is finite.

Uniform random variable.

CLASSICAL

A uniform random variable on $[a, b]$ assigns probability proportional to interval length. It can be realized as $a + (b - a)U$ with $U \sim \text{Unif}[0, 1]$. Boundedness gives all moments and exponential moments. The Fabius law is assembled from independent uniform variables with geometrically shrinking coefficients, so finite partial sums have spline densities.

Uniform spline.

CLASSICAL

A uniform spline has knots on an equally spaced grid. Cardinal B-splines are the standard examples. Although the original Fabius summands occur at different dyadic scales, a common refinement converts finite convolutions into uniform-grid splines with explicit coefficients. Centered uniform splines form the constructive approximation used in the computability theorem.

Unimodality.

CLASSICAL

A function or density is unimodal if it increases up to a mode and decreases afterward, allowing nonstrict plateaus in the basic definition. Strict unimodality requires strict inequalities away from the unique mode. The up-function is strictly unimodal with mode zero. Consequently the Fabius density rises then falls, and the CDF changes from convexity to concavity at its unique midpoint inflection.

Unique midpoint inflection.

LEAN-PROVED

The bounded Fabius CDF changes from strict convexity to strict concavity at $x = 1/2$, its unique inflection point. Through $F''(x) = 4 \text{up}'(2x - 1)$, this is equivalent to strict unimodality of the up-density. Reflection forces the midpoint location, while strict shape theorems exclude additional inflections.

Uniqueness of the periodic correction.

PROJECT / MIXED

Suppose two continuous one-periodic functions inserted into the same Fabius endpoint main term both leave an error tending to zero. Their difference, evaluated along the unbounded lower-Lambert coordinate, tends to zero. Because every phase modulo one is approached by suitable endpoint sequences, periodicity and continuity force the difference to vanish identically. Thus the correction is uniquely determined, not merely one convenient Fourier representation.

Unit-interval Fabius-up translate.

LEAN-PROVED

On the unit interval, the bounded Fabius function is an affine translate of the up-function, with the exact formula fixed by the chosen normalization. The identity is not global because F has constant tails whereas up is compactly supported and even. Every use outside the stated interval must instead employ the fold or density relation.

Unsigned Stirling numbers of the first kind.

CLASSICAL

The unsigned Stirling number $\begin{bmatrix} n \\ k \end{bmatrix}$ counts permutations of n elements having exactly k cycles. It equals $(-1)^{n-k}s(n, k)$. These numbers appear in logarithmic derivatives, rising-factorial expansions, and analytic continuations of Bell-related sequences.

Up-synthesis amplitude.

CONVENTION

An up-synthesis amplitude is the scalar multiplying one translated/dilated up-atom. It depends on the target polynomial, scale, centers, and chosen normalization of the atom. Amplitudes are not probability weights in general: they may be signed and need not sum to one unless constant reproduction forces that condition.

22 V**Valuation.**

CLASSICAL

A discrete prime valuation $v_p(r)$ records the exponent of a prime p in a nonzero rational number. It is additive on products and satisfies a non-Archimedean inequality on sums. Valuations translate divisibility into arithmetic inequalities and identify reduced denominators. The Fabius arithmetic emphasizes v_2 , but odd-prime valuations also underlie least-common-multiple denominator bounds.

Valuation histogram.

FRONTIER

A valuation histogram counts how many entries in a finite family have each p -adic valuation. For a Pascal row it refines the count of coefficients nonzero modulo p ; for rational Fabius data it records denominator depth. Generating functions of these histograms reveal digit recursions and limiting distributions.

Vandermonde determinant.

CLASSICAL

For distinct nodes $x_0, \dots, x_n$, the Vandermonde determinant is $\det(x_i^j)_{0 \leq i, j \leq n} = \prod_{i < j} (x_j - x_i)$, which is nonzero. It proves uniqueness of polynomial interpolation and invertibility of the evaluation map. On geometric nodes, its factors split into powers of two and q -Pochhammer products, explaining Gaussian interpolation weights in the dyadic formulas.

Vanishing moment.

CLASSICAL

A signed measure, sequence, or function has vanishing moments through order $m - 1$ if its integrals or sums against x^r vanish for $r < m$. Vanishing moments annihilate low-degree polynomials and improve approximation or decay. Complete Thue–Morse blocks have vanishing discrete moments up to an order determined by block length. This is the Prouhet cancellation used in Fabius finite formulas.

Varentropy.

FRONTIER

Varentropy is the variance of the information content $-\log f(X)$ for a random variable with density f :

$$V(f) = \text{Var}[-\log f(X)].$$

It refines differential entropy and appears as a second-order fluctuation quantity. Compact support and smoothness do not automatically guarantee finiteness because f vanishes at the boundary.

Variable substitution.

CLASSICAL

Variable substitution transforms an integral or equation by a change of coordinates and includes the appropriate Jacobian. Affine substitutions connect F and u ; logarithmic substitution converts dilation to translation; saddle substitution centers and scales the Bromwich contour. A flawed stationary substitution can leave an unaccounted linear term, so exact algebra and error propagation are essential.

Variance.

CLASSICAL

The variance of a square-integrable random variable is $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2$. Variances add for independent variables. It is the second cumulant and sets the Gaussian scale in central approximations. For the dyadic uniform random series, the variance is an explicit geometric sum and is encoded among the centered moments.

Vertical Bromwich line.

CLASSICAL

A vertical Bromwich line is the contour $\Re s = c$ used in inverse Laplace integration, oriented from $c - i\infty$ to $c + i\infty$. The constant c is chosen inside the transform's inversion region and to the right of relevant singularities. Saddle analysis may shift or deform this line after proving analyticity and decay. Orientation and the factor $1/(2\pi i)$ determine the sign and normalization.

Volterra integral operator.

CLASSICAL

A Volterra operator has triangular kernel

$$(V_K f)(x) = \int_a^x K(x, t) f(t) \, dt.$$

Its iterates integrate over ordered simplices, and under standard bounds its resolvent is represented by a Neumann series. Fabius integral recursions, stopped primitives, and fractional integrals are special Volterra constructions.

Volterra resolvent.

CLASSICAL

For $I - \lambda V_K$, the Volterra resolvent is the inverse operator when it exists, often written as $I + \sum_{n \geq 1} \lambda^n V_K^n$. Triangular integration frequently gives convergence without a smallness condition on λ over a compact interval because iterates acquire factorial denominators. Resolvents organize integral-equation fixed points and perturbations.

23 W

Walsh–Hadamard transform.

CLASSICAL

The Walsh–Hadamard transform replaces complex exponential characters of a cyclic group by $\{\pm 1\}$ -valued characters of the binary cube. It is computed by repeated butterfly additions and subtractions. Thue–Morse signs are a highest-order Walsh character, so Prouhet cancellation and Boolean mixed differences have a natural Walsh interpretation.

Wasserstein distance.

CLASSICAL

The p -Wasserstein distance minimizes $(\int |x - y|^p d\pi)^{1/p}$ over couplings π of two laws. In one dimension it can be computed from quantile functions. Coupling a full weighted random series with its finite truncation on the same probability space gives an immediate Wasserstein bound by the deterministic or L^p tail size.

Weak composition.

CLASSICAL

A weak composition of n into r parts is an r -tuple of nonnegative integers summing to n . There are $\binom{n+r-1}{r-1}$ such tuples. Weak compositions index repeated convolutions, coefficient extractions, Bell-polynomial expansions, and ordered distributions of derivative order.

Weak convergence.

CLASSICAL

Probability measures μ_n converge weakly to μ if $\int f d\mu_n \rightarrow \int f d\mu$ for every bounded continuous f . Characteristic-function convergence often proves weak convergence. Finite convolution laws of the truncated Fabius random series converge weakly to the infinite law because the random variables converge almost surely and remain uniformly bounded. Additional regularity yields convergence of densities or CDFs.

Weak-star convergence.

CLASSICAL

Weak-star convergence of finite measures tests convergence against continuous compactly supported functions, viewing measures as dual objects. For probability measures on a compact set it closely matches weak convergence. The repository's early measure bridge uses weak-star convergence of convolution measures before identifying the smooth density and uniform CDF limit. The star distinguishes dual-space convergence from weak convergence inside the primal function space.

Weierstrass M-test.

CLASSICAL

The Weierstrass M-test states that if $|f_n(x)| \leq M_n$ for every x and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly. On compact subsets it proves normal convergence of holomorphic series. The repository uses summable majorants for Fourier modes, global binary summands, and transform logarithms. Uniform convergence then licenses continuity and termwise operations.

Weierstrass product.

CLASSICAL

A Weierstrass product represents an entire function by factors prescribed by its zeros, with canonical exponential corrections ensuring convergence. Finite-genus products generalize polynomial factorization to infinite zero sets. The Rvachev sinc product is an especially explicit normally convergent product whose dyadic factors expose zeros and multiplicities directly; reciprocal-Gamma and theta products in the enlarged repository provide further examples.

Weighted path expansion.

CLASSICAL

A weighted path expansion unfolds a recurrence as a sum over directed paths through its dependency graph, multiplying transition weights along each path. It converts recursive definitions into finite closed sums indexed by compositions or subsets. Saddle coefficient and moment recurrences can be represented this way. In Lean, path expansions can make positivity, denominator structure, and coefficient dependence easier to prove than repeated symbolic substitution.

Weighted random sum.

CLASSICAL

A weighted random sum is $\sum a_n X_n$. Its mean, variance, support, transforms, and convergence depend on both weights and component laws. Geometric weights give self-similarity and compact support when the variables are bounded. The Fabius law is the prototypical infinite weighted sum of independent uniforms, with base-two weights producing exact dyadic structure.

Weighted-uniform law.

LEAN-PROVED

A weighted-uniform law is the distribution of a convergent sum of independent uniforms with a general summable weight sequence. It extends the geometric family and admits transform, support, expectation, variance, and continuous-linear mapping theorems under summability assumptions. Special self-similarity and exact q -identities arise only when the weights are geometric.

Well-based support.

PROJECT / MIXED

A support is well based when it contains no infinite sequence proceeding indefinitely toward larger monomials in the chosen dominance order. This condition guarantees that the coefficient of a fixed monomial in a product is a finite sum. It is the transseries analogue of the lower-bounded support of a formal Laurent series.

Well-founded recursion.

CLASSICAL

Well-founded recursion defines an object at x from values at arguments strictly smaller than x in a relation having no infinite descending chain. It generalizes ordinary induction on natural numbers. Fast dyadic evaluators, binary reductions, and recursive numerator algorithms use measures such as bit length, denominator level, or a positive grid numerator to justify termination. In formal proofs, supplying the decreasing measure is part of the algorithm rather than an informal assurance.

Well-poised basic hypergeometric series.

CLASSICAL

A basic hypergeometric series is well poised when its parameters satisfy a symmetric set of multiplicative relations that pair numerator and denominator factors. Very-well-poised series impose an additional square-root pattern. These conditions create cancellation and transformation formulas; they must be printed because the adjectives alone do not determine a convention.

Wiener or orthogonal chaos.

FRONTIER

An orthogonal chaos decomposition writes $L^2(\Omega) = \bigoplus_{n \geq 0} \mathcal{H}_n$ relative to a specified independent-coordinate or Gaussian structure. The projection J_n selects the degree- n component. For dyadic uniform coordinates the chaos basis is built from the orthogonal polynomials of the uniform law, not Gaussian Hermite polynomials.

Wigner $3j$ -symbol.

PROJECT / MIXED

A Wigner $3j$ -symbol packages angular-momentum coupling coefficients. For zero magnetic indices,

$$\int_{-1}^1 P_\ell P_m P_n \, dx = 2 \begin{pmatrix} \ell & m & n \\ 0 & 0 & 0 \end{pmatrix}^2.$$

The symbol vanishes under the same parity and triangle obstructions as the Gaunt coefficient. The square removes its sign in this specialization.

Wright omega function.

PROJECT / MIXED

Wright's omega function solves

$$\Omega(z) + \log \Omega(z) = z$$

with a chosen logarithm branch, equivalently $\Omega(z) = W(e^z)$ under compatible continuation. At large z it has a polynomial–logarithmic expansion $z - \log z + \dots$. It removes one outer exponential from many Lambert inversion problems.

24 Z**Zero-bias transform.**

FRONTIER

For a mean-zero random variable X with variance $\sigma^2 > 0$, a zero-biased variable X^* satisfies

$$\mathbb{E}[X f(X)] = \sigma^2 \mathbb{E}[f'(X^*)]$$

for an appropriate class of absolutely continuous f . The star denotes a distributional transform, not complex conjugation. Iterated zero-bias constructions require recentering and variance bookkeeping.

Zero of an entire function.

CLASSICAL

A zero of order m of an entire function F at z_0 satisfies $F(z) = (z - z_0)^m G(z)$ with $G(z_0) \neq 0$. Zeros are isolated unless the function is identically zero. The sinc product locates transform zeros at dyadically structured frequencies, with multiplicities determined by how many factors vanish.

These zeros imply exact Fourier coefficient vanishings, partitions of unity, and odd-harmonic cosine expansions.

Zero run.

CLASSICAL

A zero run is a consecutive block of zero digits or zero values. In binary recursions, long zero runs correspond to inactive dyadic scales and can be skipped by highest-set-bit algorithms. In coefficient sequences, a zero run may reflect parity, support, or finite-degree annihilation. The Fabius binary evaluator exploits bit structure rather than iterating through every zero position.

Zeta regularization.

CLASSICAL

Zeta regularization assigns finite values to certain divergent products or sums by analytic continuation of a zeta-type function. It is conceptually related to, but distinct from, the Hadamard finite part used in the Fabius Mellin analysis. The repository's Gamma–zeta coefficients arise from ordinary meromorphic continuation and residue calculus; no arbitrary assignment of a divergent original series is intended. The term is included because zeta-based regularization language appears in discussions of finite parts and must be distinguished carefully.

Zeta-regularized determinant.

FRONTIER

When a spectral zeta function is regular at $s = 0$, the zeta-regularized determinant is $\det_{\zeta} L = \exp(-\zeta'_L(0))$. It extends finite products of positive eigenvalues but is not an ordinary convergent product in general. Scale changes contribute explicit powers governed by $\zeta_L(0)$.

Zonoid.

FRONTIER

A zonoid is a Hausdorff limit of zonotopes, equivalently a convex body represented by an even measure on directions through its support function. Infinite sums of shrinking random segments suggest zonoid support sets in multivariate atomic-function generalizations.

Zonotope.

FRONTIER

A zonotope is a Minkowski sum of finitely many line segments. It is the image of a cube under a linear map. Finite sums of independent uniform segment variables have box-spline densities supported on zonotopes, linking multivariate Rvachev-type laws to spline geometry.

A Normalization and notation guardrails

The glossary is descriptive, while the live notation catalogue is normative. The following table records the distinctions most likely to corrupt a theorem when a formula is copied between documents.

Object	Canonical form	Guardrail
Bounded Fabius function	$F : \mathbb{R} \rightarrow [0, 1]$	Clamped CDF: $F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$.
Global signed extension	$\mathcal{F} : \mathbb{R} \rightarrow \mathbb{R}$	Agrees with F on $[0, 1]$; satisfies the global dilation–differential law.

Object	Canonical form	Guardrail
Rvachev up-function	$\text{up}(x) = F(1 - x)$	Even bump, support $[-1, 1]$, $\text{up}(0) = 1$; not the bounded CDF.
Ordinary Fabius inverse	$F^{-1} : [0, 1] \rightarrow [0, 1]$	Inverse of the strictly increasing restriction $F _{[0,1]}$.
Clamped Fabius inverse	$F^{-1}(\min\{1, \max\{0, y\}\})$	Global real input, constant outside the unit ordinate interval.
Thue–Morse bit/sign	$t_n = s_2(n) \bmod 2, \varepsilon_n = (-1)^{t_n}$	Bit values 0, 1 and signs ± 1 are different types.
Radian sinc	$\text{sinc } z = \sin z / z$	The argument already contains any required factor of π .
2π -Fourier transform	$\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \xi} dx$	Integer sampling and frequency lattices use this convention.
Angular Fourier transform	$\widehat{f}(\omega) = \int f(x) e^{-i\omega x} dx$	Related by $\widehat{f}_{2\pi}(\xi) = \widehat{f}_{\text{ang}}(2\pi\xi)$.
Finite q -Pochhammer	$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j)$	Length $n \in \mathbb{N}$; the empty product at $n = 0$ is one.
Infinite q -Pochhammer	$(a; q)_\infty = \prod_{j \geq 0} (1 - aq^j)$	Analytic product for $ q < 1$; zeros and logarithm branches are tracked.
Gaussian coefficient	$\begin{bmatrix} n \\ k \end{bmatrix}_q$	Canonical polynomial continuation, not merely a quotient away from roots of unity.
Geometric-uniform law	$(1 - q) \sum_{j \geq 0} q^j U_j$	Support-normalized on $[0, 1]$; centered and standardized variants are separate.
Legendre polynomial	$P_n(1) = 1$	Classical, neither monic nor orthonormal; norm squared $2/(2n + 1)$.
Lambert branches	W_0 and W_{-1}	Principal and lower real branches; endpoint saddles require the branch stated by the theorem.

B Representative Lean crosswalk

This table is intentionally representative rather than a substitute for the 675-module generated inventory. It gives high-value entry points from mathematical terminology to active module families. Exact declaration names and imports remain controlled by the repository facade and generated census.

Module	Mathematical interface
<code>Basic.lean</code>	<code>IsFabius</code> , foundational tails/reflection/differential hypotheses
<code>Differential.lean</code>	iterated derivative scaling and dyadic derivative formulas
<code>Arithmetic.lean</code>	exact Fabius arithmetic and recurrence infrastructure
<code>Existence.lean</code>	existence of the canonical bounded Fabius object
<code>OriginalUniqueness.lean</code>	original compact-support predicate and uniqueness bridges
<code>FabiusInverse.lean</code>	<code>fabiusInv</code> , strict monotonicity, order isomorphism
<code>GlobalExtension.lean</code>	global signed extension, translate cells, global calculus
<code>FourierProduct.lean</code>	infinite sinc-product identity
<code>FourierAnalytic.lean</code>	analytic Fourier consequences and zero structure
<code>ProbabilityRepresentation.lean</code>	random-series law and distributional representation
<code>StepApproximationLimit.lean</code>	finite spline/step approximants and convergence
<code>FabiusComputable.lean</code>	effective evaluation and computability interface
<code>FabiusLogScale.lean</code>	endpoint logarithmic scale
<code>FabiusLogSquaredAsymptotic.lean</code>	quadratic logarithmic endpoint law

Module	Mathematical interface
NegativeLaplace.lean	negative-Laplace product and decomposition
MellinBose.lean	Mellin/Bose kernel and finite-part infrastructure
PeriodicCorrection.lean	period-one endpoint correction
FabiusLambertPhase.lean	Lambert phase coordinates
BromwichSaddle.lean	Bromwich inversion and saddle localization
FabiusSharpAsymptotic.lean	sharp endpoint asymptotic
FabiusFullAsymptoticExpansion.lean	all-orders endpoint expansion
ThueMorseAutocorrelation.lean	finite-window and dyadic autocorrelations
ThueMorseAutocorrelationLimit.lean	normalized limit recursion
QPochhammerEntire.lean	finite/infinite q -product analytic structure
GeometricPochhammerNormalConvergence.lean	normal convergence on compact sets
QPochhammerDissection.lean	residue-class product dissections
QPochhammerInfinite.lean	infinite q -Pochhammer identities
GaussianBinomialPolynomialStructure.lean	polynomiality and coefficient structure
GaussianBinomialPalindromic.lean	reciprocal-base palindromy
GaussianBinomialAtNegOneDerivative.lean	value and derivative phenomena at $q = -1$
QBinomialTheoremInfinite.lean	analytic infinite q -binomial theorem
BasicHypergeometricSeries.lean	normalized ${}_r\phi_s$ series
HeineTransformation.lean	Heine's transformation
QGaussSummation.lean	q -Gauss summation
QPfaffSaalschutz.lean	terminating balanced ${}_3\phi_2$ summation
JacksonIntegral.lean	Jackson integral and q -calculus identities
ThetaQuasiPeriodicity.lean	multiplicative theta transformation law
JacobiTripleProduct.lean	bilateral series/product identity
QLucas.lean	root-of-unity digit reduction
CyclotomicDivisibility.lean	cyclotomic exponents and divisibility
QBetaIntegral.lean	q -beta integral

C Coverage map and edition delta

The previous live glossary exposed 437 headwords. This edition exposes 976, a net increase of 539. The increase is not a mechanical declaration dump: synonyms and theorem-name variants are merged, while materially different normalizations, branches, equality modes, and research statuses receive separate entries.

Broad domain	Headwords
Foundations, formalization, notation, and general analysis	320
Thue–Morse, dyadic arithmetic, and coefficient calculus	134
Probability, moments, operators, and information	99
Approximation, splines, orthogonal systems, and representations	92
Asymptotics, inversion, transseries, and dynamics	87
q -series, special functions, and geometric calculus	87
Fourier, transforms, sampling, and spectra	82
Fabius, inverse Fabius, and Rvachev core	75
Total	976

C.1 Status distribution

Status	Headwords
Classical	468
Lean-proved	179
Project/mixed	151
Frontier	120
Convention	56
Historical	2

D Audit method and maintenance contract

The update was produced by comparing the prior glossary against the current repository facade, generated declaration census, notation catalogue, primary exposition, Lean walkthrough, active frontier synthesis, and recursive active TeX inventory. The audit normalized aliases before counting concepts. It then screened each retained or added entry for the following fields:

1. exact mathematical object and argument types;
2. domain, codomain, support, and endpoint conventions;
3. characteristic definition or formula;
4. normalization, transform convention, and branch data;
5. pointwise, almost-everywhere, formal, or asymptotic equality mode;
6. current proof status and the boundary between Lean and frontier prose;
7. collisions with the bounded/global/up triad, bit/sign conventions, Fourier normalizations, and overloaded one-letter symbols.

A future glossary update is incomplete unless it reruns the module/declaration census, the strict notation audit, the active-TeX inventory, alias checks, and rendered-PDF inspection. Numerical evidence and formal transseries algebra must never be silently promoted to an analytic or kernel-checked theorem.

E Repository sources inspected

All repository links below were accessed on 3 September 2026 from the main branch. The formal module/declaration census was the repository's 1 September 2026 checkpoint; the active TeX and notation counts were re-derived by the notation catalogue on 3 September 2026.

Source family	Location
Live project root	https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction
Current documentation root	https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs
Prior glossary	https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/FabiusFunction_Mathematical_Glossary
Mathematical notation catalogue	https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/FabiusFunction_Mathematical_Notation_Catalogue

Source family	Location
Primary exposition	https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up
Lean walkthrough	https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/fabius_lean_walkthrough
Semi-formalized frontier synthesis	https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers

F Selected external reference works

References

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- [7] C. de Boor, *A Practical Guide to Splines*, revised edition, Springer, 2001.
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Snapshot limitation. The repository is a living project. This glossary is complete relative to the active main-branch corpus and generated inventories retrieved on the dates printed above. Later modules or documentation changes require a fresh audit; archived terminology remains historical unless promoted back into the active tree.
